

Drilling Engineering Introduction

02 May 2026

- 1 Drilling History
- 2 Drilling and Rig Type
- 3 Well Construction
- 4 Drilling Cost and Procurement
- 5 Drilling Risk and Problem
- 6 Exercises



1. Just want to know Drilling
2. I would like become Drilling Engineer → **MANAGER**
3. Why drilling **is easy (?)** but drilling cost so expensive?
4. What are Drilling needs and how to **PROCURE?**
5. I would like to marry a Driller



To Be Drilling Engineer

Drilling Engineering

- Casing & Wellhead design
- Directional Drilling, BHA, T & D
- Bit Study and Design
- Basic Rig Selection
- Drilling Program
- Drilling AFE
- AFE Close Out

Drilling Operation

- Drilling Operation sequences and procedure
- Drilling Risk Assessment
- Post Drilling Review
- Final Drilling Report

Drilling Support

- Drilling Equipment and Services
- Market Survey
- Bid document template
- Tender Evaluation
- Price negotiation

To Be Drilling Specialist

- Extended Reach Well design
- Advance casing design
- Offshore well design
- Rig Sizing and Selection
- Oil and Gas and Geothermal Design
- Publish drilling papers
- Work Over project

- Critical Drilling and WO rig layout
- Drilling personnel networking
- Stuck Pipe Prevention study
- UBD - MPD
- Managing Drilling Fluid
- Drilling Waste handling
- Drilling Social Engineering

- Most efficient drilling cost structure
- Effective drilling procurement method
- Good relationship with main Drilling Contractor
- Good relationship with main Regulator

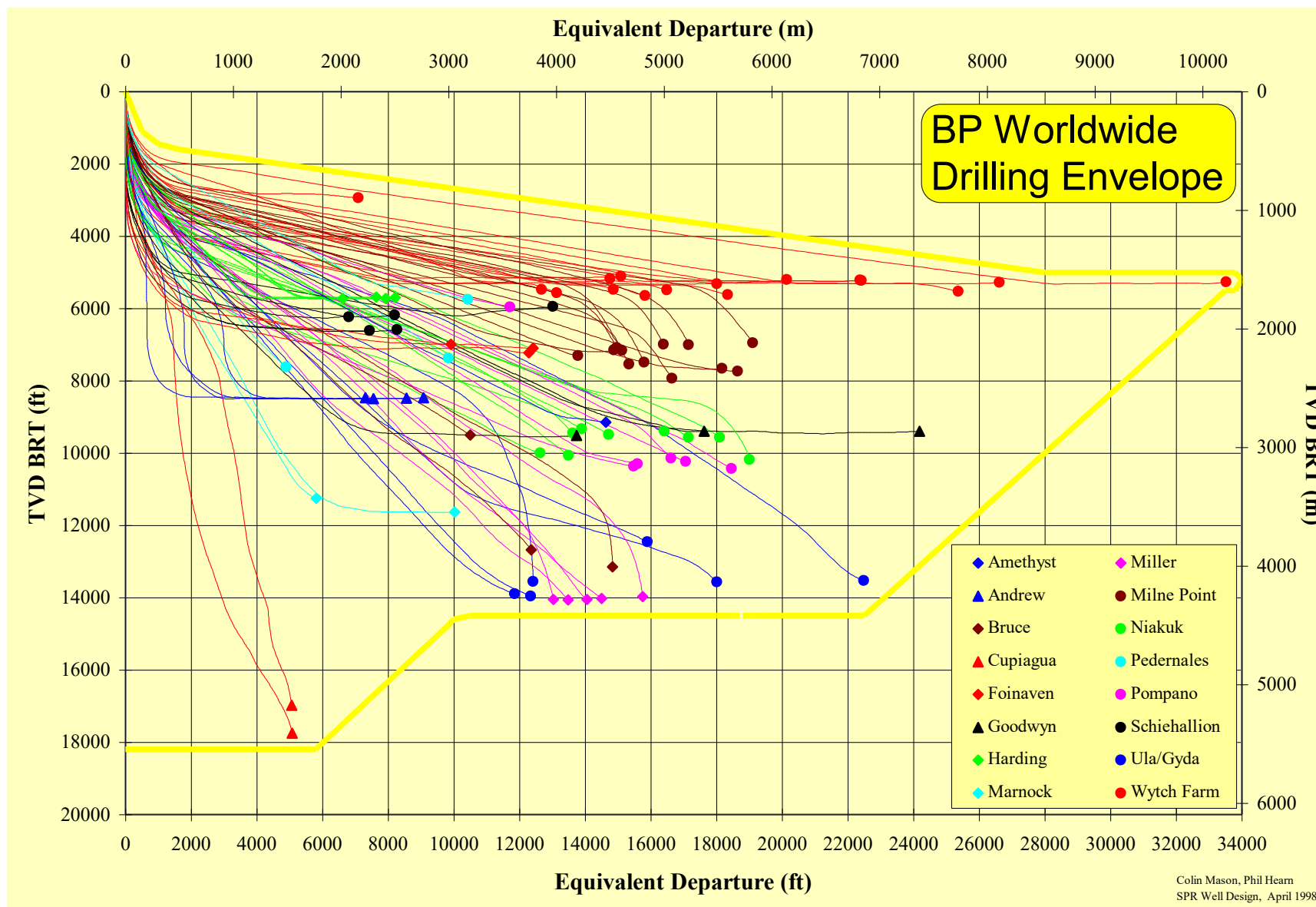
Tahun	Sejarah Perminyakan Indonesia
17 Nov 1884	Zijlker melakukan pengeboran 1 sumur Telaga Tiga berlangsung sekitar dua bulan, minyak yang diperoleh hanya sekitar 200 liter. Semburan gas yang cukup tinggi dari sumur Telaga Tiga, membuyarkan harapan untuk mendapatkan minyak yang banyak.
1885	Pemboran sumur ke 2, sumur Telaga Tunggal I : Saat pengeboran mencapai kedalaman 22 meter, berhasil diperoleh minyak sebanyak 1.710 liter dalam waktu 48 jam kerja. Saat mata bor menyentuh kedalaman 31 meter, minyak yang dihasilkan sudah mencapai 86.402 liter. Jumlah itu terus bertambah hingga pada 15 Juni 1885, ketika pengeboran mencapai kedalaman 121 meter, tiba-tiba muncul semburan kuat gas dari dalam berikut minyak mentah dan material lainnya dari perut bumi. Semburan minyak dari Sumur Telaga I jadi MOMENTUM PERTAMA KEBERHASILAN PENAMBANGAN MINYAK DI INDONESIA
1934	Telaga Tunggal I itu sendiri akhirnya berhenti operasi setelah habis minyaknya diproduksi oleh pemerintah Belanda yang mengelola ladang minyak ini melalui perusahaan Bataafsche Petroleum Matschappij (BPM)

1. Drilling History

Tahun	Sejarah Perminyakan Indonesia
1871	Ekplorasi di daerah Cirebon – tidak berhasil
1881	Ekplorasi di North Sumatra, Deli – berhasil
1885	Produksi minyak pertama, dari sumur Telaga Tunggal
1890	Pembangunan Cepu dan Wonokromo Refinery
1892	Pembangunan Pangkalan Brandan Refinery
1894	Pembangunan Balikpapan Refinery
1907	Royal Dutch Co & Shell Transport and Trading merger menjadi Bataafsche Petroleum Maatschapij (BPM), satu-satunya perusahaan minyak di Indonesia sampai tahun 1911
1912	Stanvac beroperasi di Talang Akar dan Pendopo, Sumatra Selatan
1930	Caltex beroperasi di Minas dan Duri
1935	NV NIAM (perusahaan Belanda) beroperasi
1945	P. Brandan & Aceh (ex BPM) dikuasi pemerintah Indonesia dan dioperasikan oleh PT MNRI
1954	PT MNRI berubah menjadi TMSU (Tambang Minyak Sumatra Utara). 1961: TMSU → PN Permina
1960	NV NIAM berubah menjadi PT Perminco. 1961: PT Perminco → PN Pertamina
1968	PN Permina dan PN Pertamina bergabung menjadi PN Pertamina

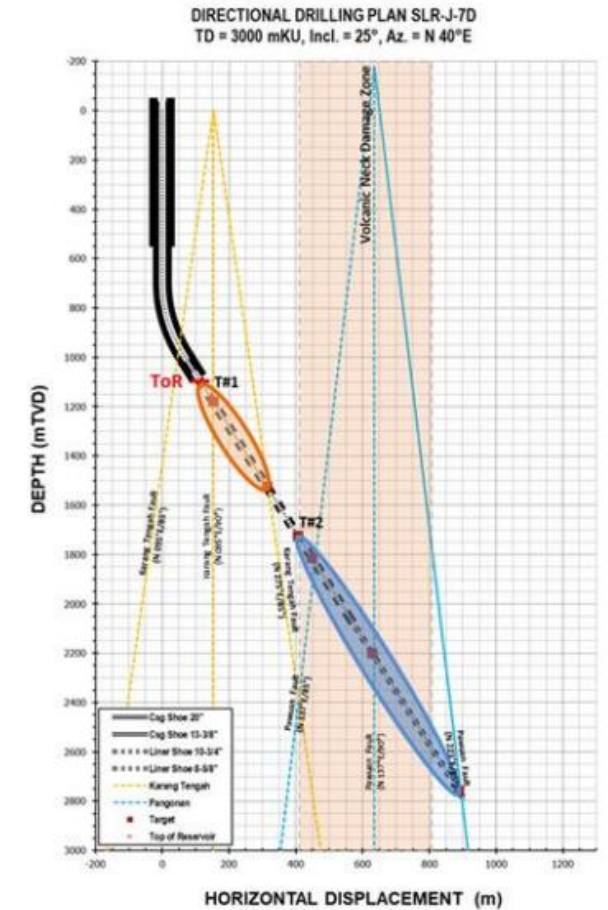
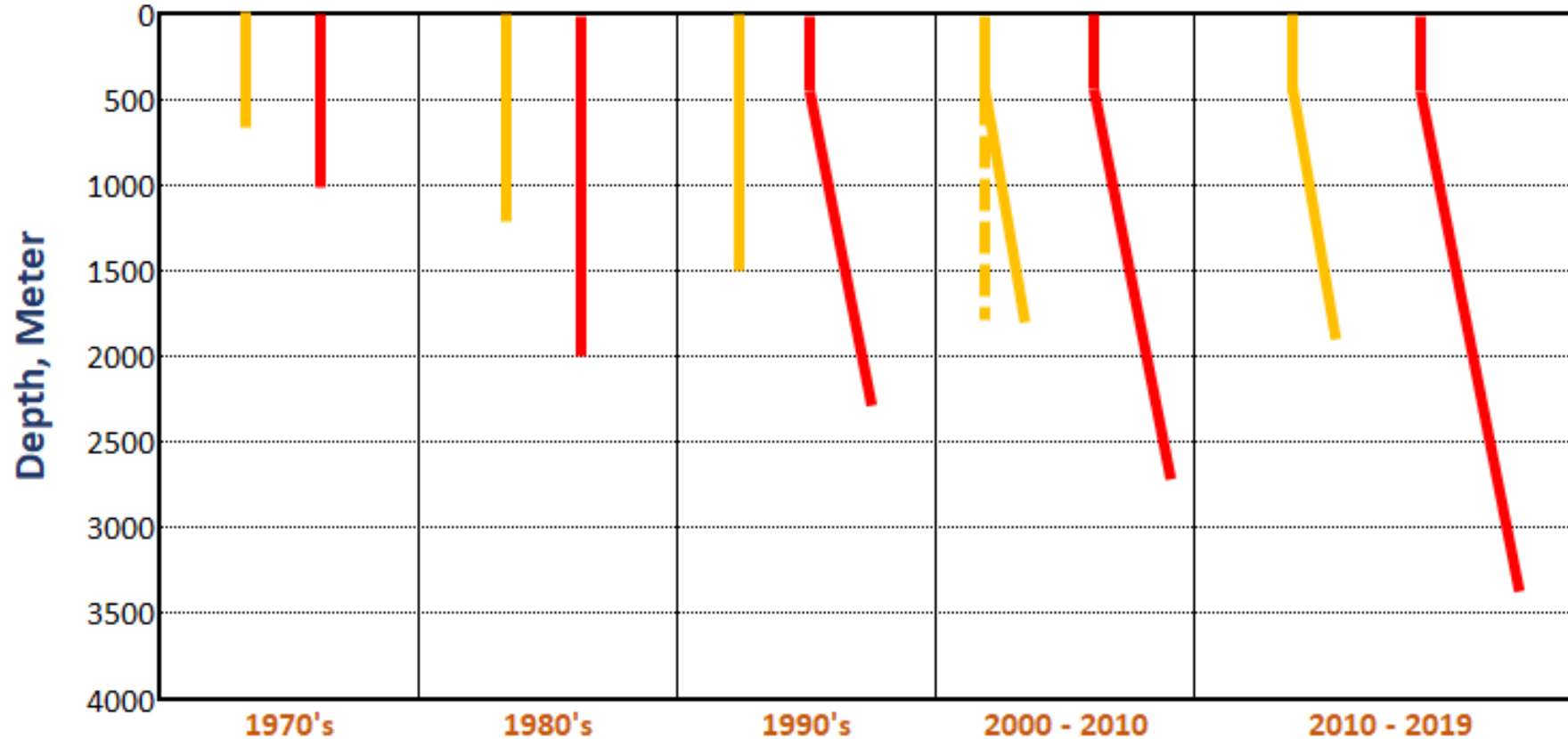
Tahun	Sejarah Geothermal Indonesia
1918	Usulan JB Van Dijk untuk memanfaatkan sumber energi panasbumi di daerah kawah Kamojang, Jawa Barat, merupakan titik awal sejarah perkembangan panasbumi di Indonesia
1926 - 1928	Lapangan panasbumi Kamojang, dengan sumurnya bernama KMJ-3, yang pernah menghasilkan uap pada tahun 1926, merupakan tonggak pemboran eksplorasi panasbumi pertama oleh Pemerintah kolonial Belanda. Sampai sekarang, KMJ-3 masih menghasilkan uap alam kering dengan suhu 140C dan tekanan 2,5 atmosfer (atm).
1972	Telah dilakukan pemboran pada enam buah sumur panasbumi di pegunungan Dieng, dengan kedalaman mencapai 613 meter. Tidak ditemukan uap
1982	Kegiatan Panas Bumi Lahendong
1982	Pertamina menandatangani kontrak pengusahaan panasbumi dengan Unocal Geothermal of Indonesia (UGI) untuk sumur panasbumi di Gunung Cisalak, Jawa Barat
1994	Ditandatangani kontrak pengusahaan panasbumi antara Pertamina dengan empat perusahaan swasta: Wayang Windu, Jawa Barat (PT Mandala Nusantara), Karaha, Jawa Barat (PT Karaha Bodas Company), Dieng, Jawa Tengah (PT Himpurna California Energy), dan Patuha, Jawa Barat (PT Patuha Power Limited)
1995	Penandatanganan kontrak (JOC & ESC) Pertamina Bali Energy Limited dan PT PLN (Persero) untuk pengusahaan dan pemanfaatan panasbumi di daerah Batukahu, Bali

1. Drilling Record - Oil and Gas Directional Well



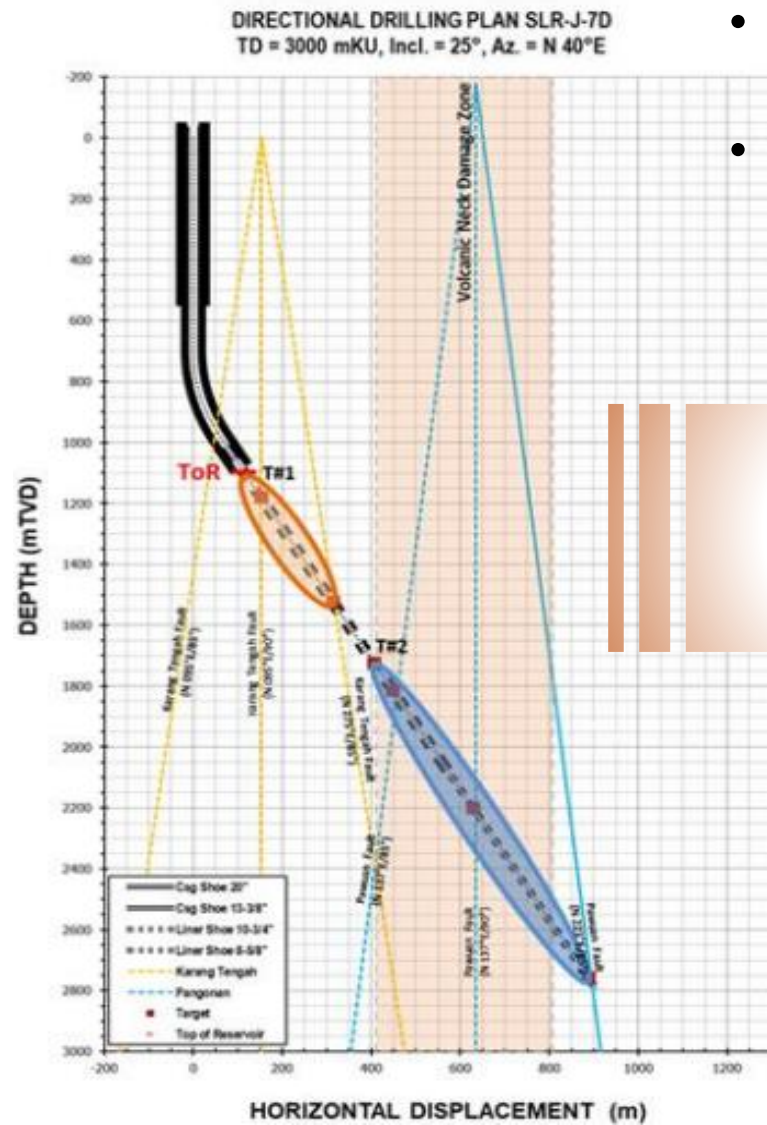
1. Drilling History – Geothermal Drilling Depth in Indonesia

Well Depth and Profile



SLR-J-7D
TD 3009 m

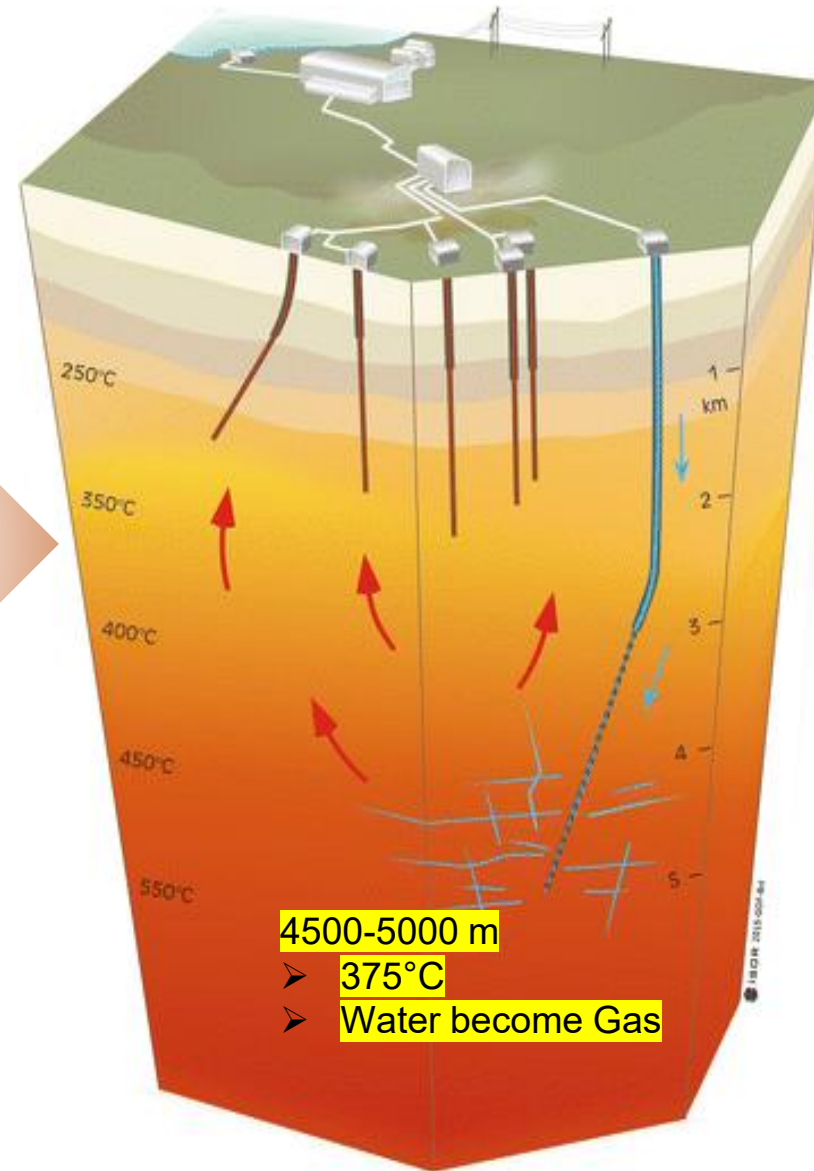
1. Future Deep Geothermal Drilling for Supercritical System



SLR-J-7D: TD 3009 m, 340°C

- Steam and Brine
- Scale problem

Future Prospect

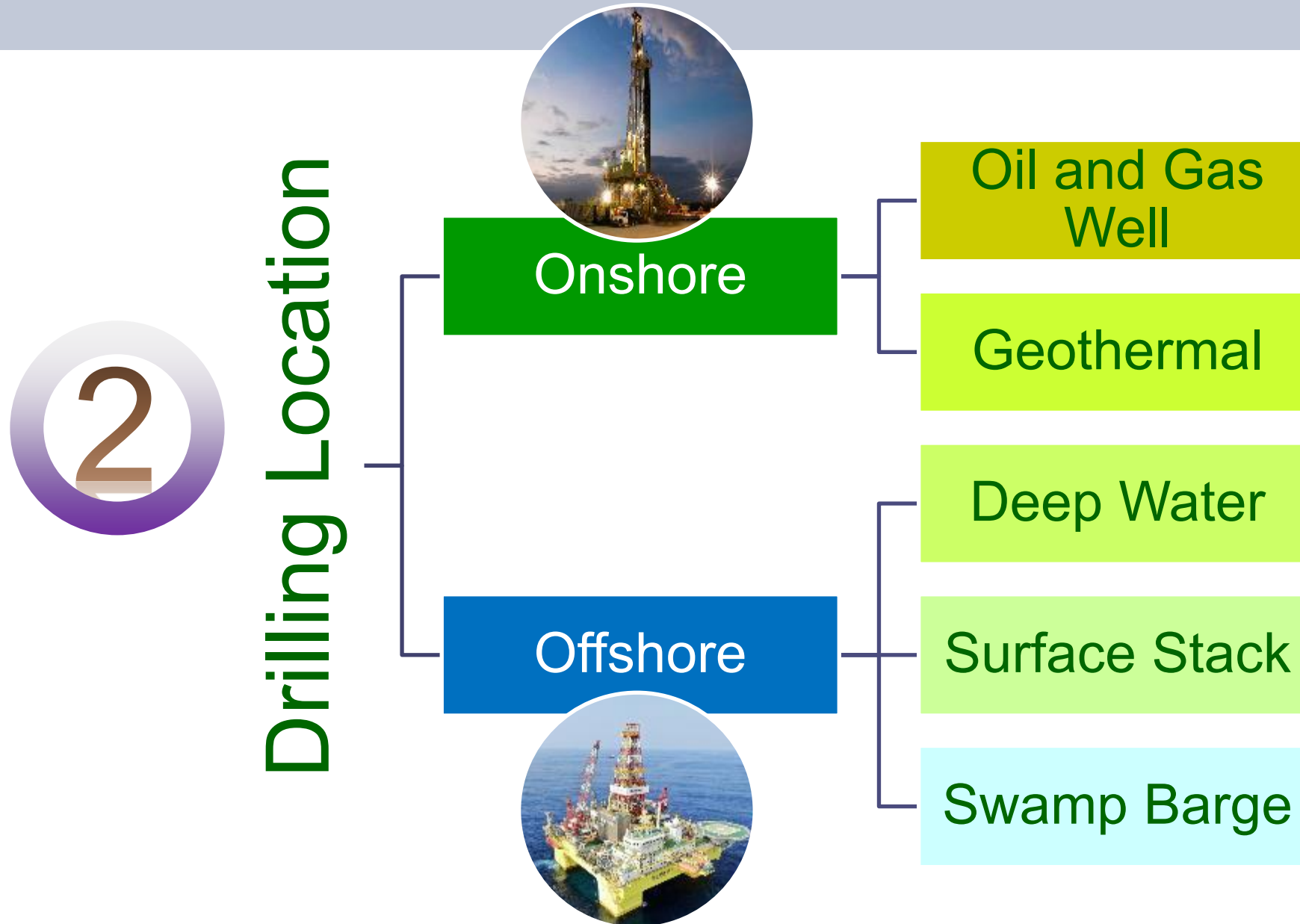


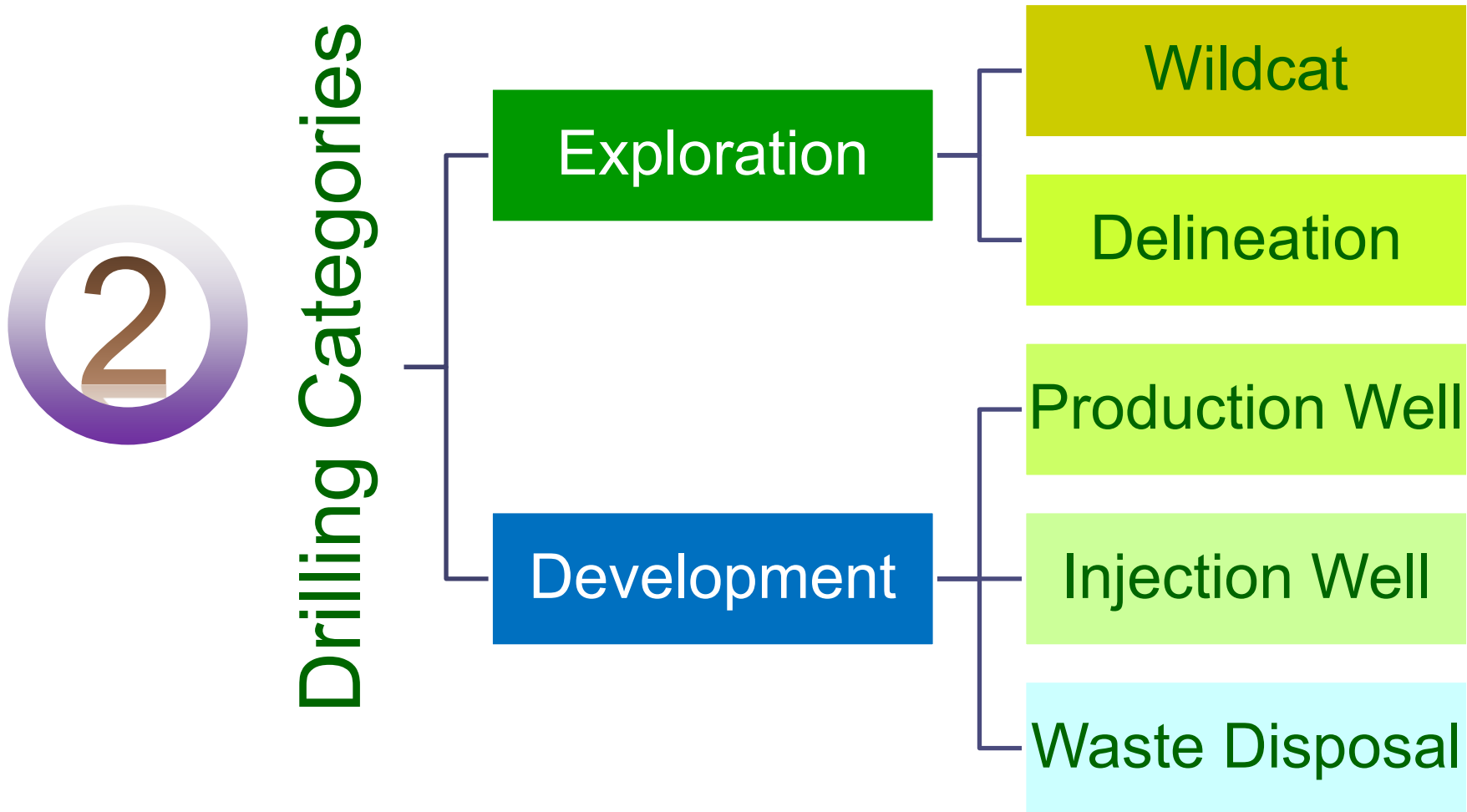
4500-5000 m

➤ 375°C

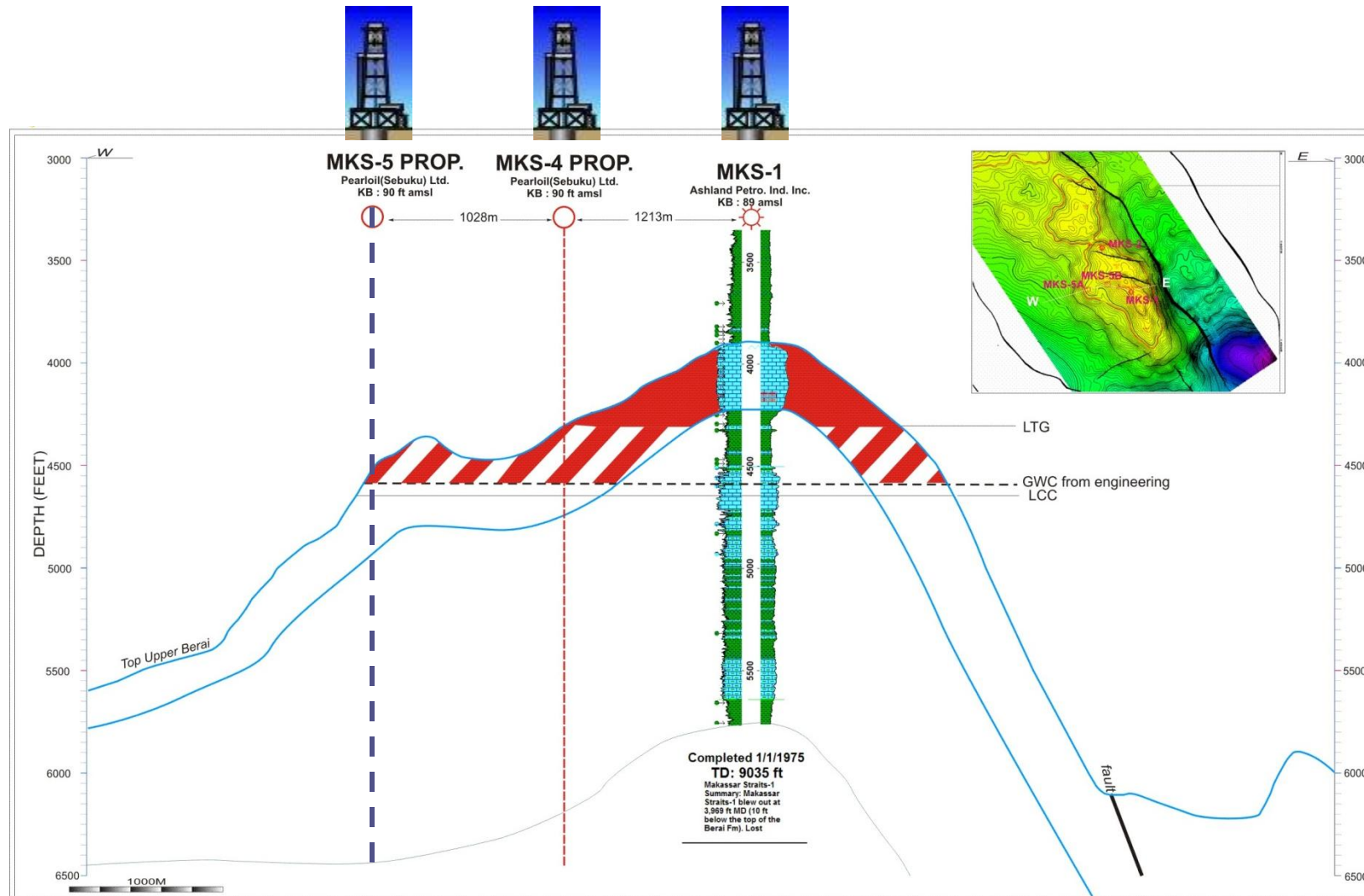
➤ Water become Gas

- Water Gas
- No scale problem



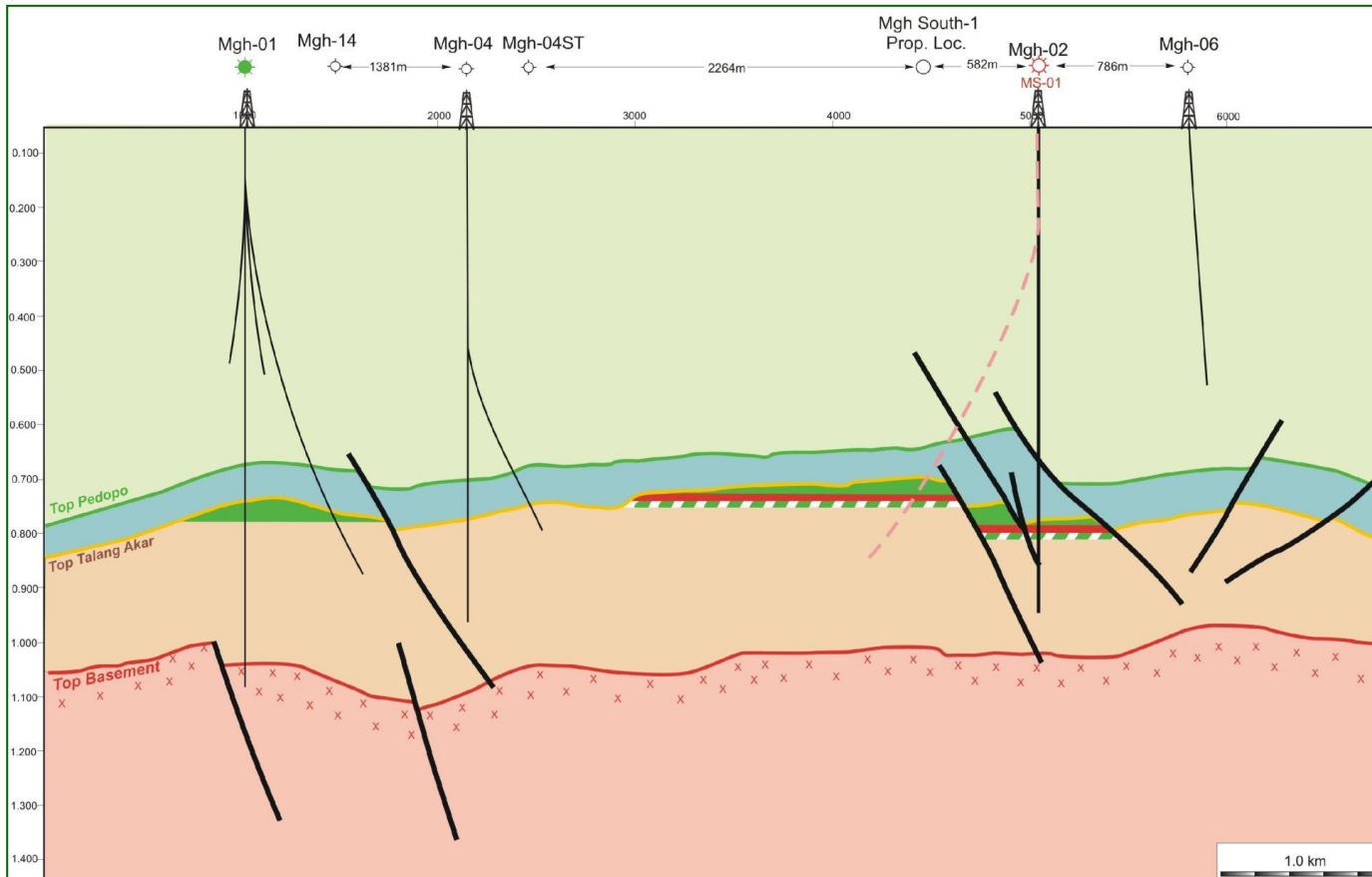


2. Drilling Type: Exploration → Wildcat and Delineation

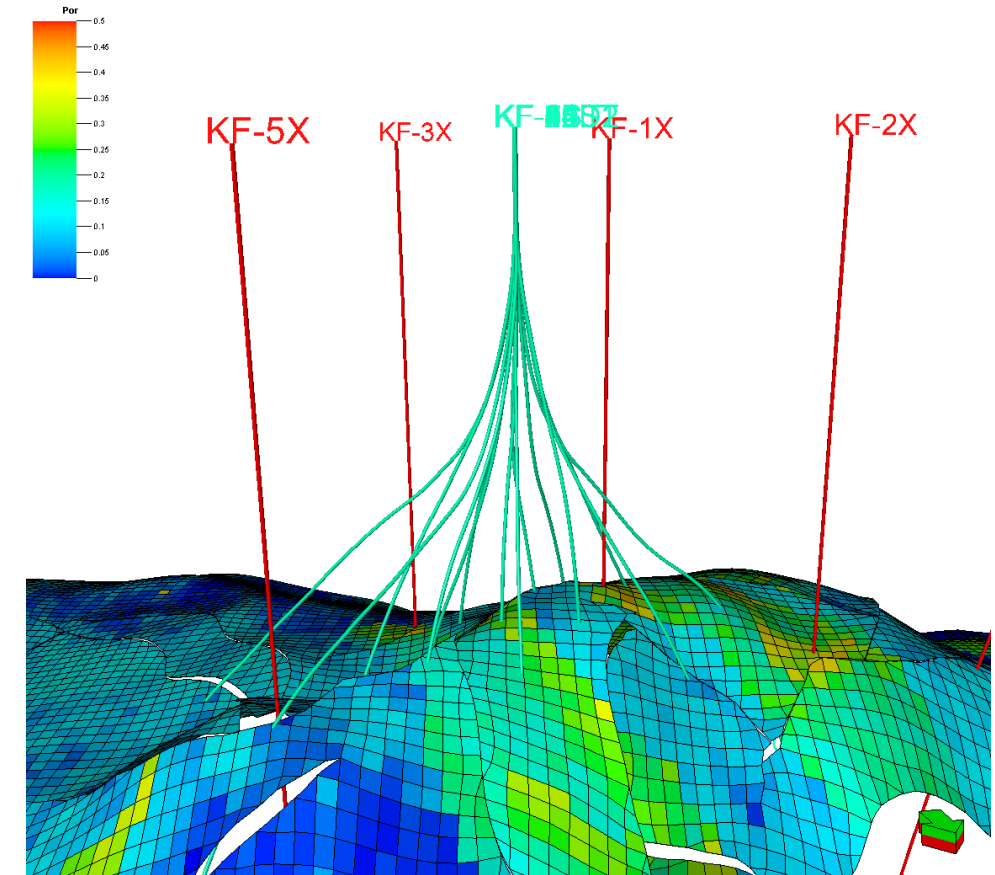


2. Drilling Type: Development Wells

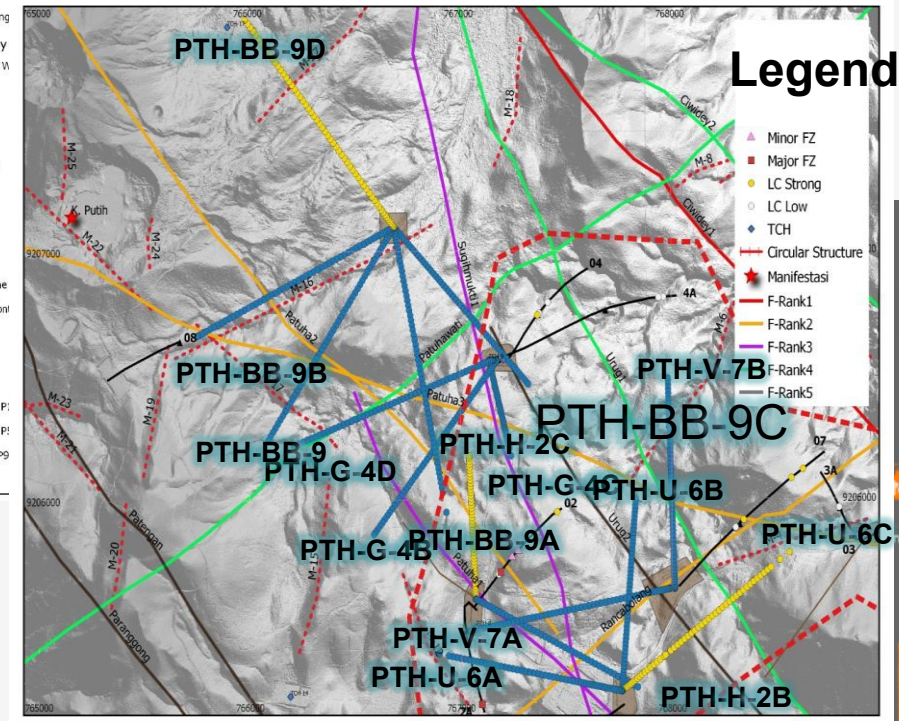
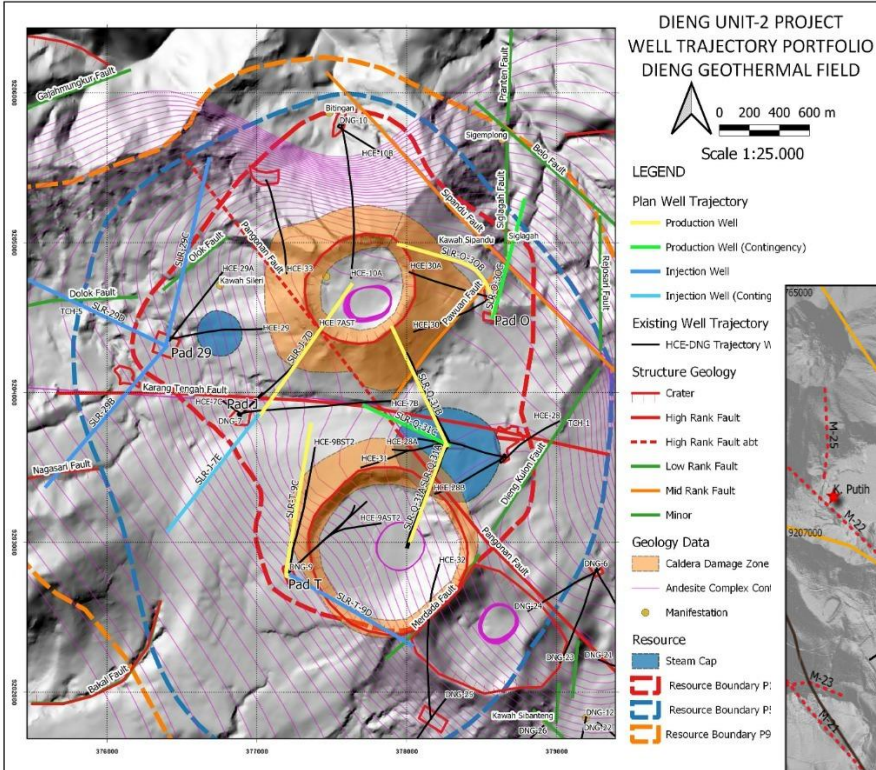
Onshore Oil Field



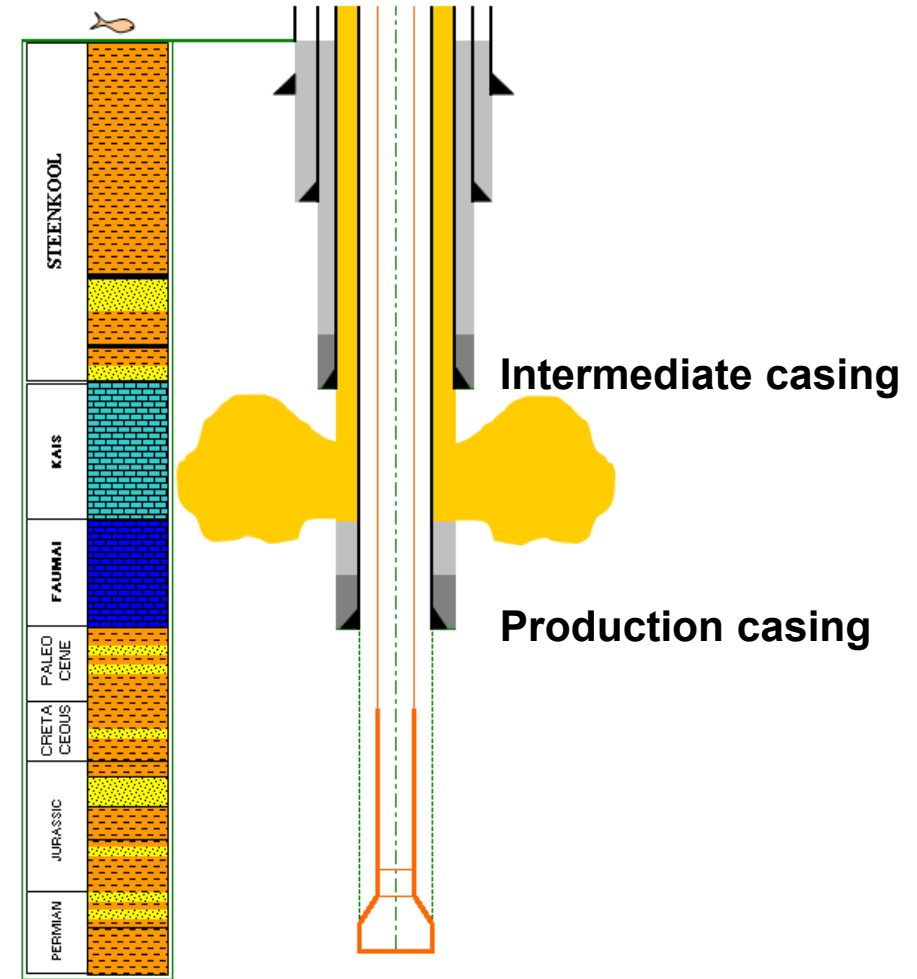
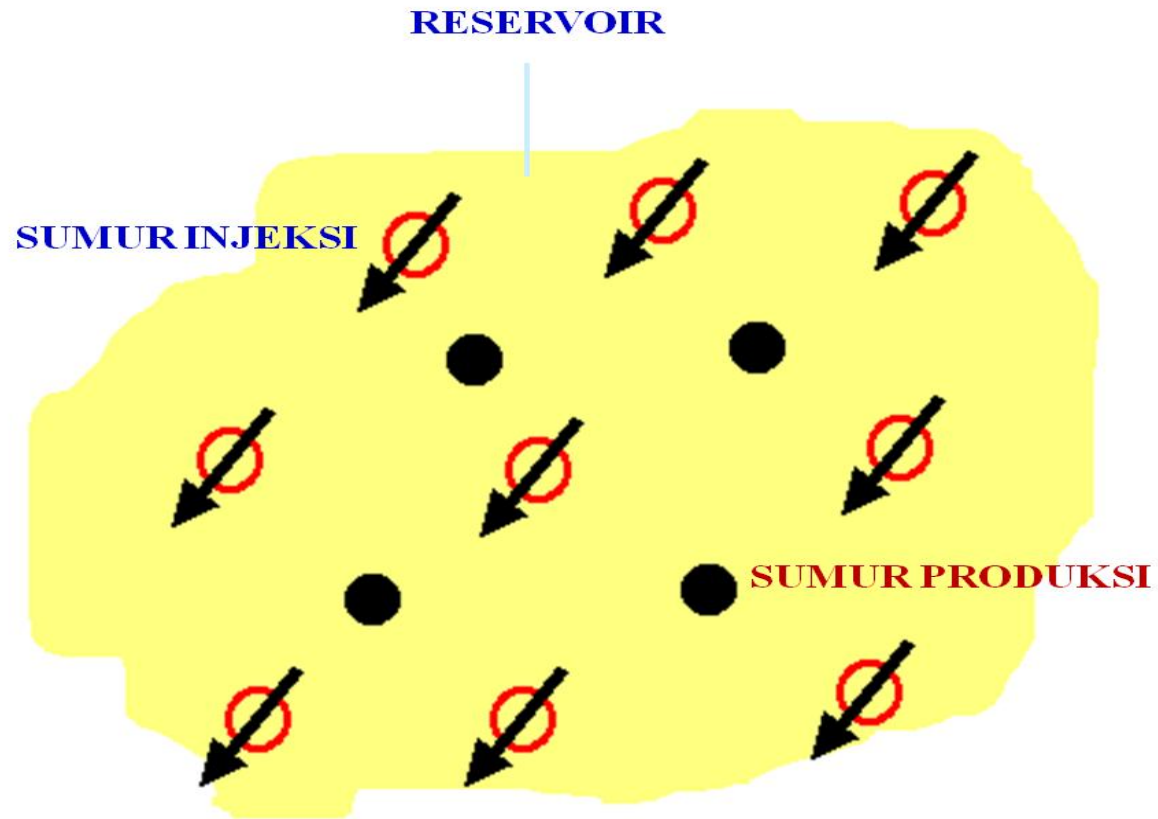
Offshore Oil and Gas Field



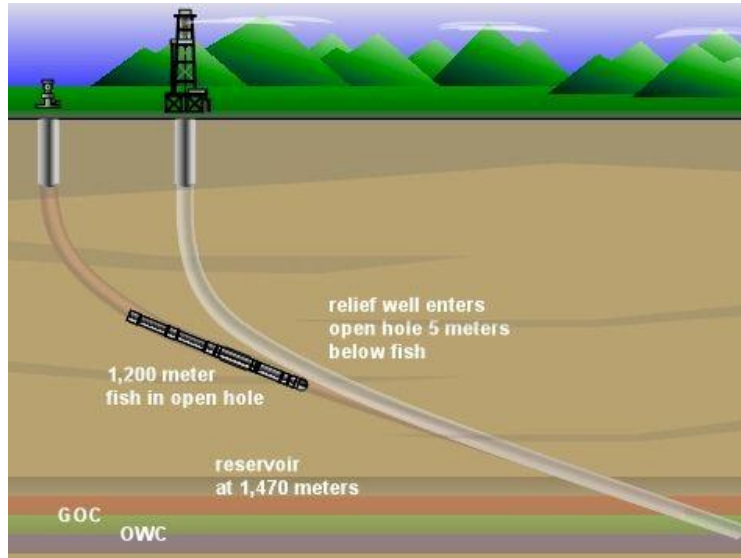
Development GEOTHERMAL Field



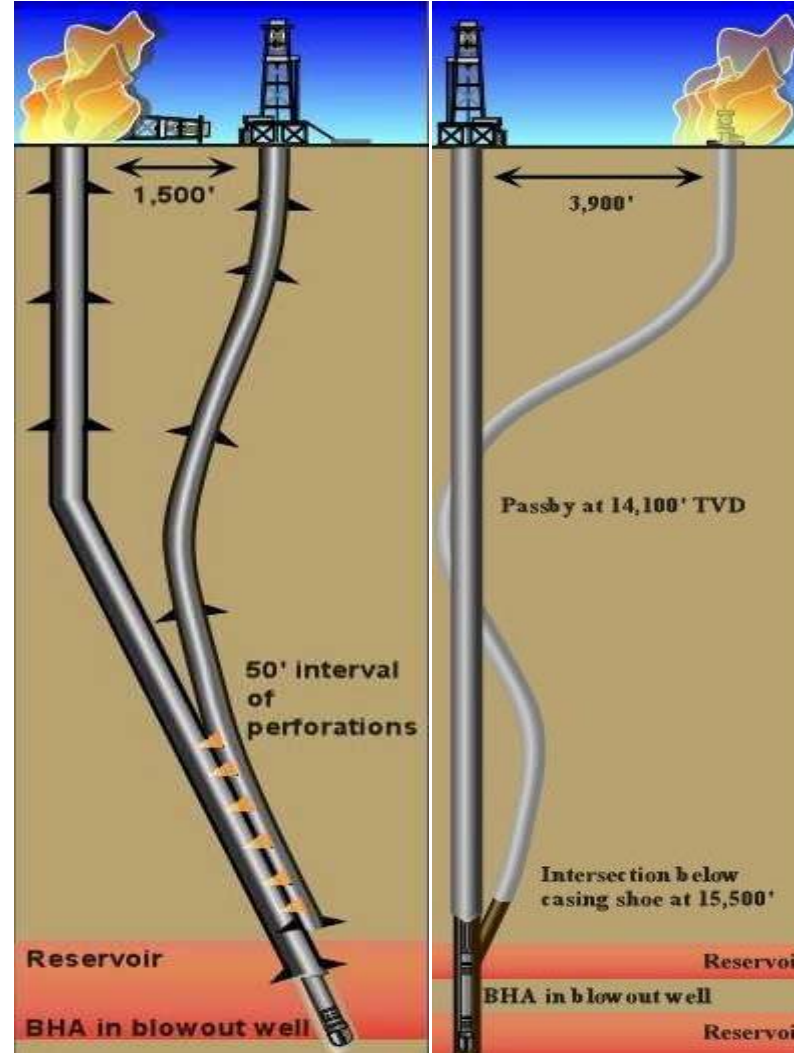
2. Drilling Type: Injection well and Waste Disposal Well



2. Drilling Type: Relief Well



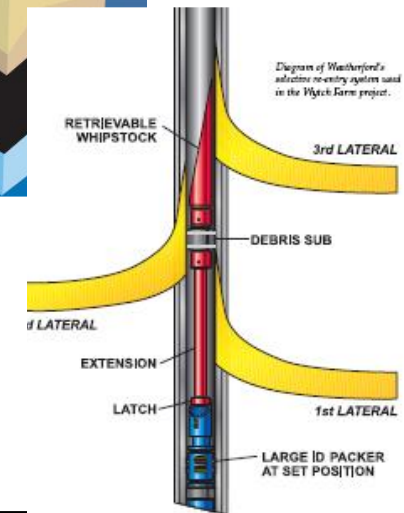
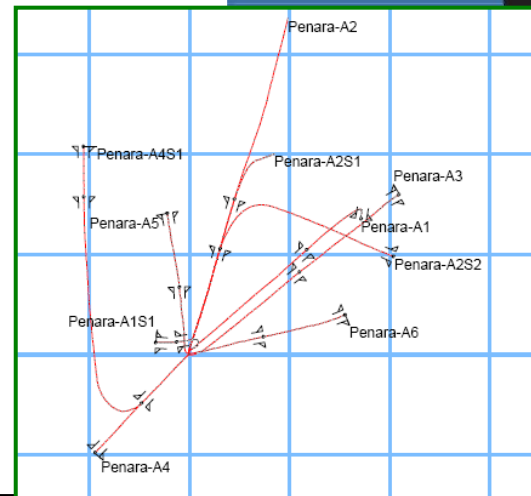
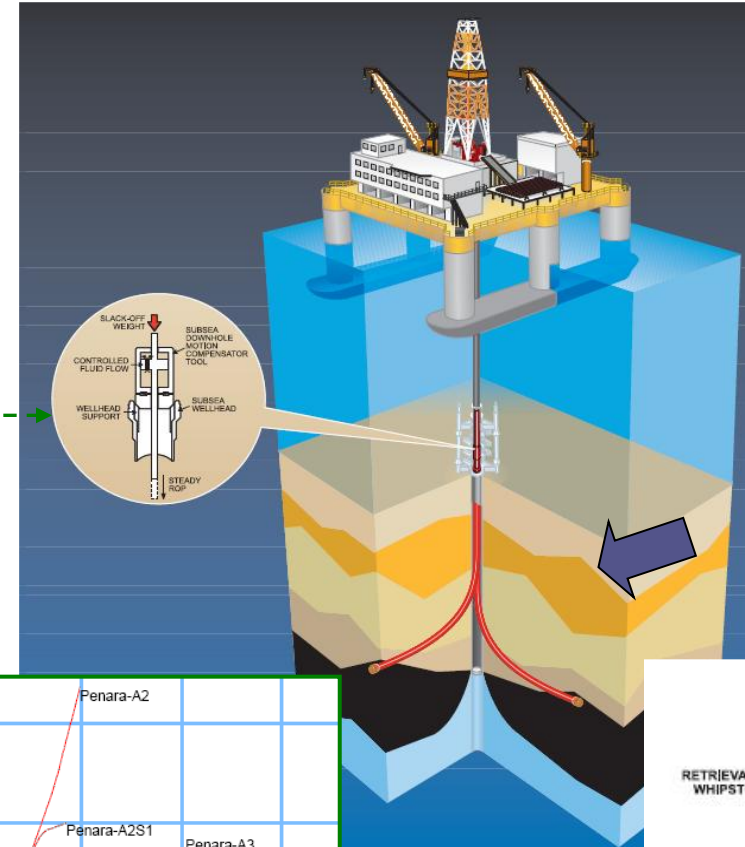
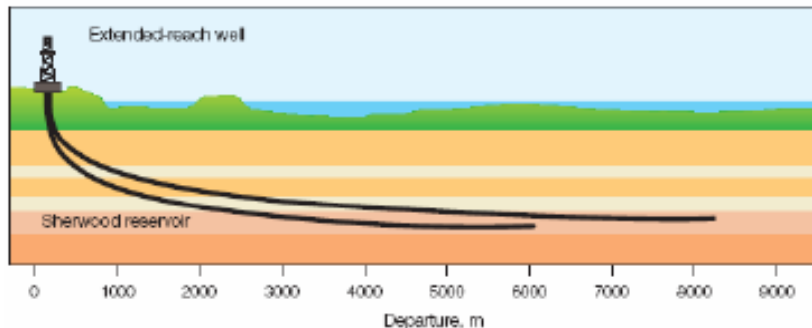
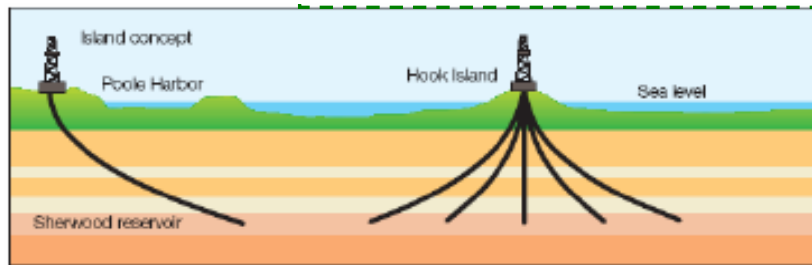
Drill relief well toward to blowout wellbore and injecting heavy mud to reduce blow out rate and finally kill the blow out well.



Drilled relief well to kill blow out well,
@ Mobil Oil Indonesia, at
Lhoksukon, Aceh

2. TYPE OF DRILLING REFER TO WELL GEOMETRY

- Vertical (Lurus)
- Directional (Berarah)
- Horizontal (Datar)
- Multi Lateral (Bercabang)



2

Drilling Rig Type

2. Drilling Rig Type for Oil and Gas



Semi Submersible



Drillship



Jack Up Rig



Swamp barge



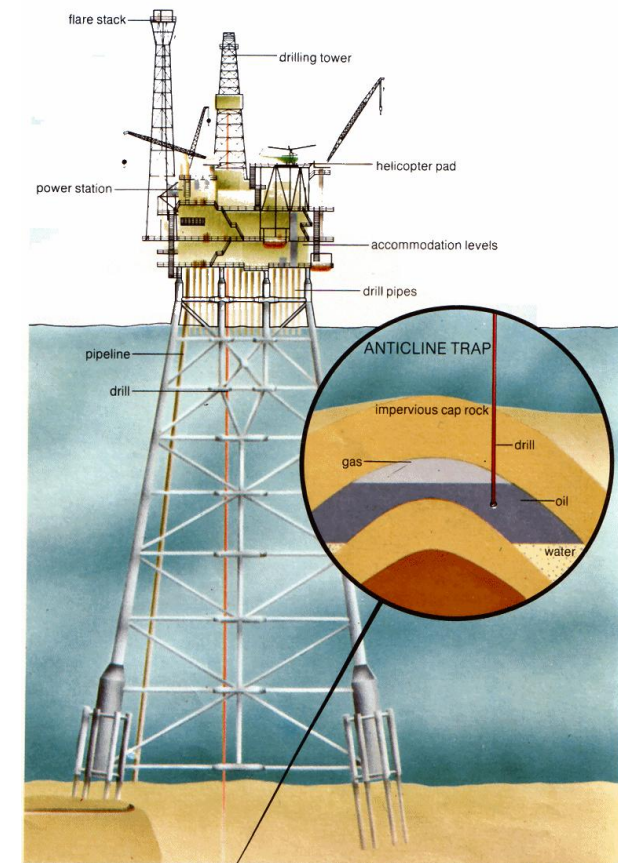
Land Rig

2. Drilling Type: Offshore Drilling Platform

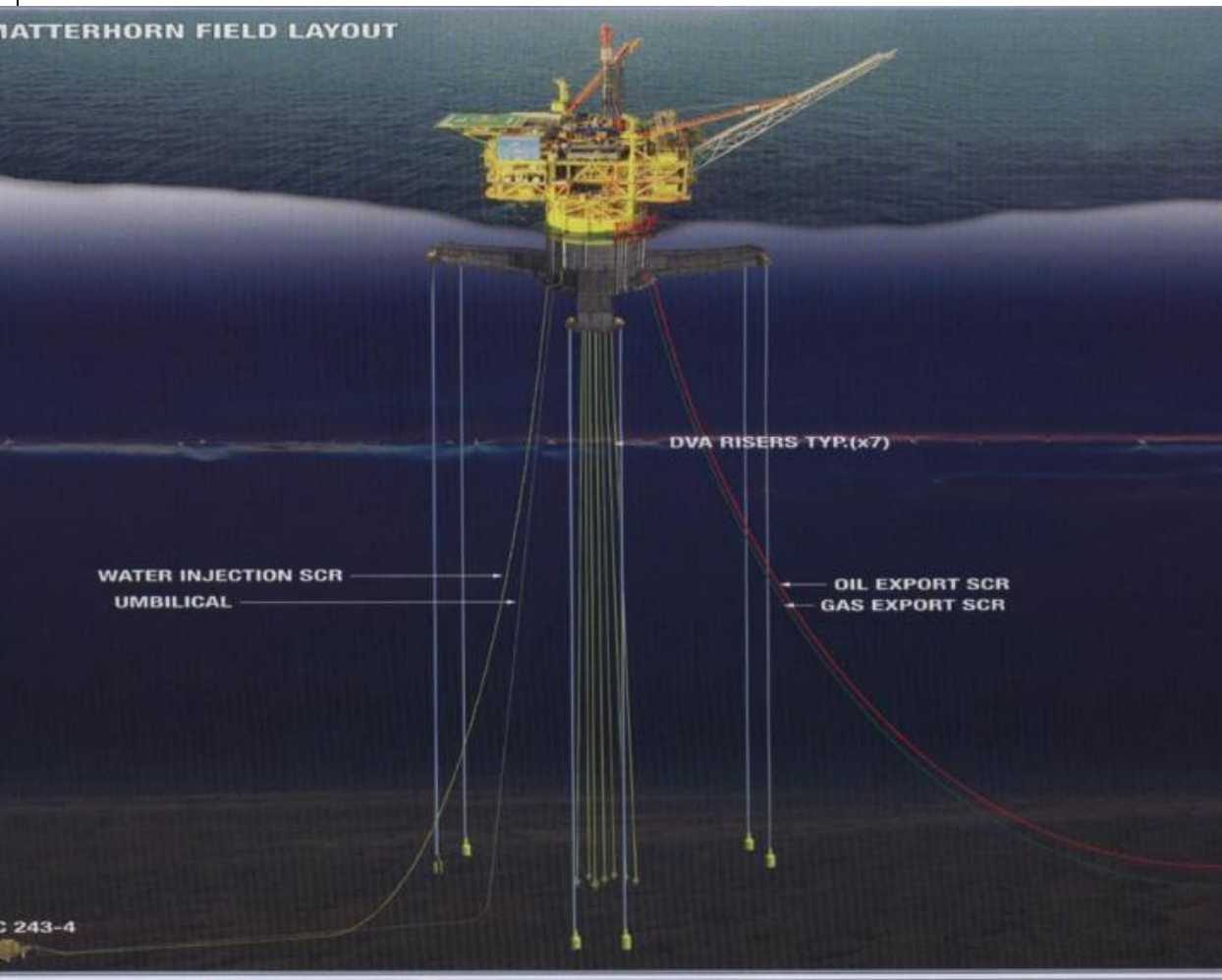


1. Grounded
2. Floating

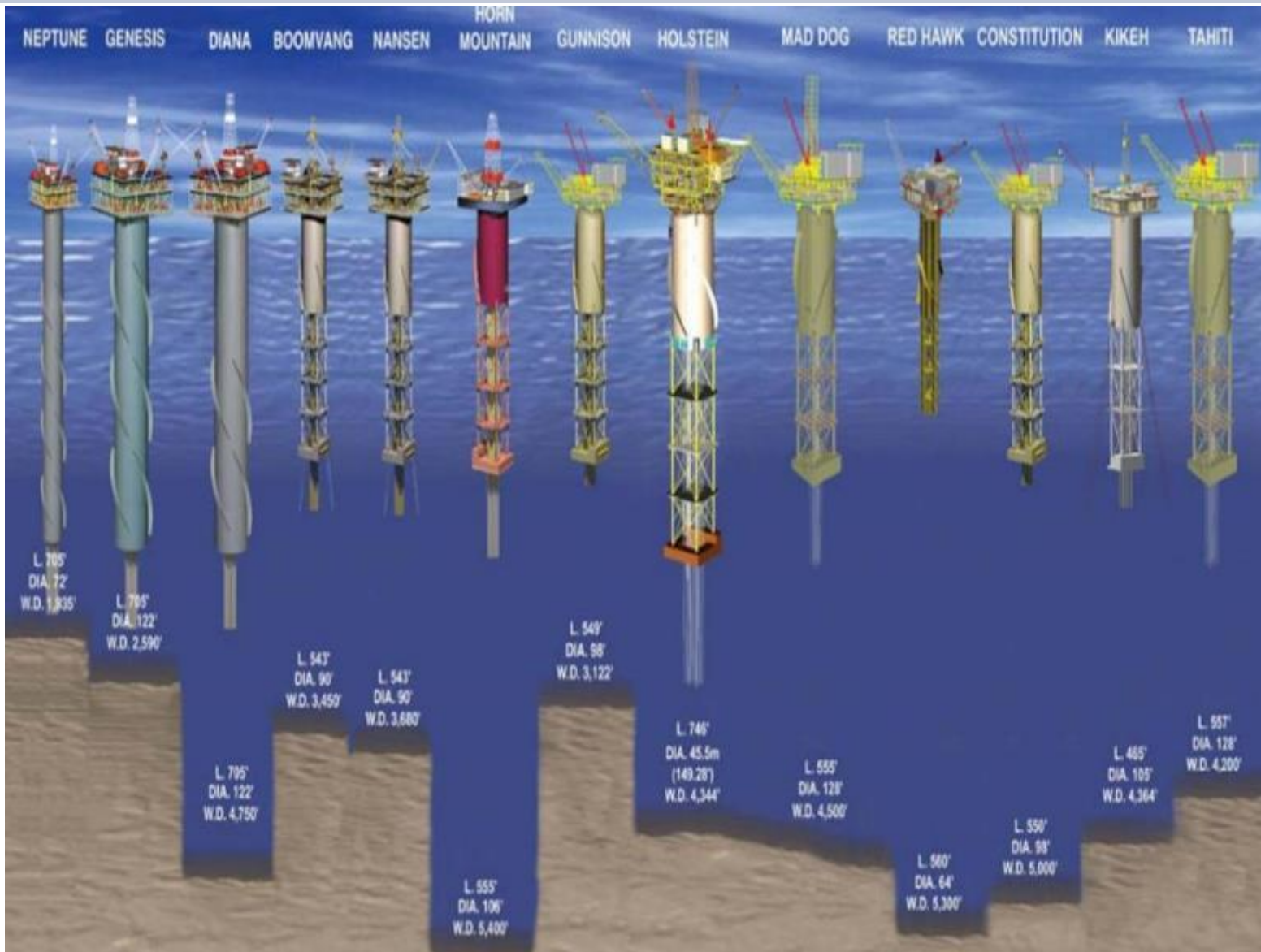
OIL PRODUCTION PLATFORM



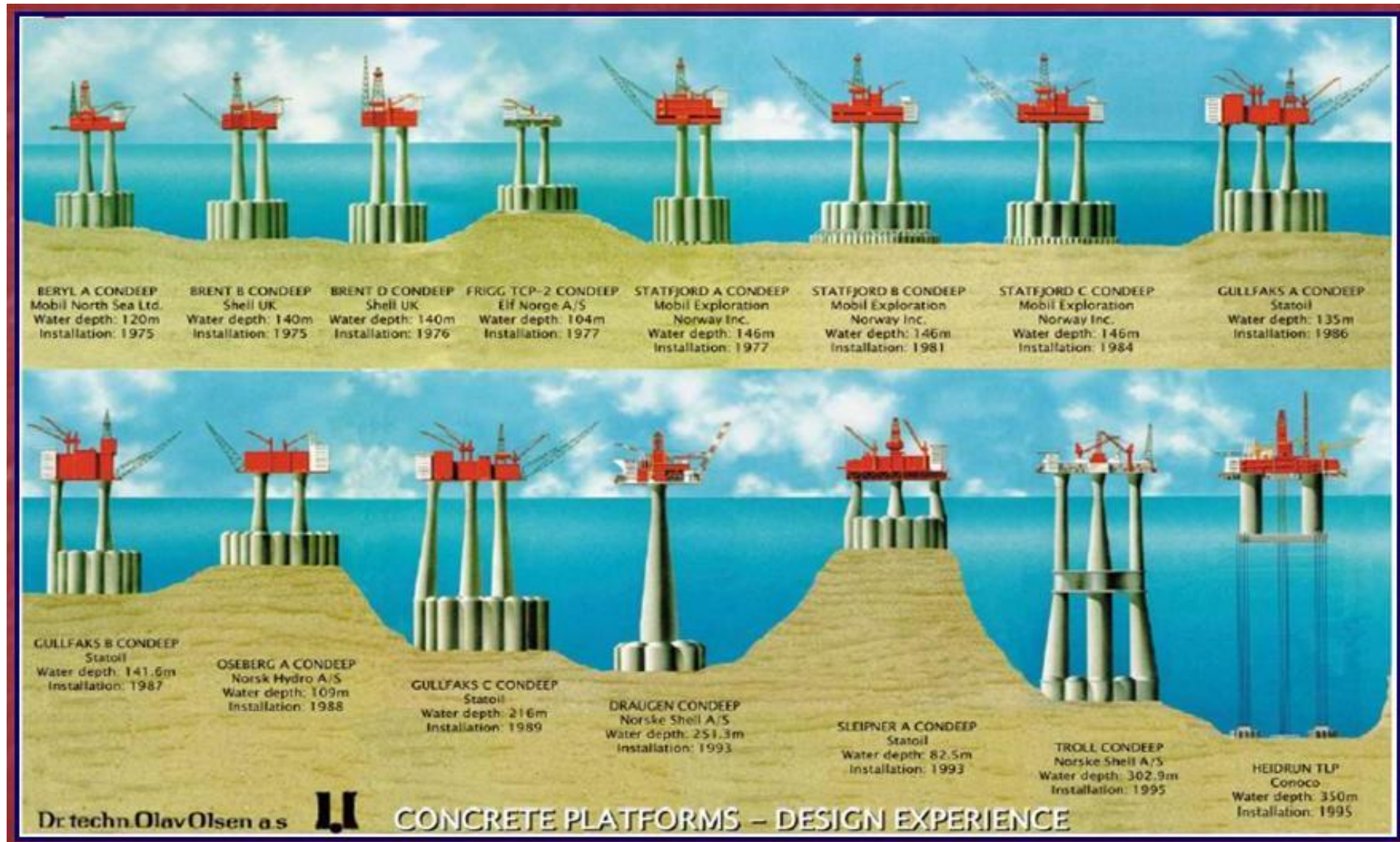
2. Drilling Type: Floating - TENSION LEGS PLATFORM



2. Drilling Type: Floating - SPAR PLATFORM



2. Drilling Type: Grounded - CONCRETE GRAVITY PLATFORM



2. Drilling Type: Grounded - Swamp Barge



- Cilipgo-1 Well
- Salawati Block
- Drilled 2008



OFFSHORE RIG MOBILIZATION

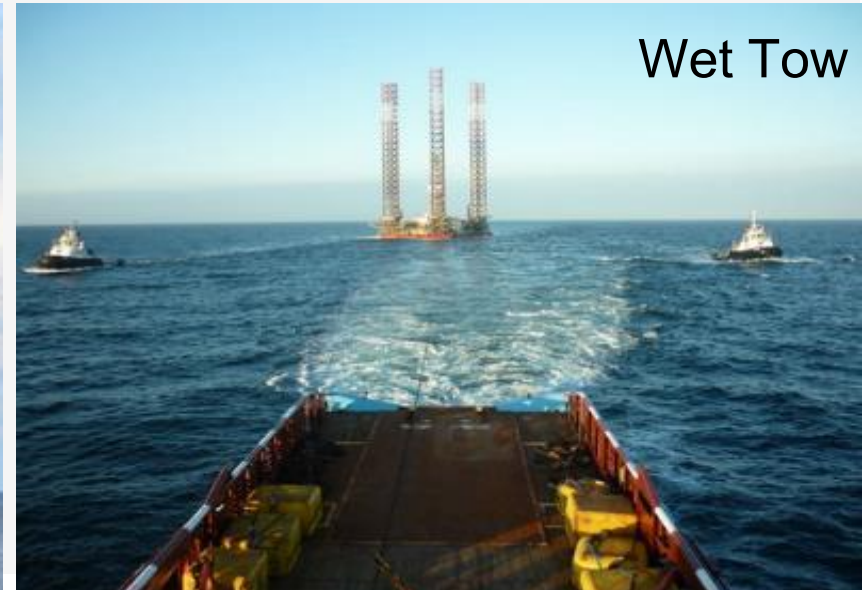
Dry Tow



Dry Tow



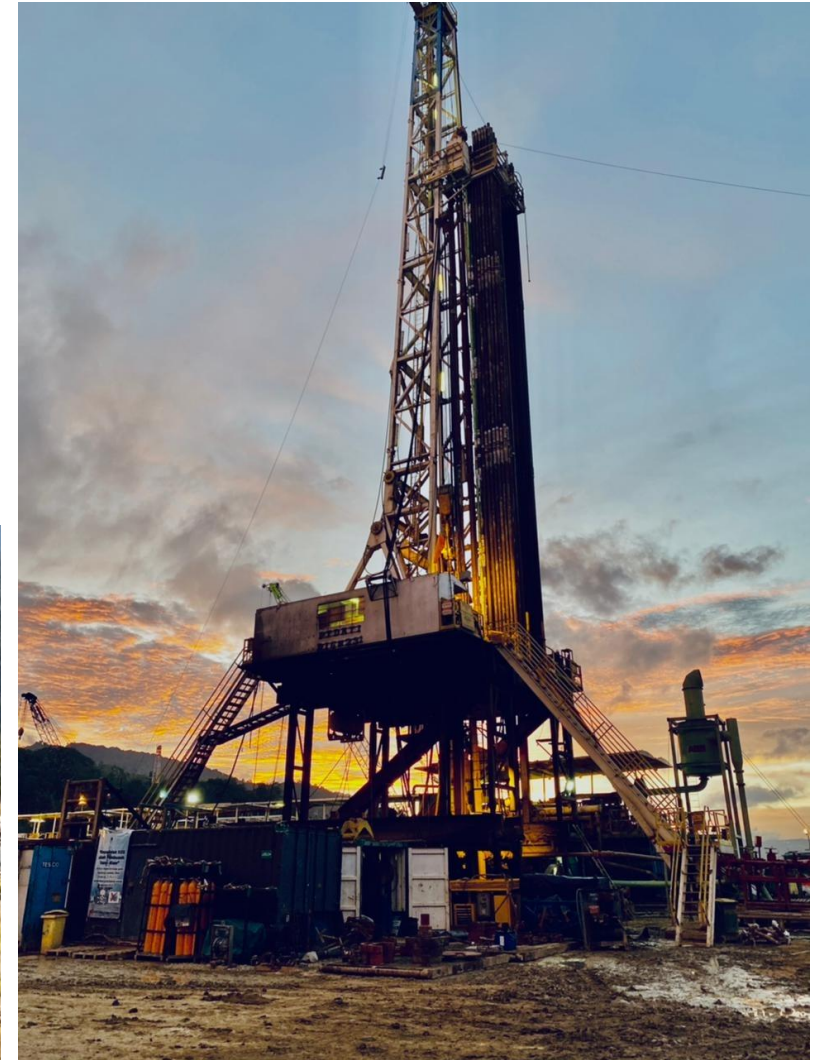
Wet Tow



2. Drilling Type: Offshore Equipment



2. Land Drilling Rig Type for Geothermal



2. Drilling Rig Mobilization, Rig Up, Rig Moving, Rig Down and Rig Demobilization

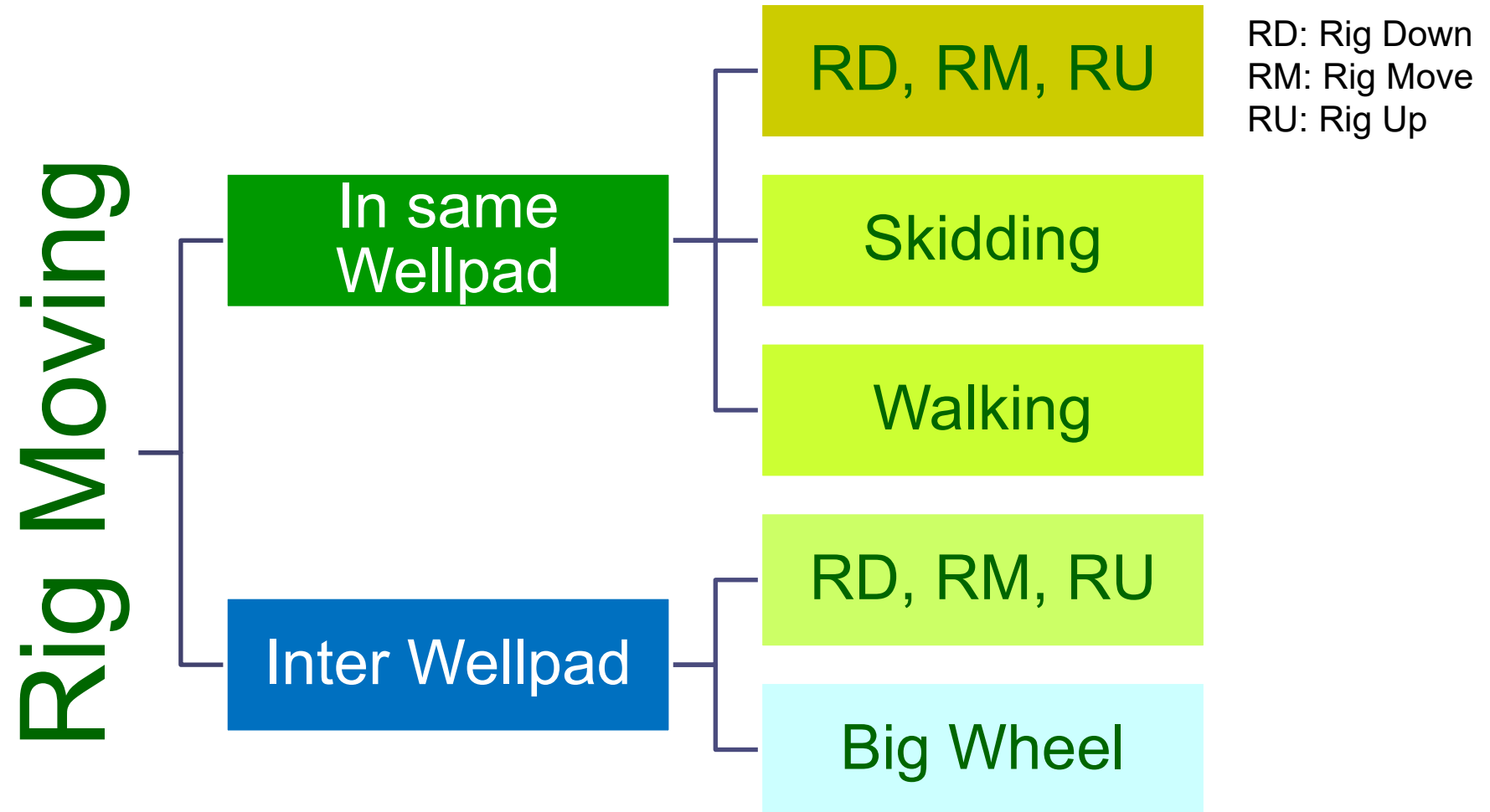
Rig mobilization is the structured process of preparing, transporting, and assembling a drilling rig at a new site (rig up), ensuring safety, efficiency, and operational continuity.

Rig Mobilization Lump-Sum fee is including all preparation cost, personnel, inspection, fuel, any upgrading (if needed), completing all spare parts needed and mobilized the Rig and the other equipment and personnel to the first well location, customs clearance, have all rig equipment **100% rigged up**, all major and critical rig third party inspector closed, and all administration and inspection requirement is completed as per EBTKE / Government of Indonesia Regulation prior commencing the drilling operation

Rig demobilization is the process of safely and efficiently removing equipment, materials, and personnel from a drilling rig site after operations are completed

Demobilization Lump-Sum shall be inclusive of all cost, including but not limited to: insurance, all life support services on site during demobilization process, fuel, any standby rate, dismantling, packing, logistic cost such as freight, customs clearance & trucking and associated cost during transfer process from COMPANY project site to any other project/location chosen by CONTRACTOR. Contractor responsible to do site reclamation.

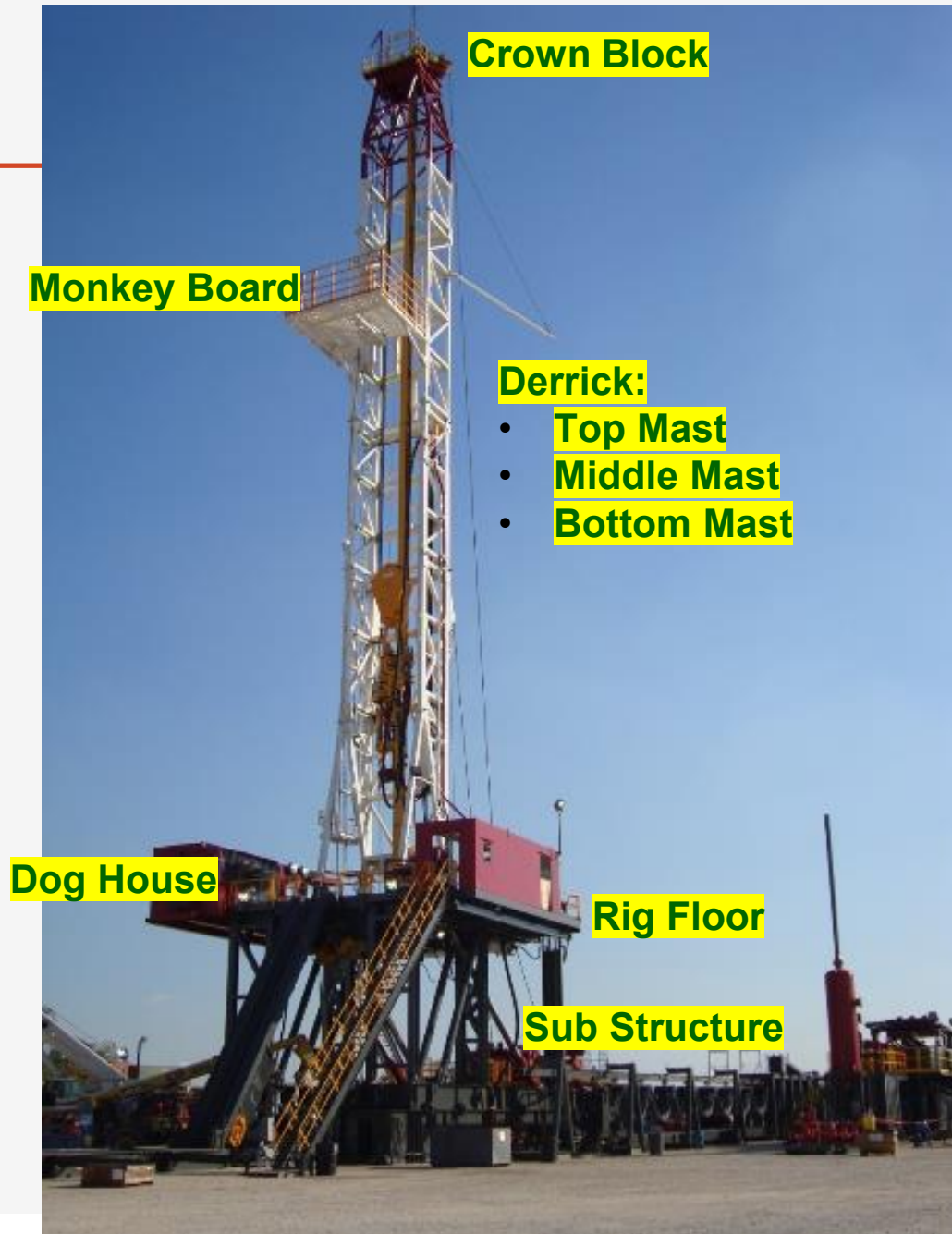




2. Heavy Equipment for Rig Moving



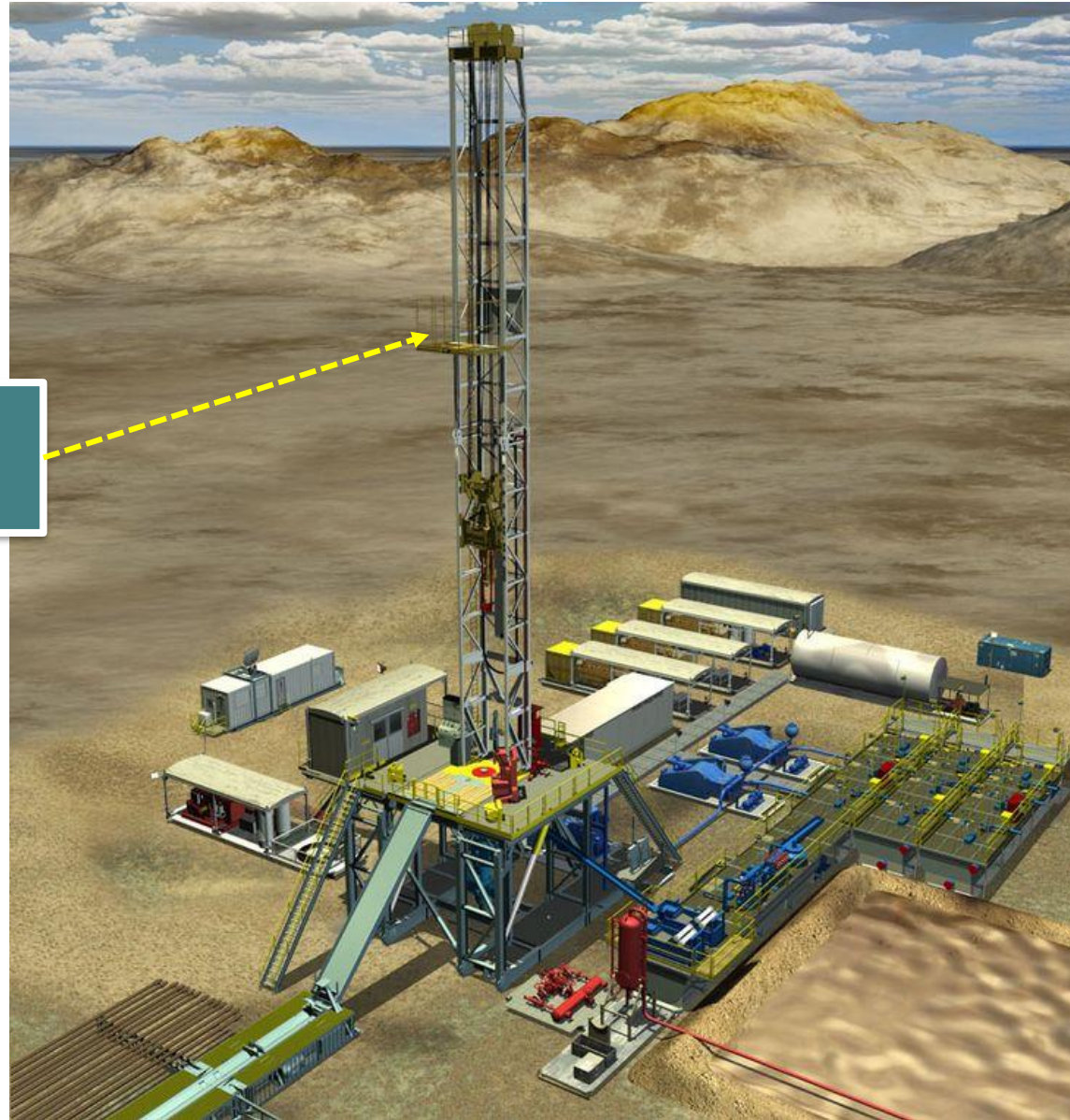
Rig Structure



2. Land Rig Layout and Equipment

Rig Structure

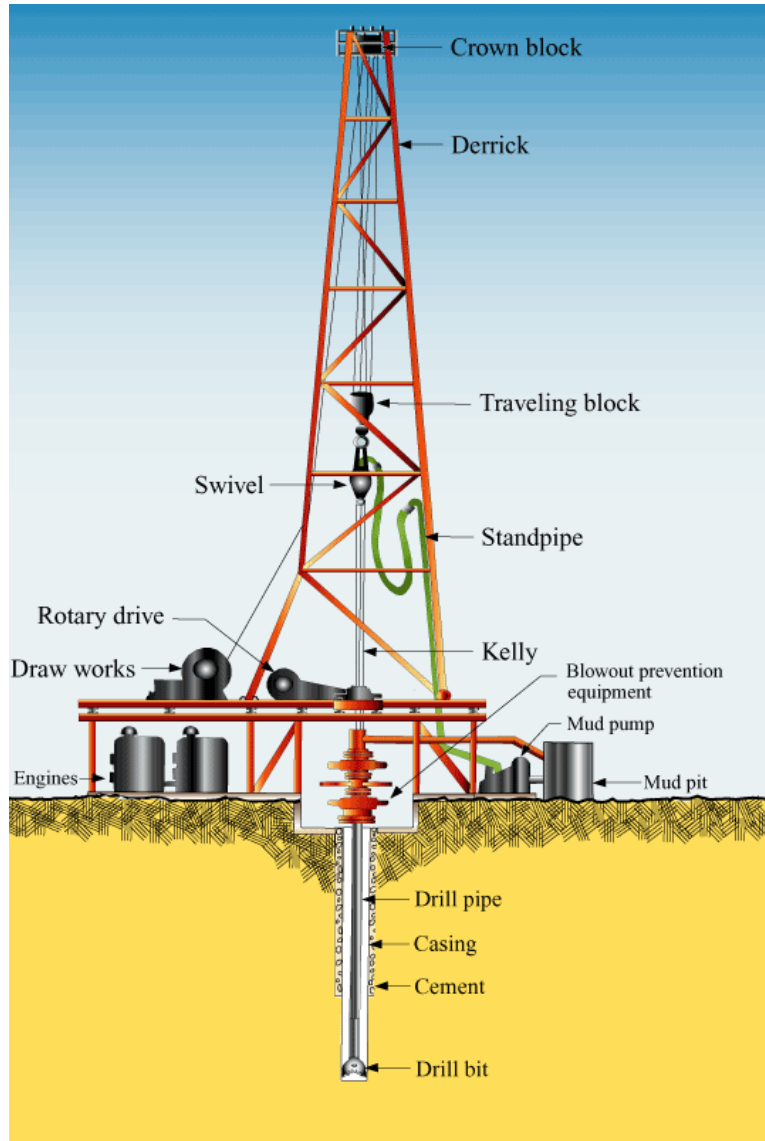
1. Crown Block
2. Travelling Block
3. Top Drive System
4. Derrick
5. Rig Floor
6. Drawwork
7. Dog-House
8. **Monkey-Board**
9. Sub Structure
10. V Door
11. Cat-Walk
12. Skid System



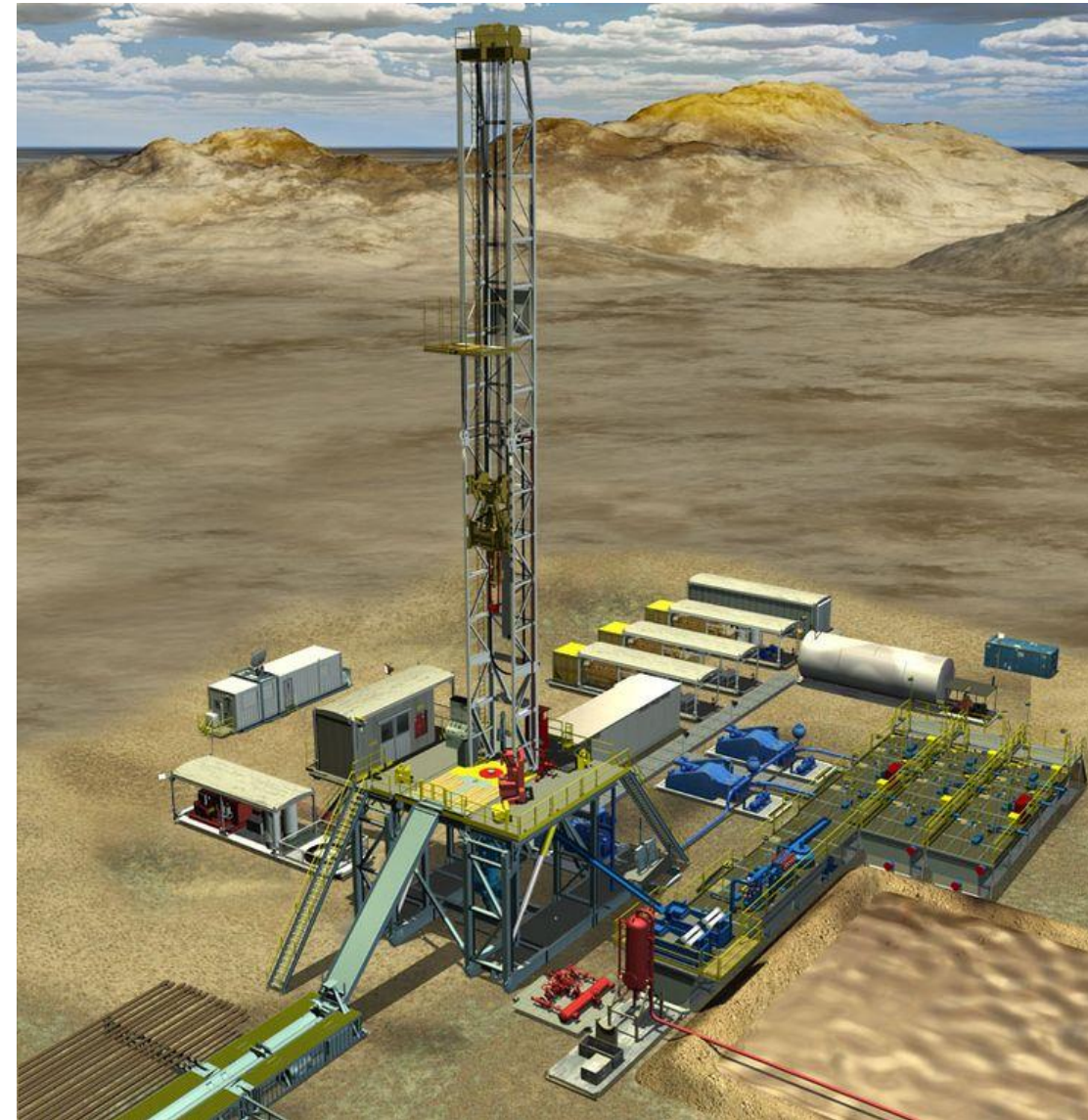
Rig Equipment

1. BOP
2. Koomey and Accumulator
3. Mud Pumps
4. Mud Tanks
5. Shale Shakers
6. Desilter
7. Desander
8. Gas-Degaser
9. Genset
10. SCR or VFD
11. Fuel Tanks
12. Mini Camps and Office
13. Warehouse
14. Automatic Driller

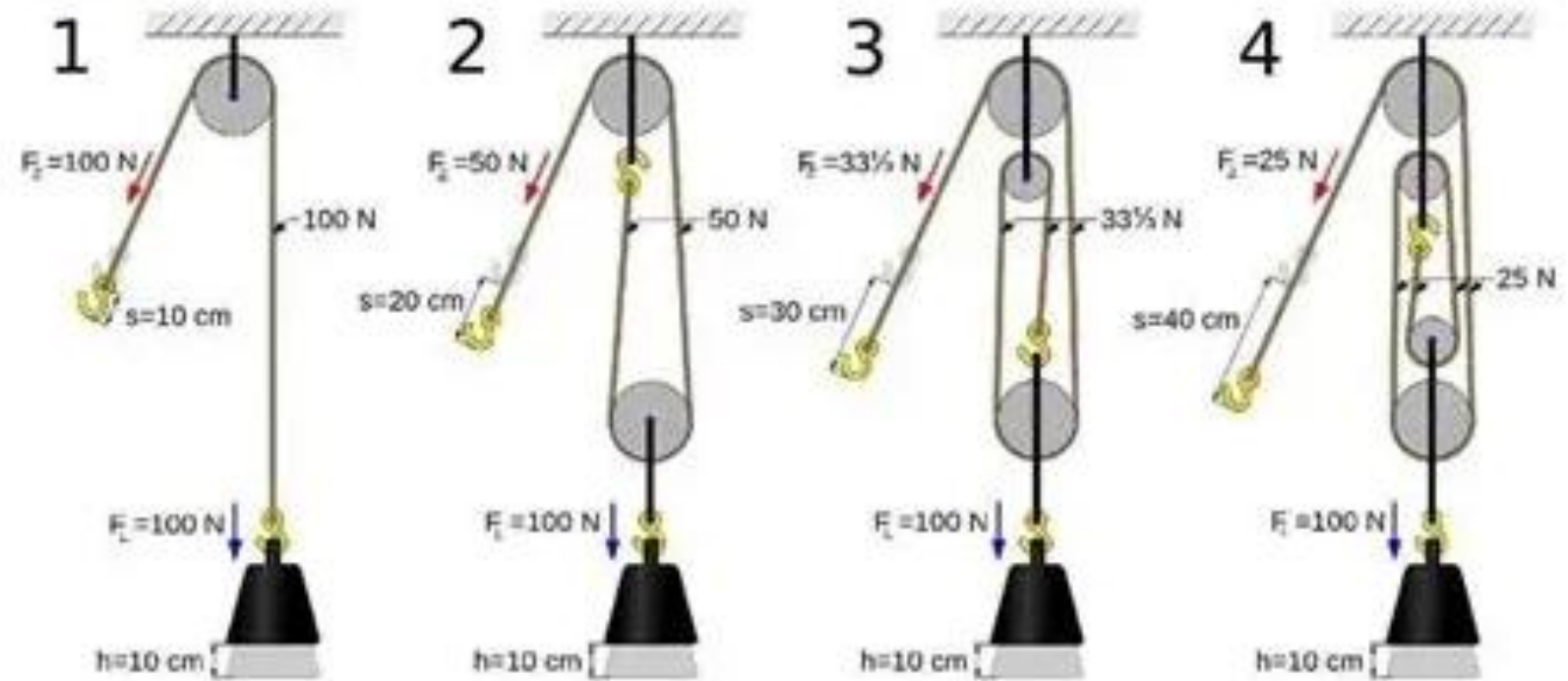
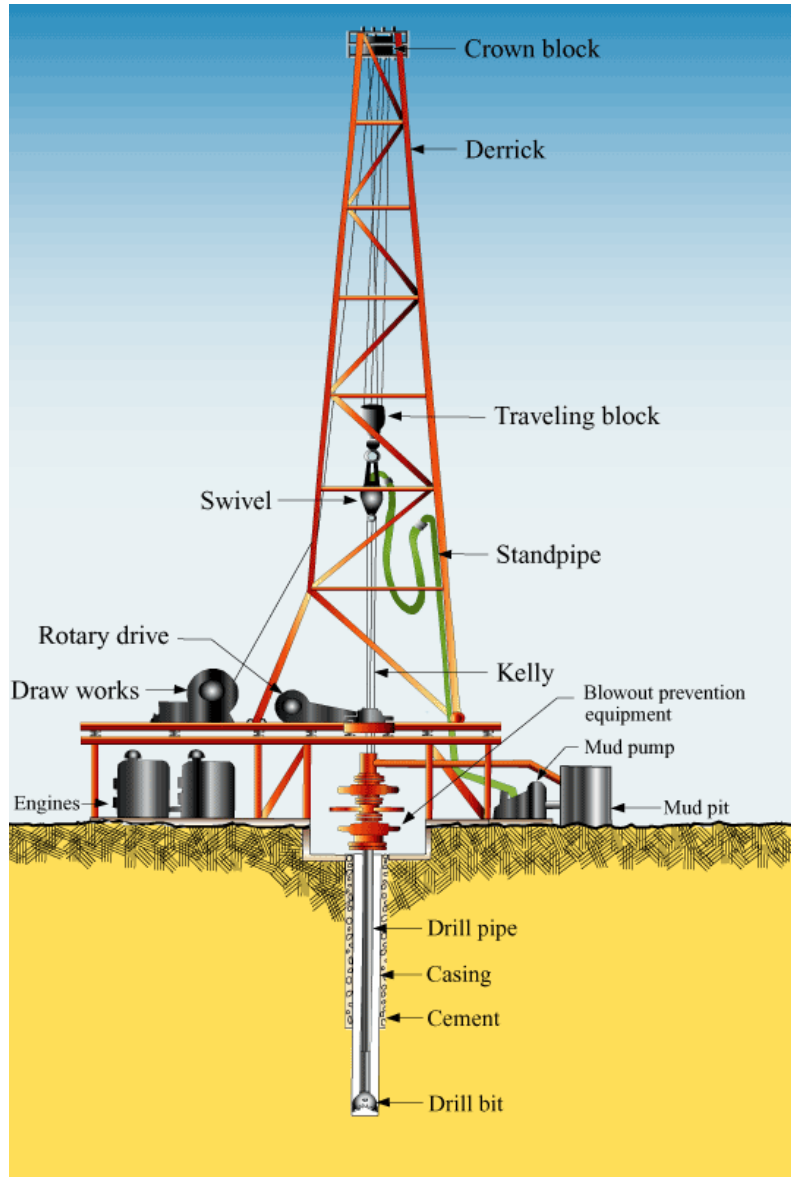
2. Drilling Rig Component



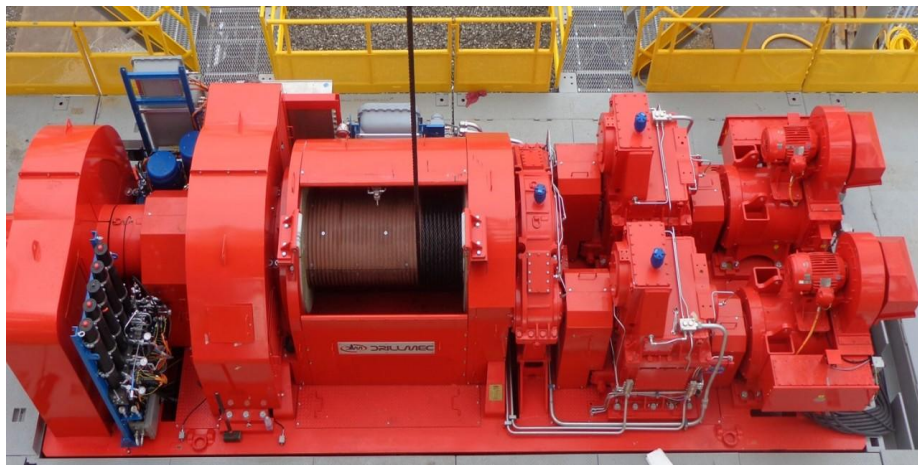
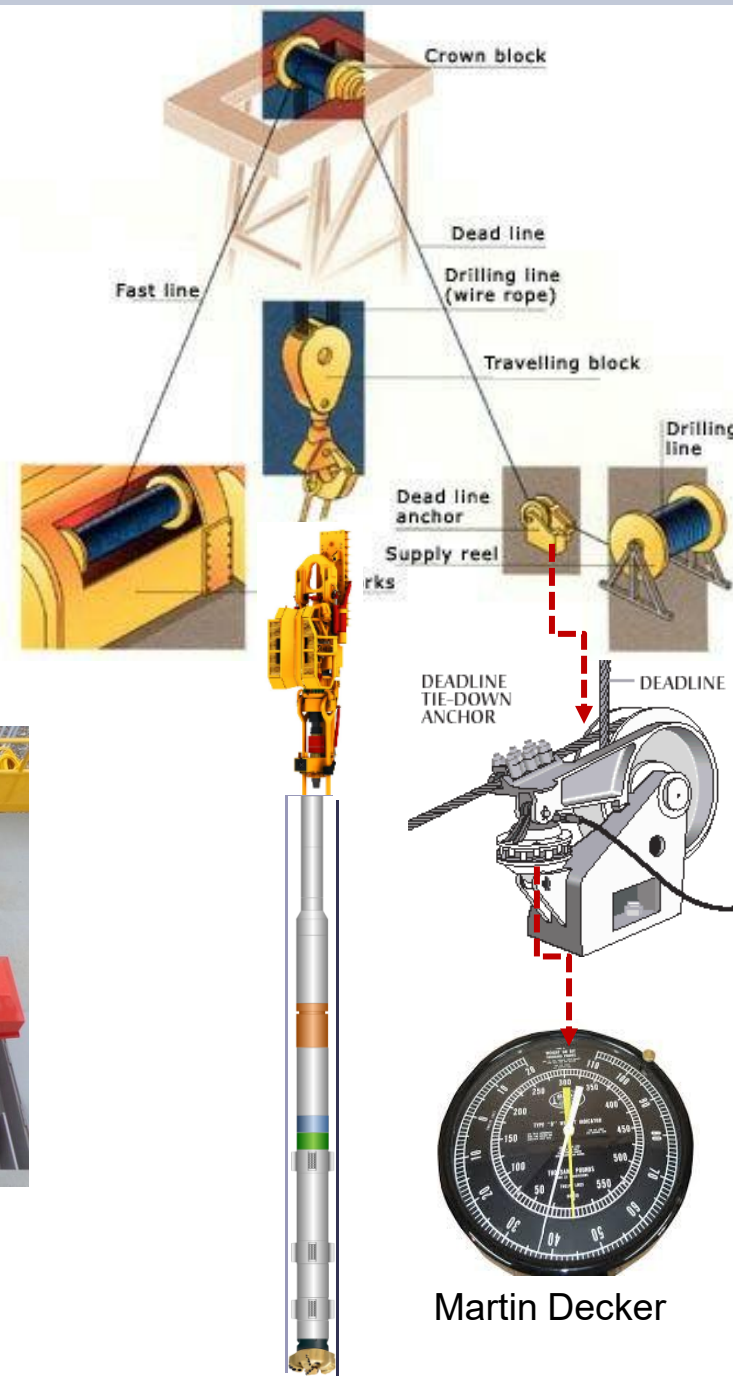
1. Rig Hoisting System
2. Rig Rotating System
3. Rig Circulation System
4. Well Control Equipment
5. Drill String



2. Rig Hoisting Pulley Basic Concept



2. Drilling Rig Hoisting System

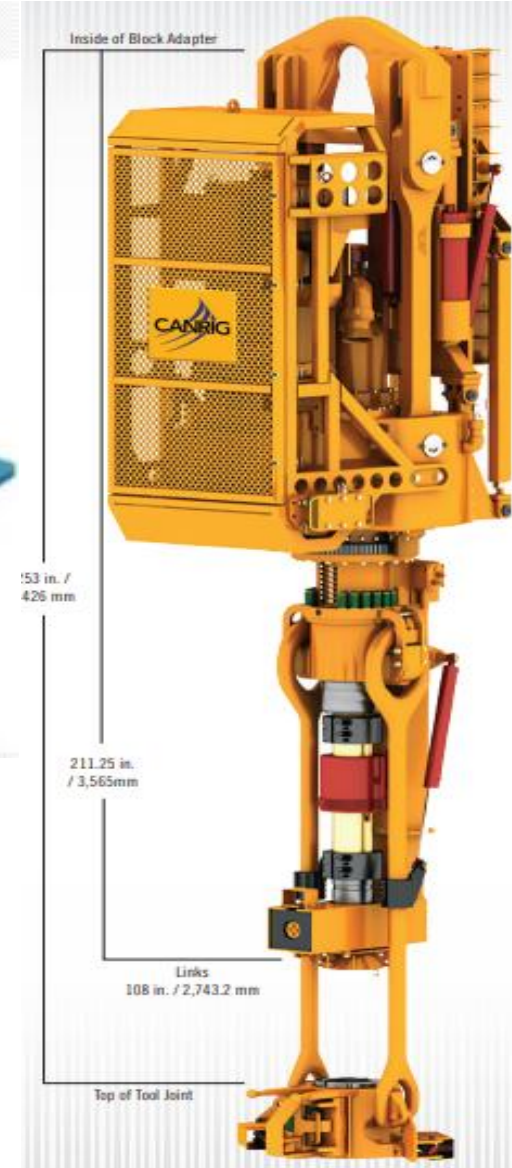
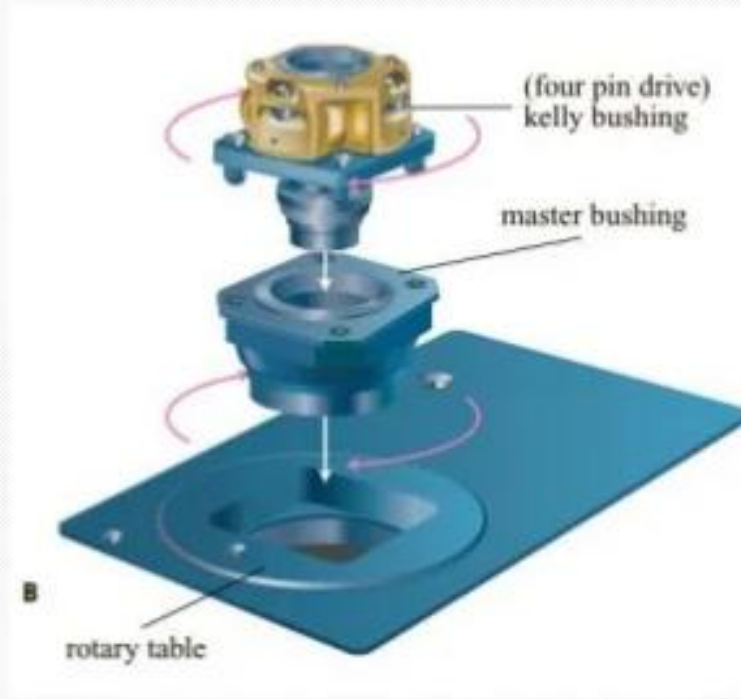


Martin Decker

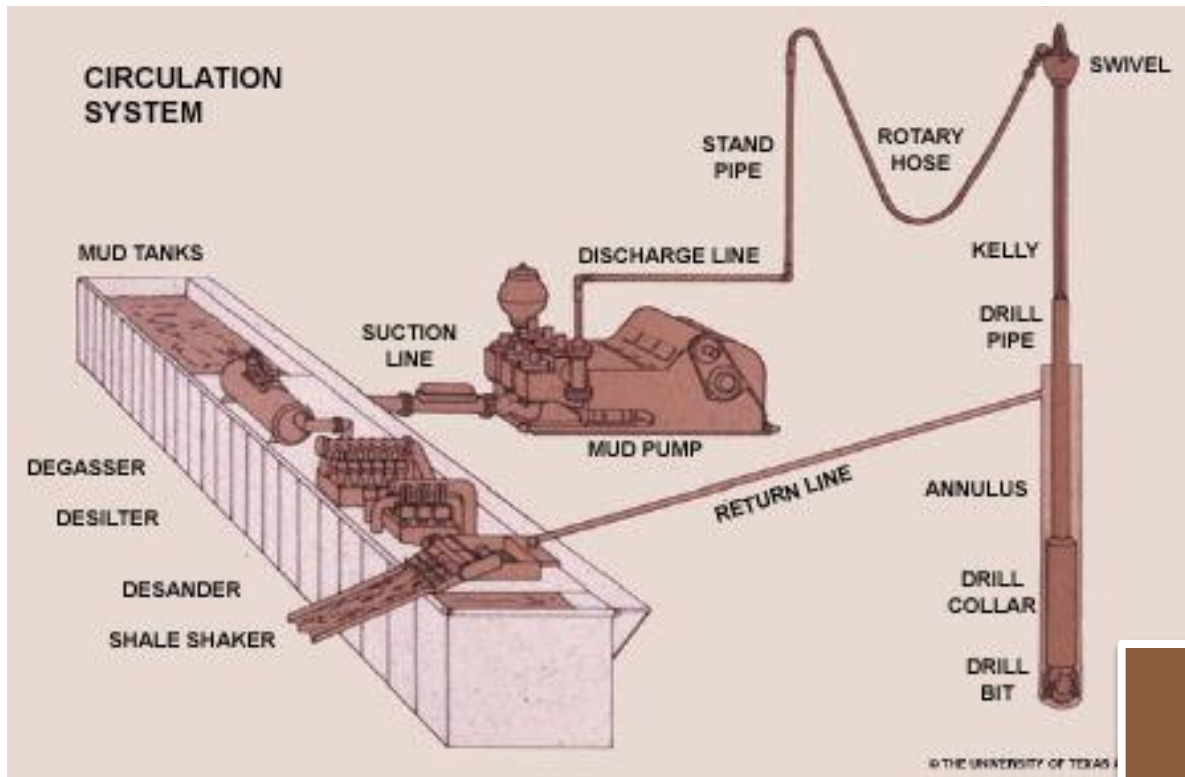
2. Drilling Rotating System

6. Rotary table

- ❖ The rotary table makes the drill string rotate and supports its weight during operations or during the connection of a new drill pipe, when it cannot be borne by the hook

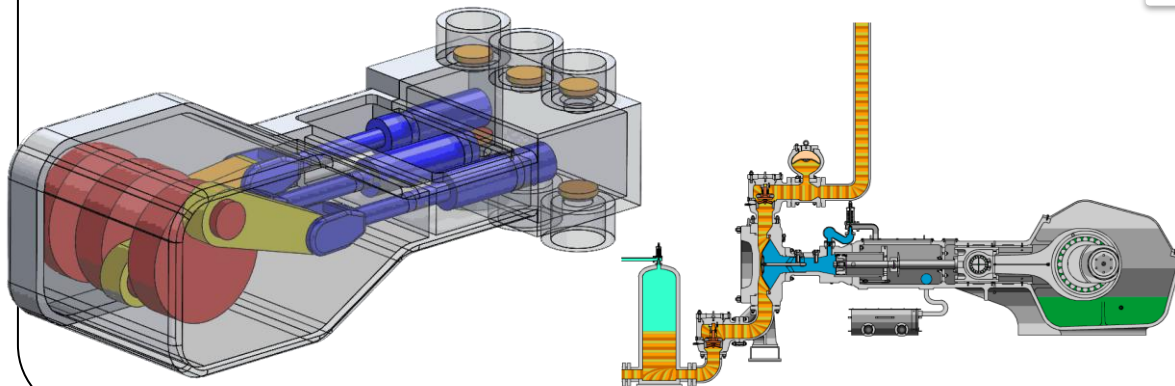


2. Drilling Circulation System

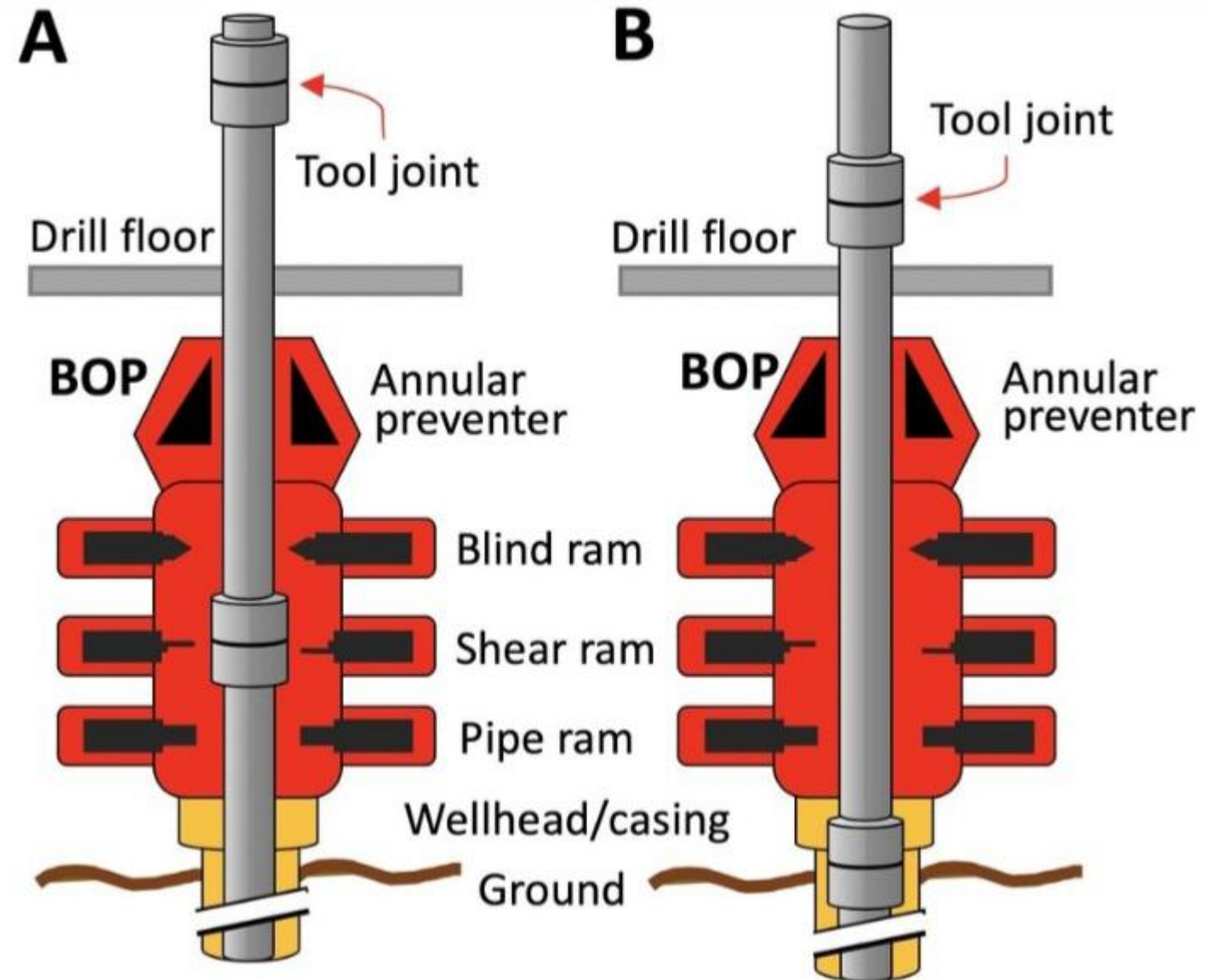


The mud circulation system in drilling rigs: circulating the fluid, cleaning cuttings, and controlling formation pressure and consists of several key components:

1. Mud Pumps
2. Swivel and Standpipe
3. Shale Shaker: sifts out small particles and cuttings
4. Desanders and Desilters: remove fine particles and solids.
5. Degasser: removes gas from the fluid.
6. Mud tanks.



2. Drilling Rig Well Control Equipment

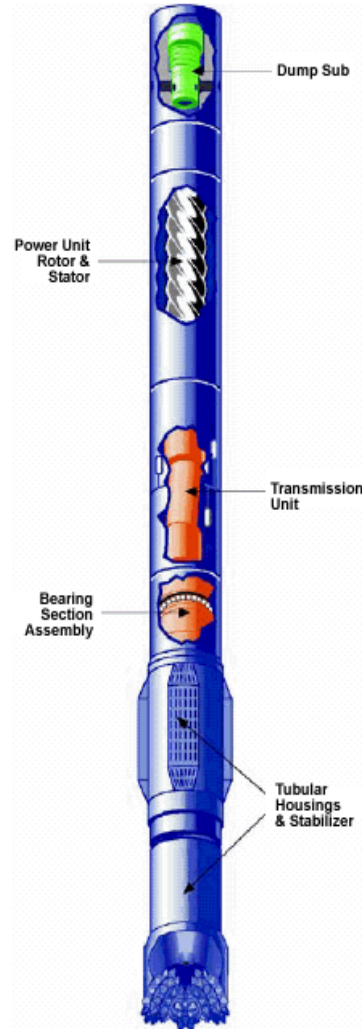


3

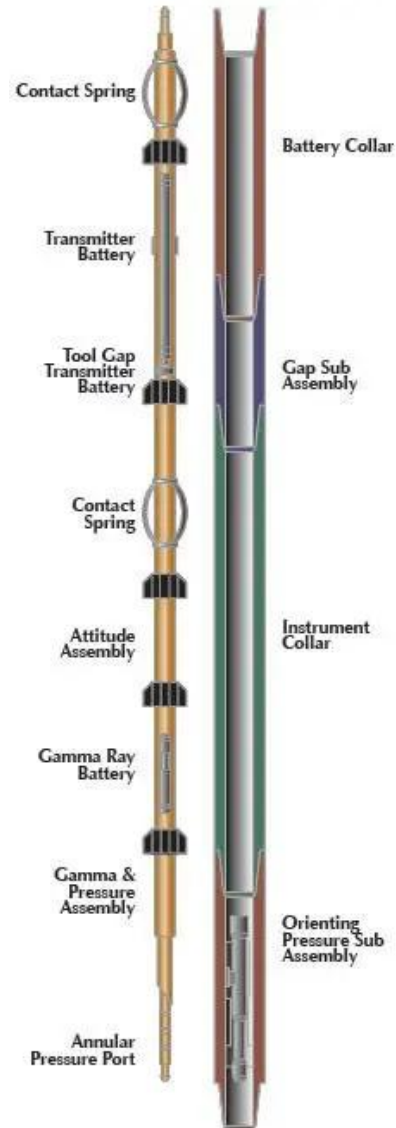
Drilling String and Bit

3. Well Construction – Bottom Hole Assembly

Mud Motor



Drilling Bit



MWD tool

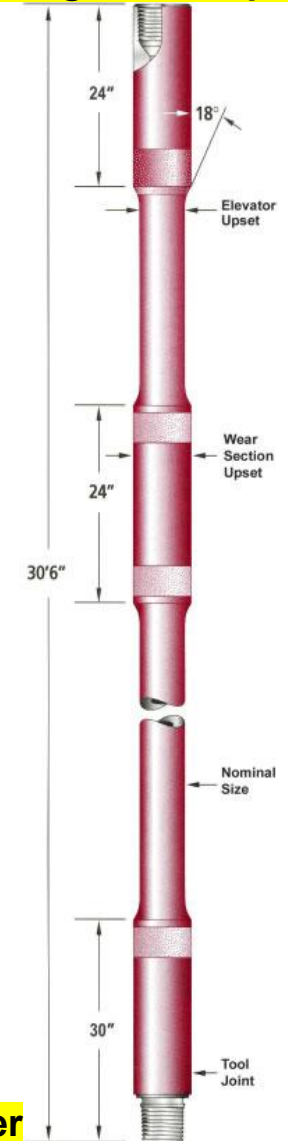


Drill Collar



IBS
Integrated String Stabilizer

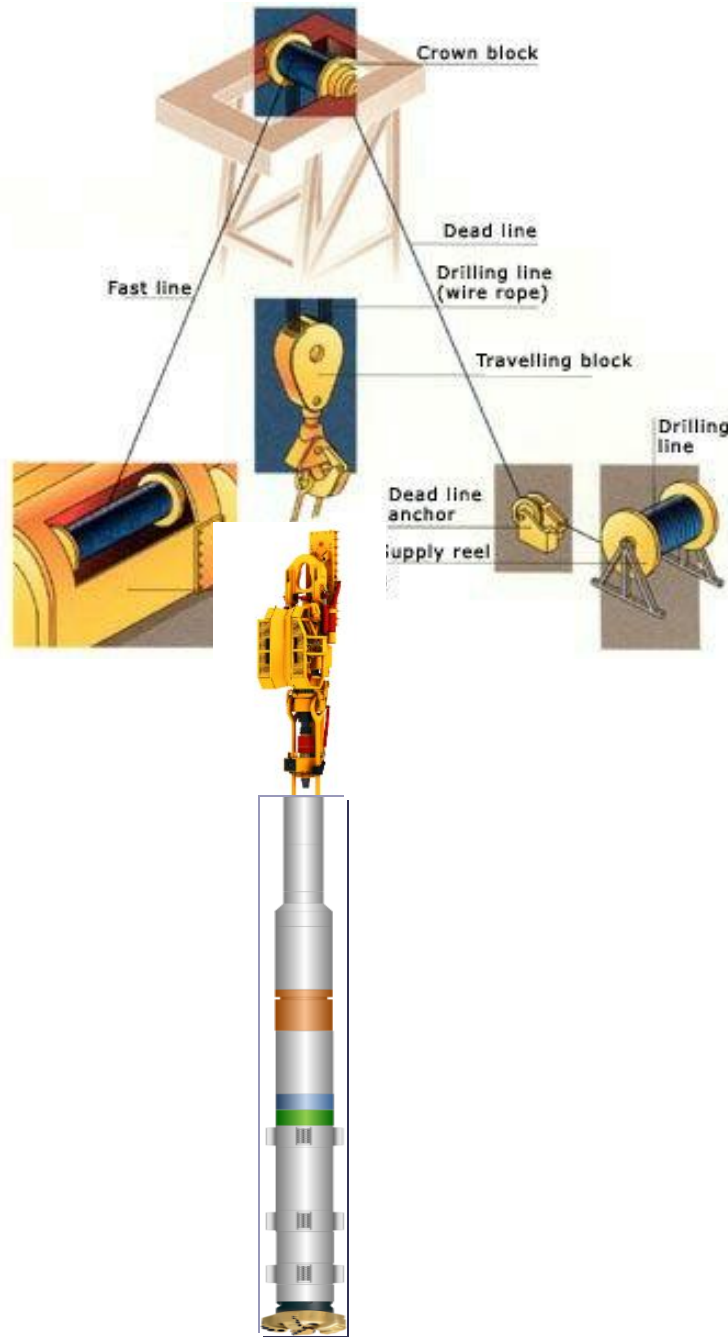
HWDP
Heavy Weight Drill Pipe



DP
Drill Pipe



3. Rig Weight Indicator



Drilling Loads:

- Travelling Block = ± 30 Klbs
- Top Drive System = ± 25 Klbs
- BHA Weight = 150 Klbs

Weight Indicator Reading

$$= 25 + 30 + 150 = 205 \text{ Klbs}$$



Martin Decker Weight Indicator

3. Well Construction – Bottom Hole Assembly

BHA and WOB Spreadsheet

Weight
Above
Jars



Weight
Below
Jars

12-1/4" BHA DESIGN w/ Motor

No	Description	Length Ft	Wt/Ft Lb/Ft	Wt in Air Lbs	Wt in Mud Lbs	Remark
1	12-1/4" Insert Bit	1	217,5	217,5	184,9	
2	9-5/8" Mud Motor	35	217,5	7.612,5	6.470,6	
3	9-1/2" Float Sub	6	217,5	1.305,0	1.109,3	
4	8" X/O Sub	10	150,0	1.500,0	1.275,0	
5	11-1/2" IB Stabilizer	10	150,0	1.500,0	1.275,0	
6	1x 8" NMDC	31	150,0	4.650,0	3.952,5	
7	8" HOS MWD	25	150,0	3.750,0	3.187,5	
8	6x 8" Spiral DC	186	150,0	27.900,0	23.715,0	
9	6-1/2" X/O Sub	10	60,0	600,0	510,0	
10	4x 5" HWDP	124	49,3	6.113,2	5.196,2	
11	6-1/2" Jars	35	92,0	3.220,0	2.737,0	
12	26x 5" HWDP	806	49,3	39.735,8	33.775,4	
String Weight Below Jars				48.435,0	46.876,0	
String Weight Above Jars				49.669,0	36.512,4	
String Weight Total				98.104,0	83.388,4	

Weight On Bit ?

2. Drilling Rig Specification

RIG WBR # 01

GENERAL INFO

Rig Name	: WBR # 01
Rig Type	: Skid Mounted
Year Built	: 2001
Rig Power Rating	: 1,000 HP
Drilling Depth	: 10,200 ft with 5" DP
Rig Loads	: 100 Loads
Camp Loads	: 40 Loads

MAST

Make/Type	: Baoji Petroleum Manufacture Co., Ltd
Clear Height	: 144.3 ft (44 meter)
Mast Base	: 10.9 x 15.6 m
Static Hook Load	: 221.4 Ton (2206.32 KN)
Racking Capacity	: 5" DP = 2000 m; 3-1/2" DP = 2000 m;

SUBSTRUCTURE

Make/Type	: Baoji Petroleum Manufacture Co., Ltd
Rig Floor Dimension	: 13.06 m x 10.31 m
Height	: 7.5 m
Rotary Load	: 229.4 Ton (2250 KN)
Setback Load	: 140 Ton (1394 KN)

DRAWWORKS

Make/Model	: Baoji Petroleum Manufacture Co., Ltd
Rated Input Power	: 1200 kW
Drilling Line	: 32 mm (1-1/4")
Auxiliary Brake System	: Elmago Eddy Current Brake DWS40

TOP DRIVE

Make/Model	: NOV TDS10SA
Rated Load Capacity	: 250 Ton
Drilling Torque	: 21,000 ft-lbs
Power Rating	: 350 HP

ROTARY TABLE

Make/Model	: ZP 275
Rate Load Capacity	: 4,500 KN
Table Opening	: 27-1/2"
Drive Type	: Mechanical

TUBULAR

: DP 5", 19.5 PPF, 2000 ft.
: DP 3-1/2", 15.5 PPF, 2000 ft.
: HWDP 5"
: HWDP 3-1/2"
: DC 8"
: DC 6-1/4"



POWER SUPPLY SYSTEM

Engines Make/Model	: 3 x CAT D-3512C
Engines Power	: 3 x 1,476 HP @ 1,200 RPM
Power Generators	: 3 x Leroy Sommer
Electric Power	: 3 x 1700 Kva @ 1,200 RPM
Emergency Power	: Volvo TAD1641GE 473 hp @ 1,500 rpm
Rig Power Voltage	: 380V/400V, 50 Hz

MUD SYSTEM

Mud Pumps Make/Model	: 3 x BOMCO F1300
Max Rated Input	: 1,300 HP AC
Standpipe Assembly	: 1 x 5" 5,000 Psi
Shale Shakers	: 3 x Kemtron
Desander	: Derrick SCM-SPG-3P
Desilter	: Derrick SCM-SPG-3P
Mud Cleaner	: Derrick SCM-SPG-3P
Mud Degasser	: Mf SWACO CD1400
Mud Gas Separator	: 9.73 mm(T)x1.223 mm(OO)x3,400 mm(SIS)

WELL CONTROL EQUIP.

BOP Annular	: 1 unit 21-1/4" x 2,000 Psi
	: 1 unit 13-5/8" x 5,000 Psi
	: 1 unit 7-1/16" x 5,000 Psi
BOP Rams	: 1 unit 13-5/8" x 5,000 Psi
	: 1 unit 7-1/16" x 5,000 Psi
Choke Manifold	: 4-1/16" x 5,000 Psi

CAMP CAPACITY

: 150 Person

RIG #.231 / 1000 HP Land Drilling Rig

Rated Drilling Depth :

5 in Drill Pipe : 13,000 ft (3,962 M)

Mast

Make	: Dresco
Height	: 136 ft (41 M)
Max Hook load	: 600,000 Lbs
Racking Capacity	: 5 in. Drill Pipe – 12,500 ft (3,810 M)

Drawworks

Make	: Continental Emsco D-3
Rated Input	: 1,000 Hp (746 kW)
Input Power	: Two (2) GE- 752 DC Traction Motor Rated 1,000 HP (746 Kw) Each
Drilling Line	: 1-1/4 in (31.75 mm)

Traveling Equipment

Traveling Block	: Continental Emsco
Max Load	: 700,000 Lbs
Swivel	: Ideco LB – 400
Max Load	: 800,000 Lbs

Substructure

Make & Type	: Dresco Slingshot
Clear Working Height	: 26.9 ft c/w Skid Beam
Setback Load	: 350,000 Lbs

Rotary Equipment

Top Drive	: Tesco 500 Ton
Input Power	: Caterpillar D-3412 engine 750 Hp (559 kW)
Rotary Table	: Ideco LR 275 27 1/2 in (698.5 mm)
Input Power	: Driven by Drawworks
Static load Rating	: 570,000 Lbs

BOP System

Annular Preventer	: One (1) 21 1/4" in Hydril 2,000 Psi
Ram Preventer	: Two (2) 20 3/4" Hydril Single Ram 3000Psi



Gensets

Engines	: Four (4) Cat D 398 Diesel Engines, 900 Hp (671 kW) @ 1,200 Rpm each
Generators	: Four (4) Kato 800 kW (1,000 kVa) each generator

Power Distribution

Gen Control	: Four (4) Ross Hill Model 1400 Three (3) Ross Hill Model 1400 SCR bays
MCC	: 600 V, 480 V, 230 V, 127 V

Mud Pumps

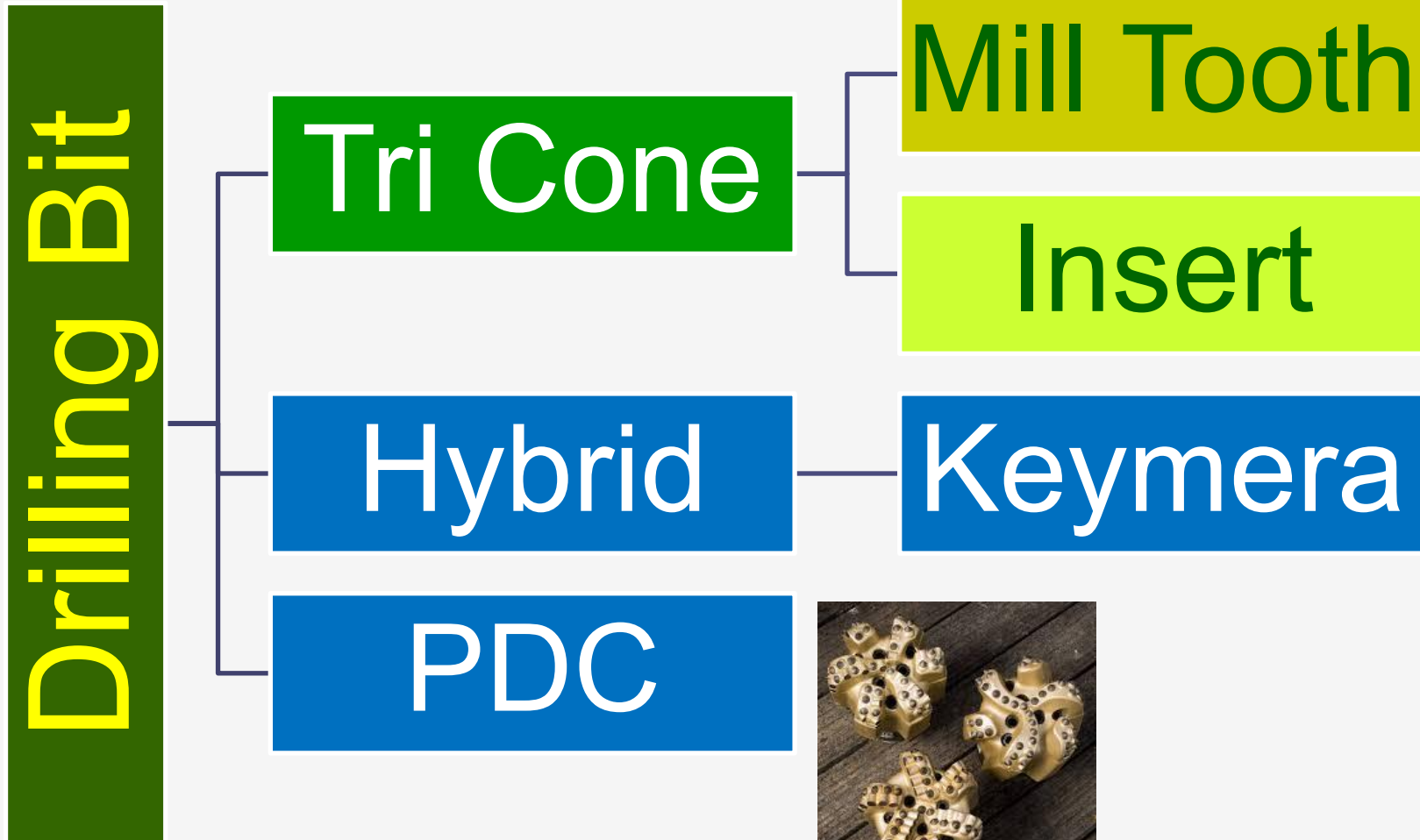
Pumps	: Four (4) Continental Emsco F-1000 Triplex pump, Rated 1,000 Hp (746 kW) each
Input Power	: Two (2) F-1000 each Driven two (2) GE 752 Traction motor Rated : 1,000 Hp (746) each
	: Two (2) F-1000 Driven by Cat D- 398 Diesel engine each Rate 900 Hp (671 KW)

Mud Handling / Solid Control

Active Mud system	: 1,200 bbls (191 m3)
Shale shaker	: Three (3) Derrick High speed Linear Motion, each 500 GMP
Mud Cleaner	: Derrick Super G 4- panel dryer / Mud Cleaner w/ 10" x 3 Cone Desander and 4" x 20 Cone Desilter
Degasser	: Derrick Vacu – Flo 1200 ; 1,200 GPM

2. How to determine Rig Sizing

DRILLING Categories Based On LOCATION



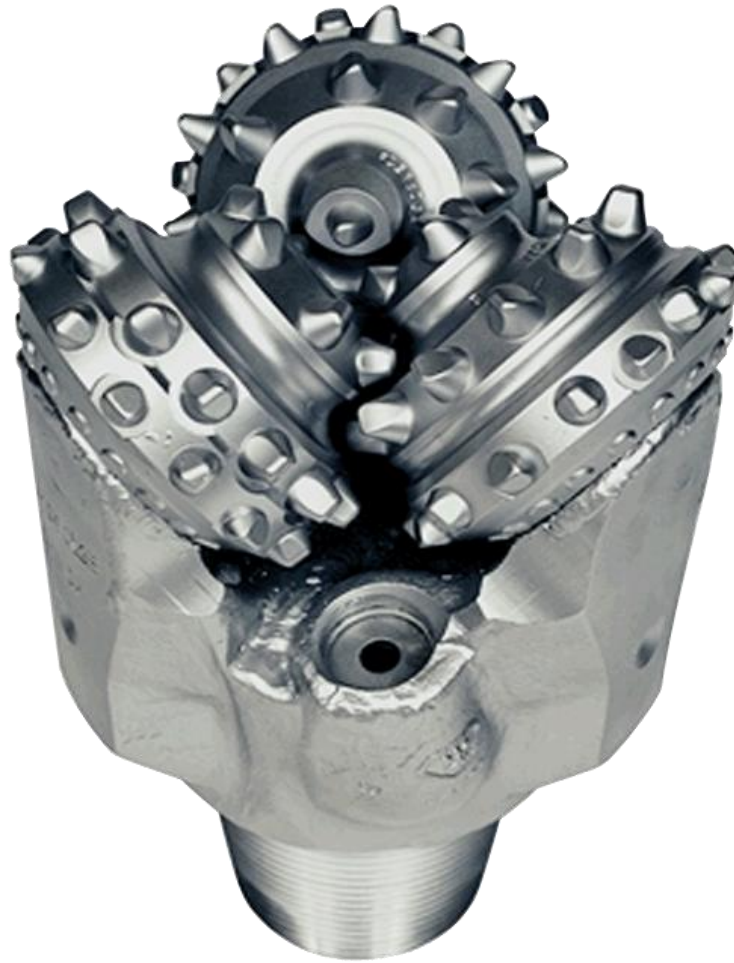
3. Well Construction – Drilling Bits

Milled steel teeth are hardfaced
with tungsten carbide



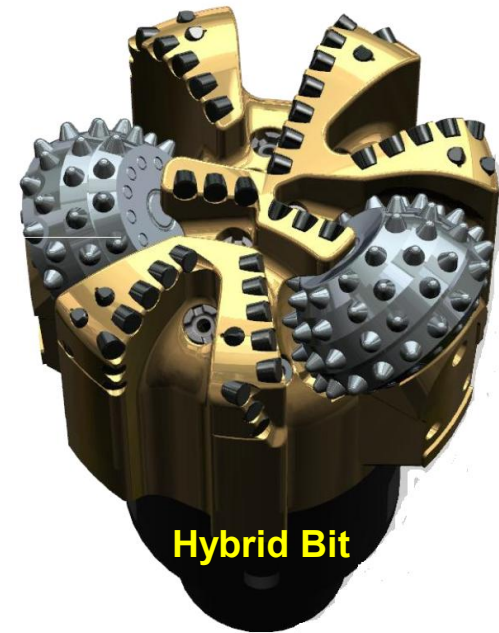
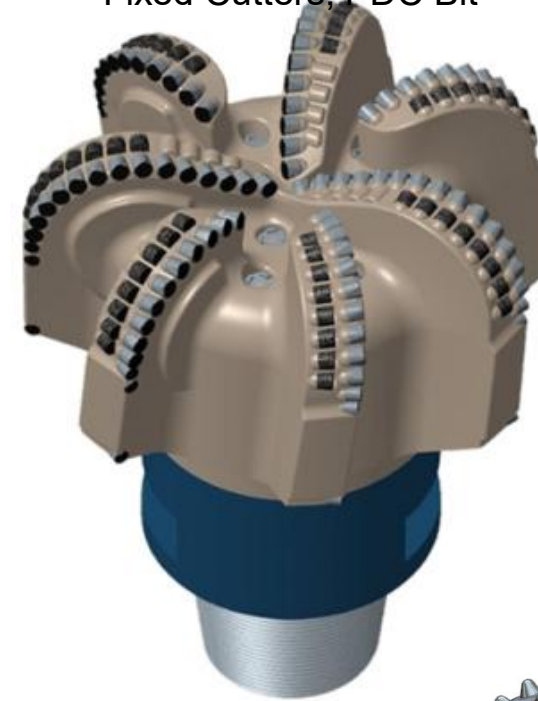
Steel Tooth

Inserts are press fit into the cone



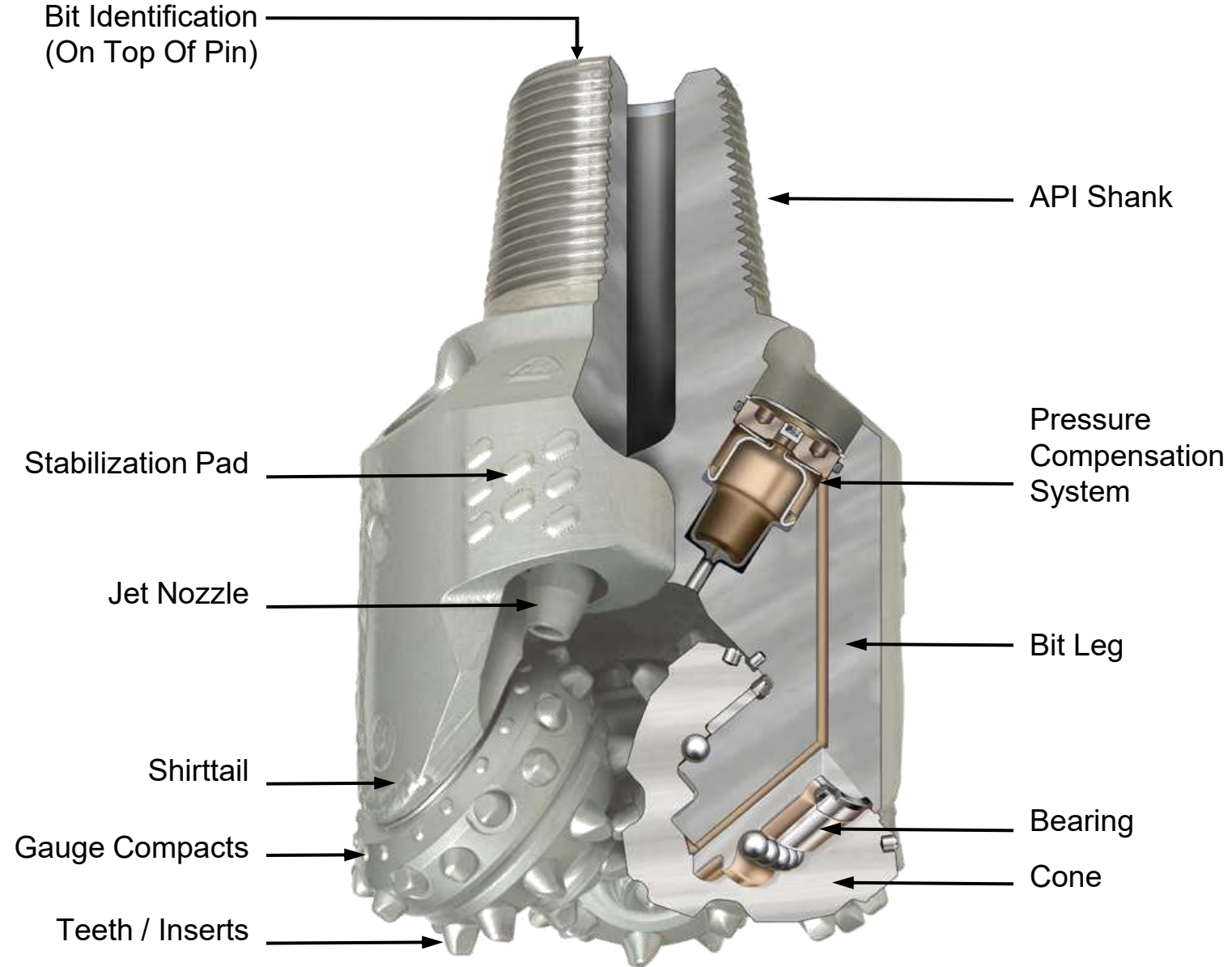
Tungsten Carbide Insert (TCI)

Fixed Cutters, PDC Bit

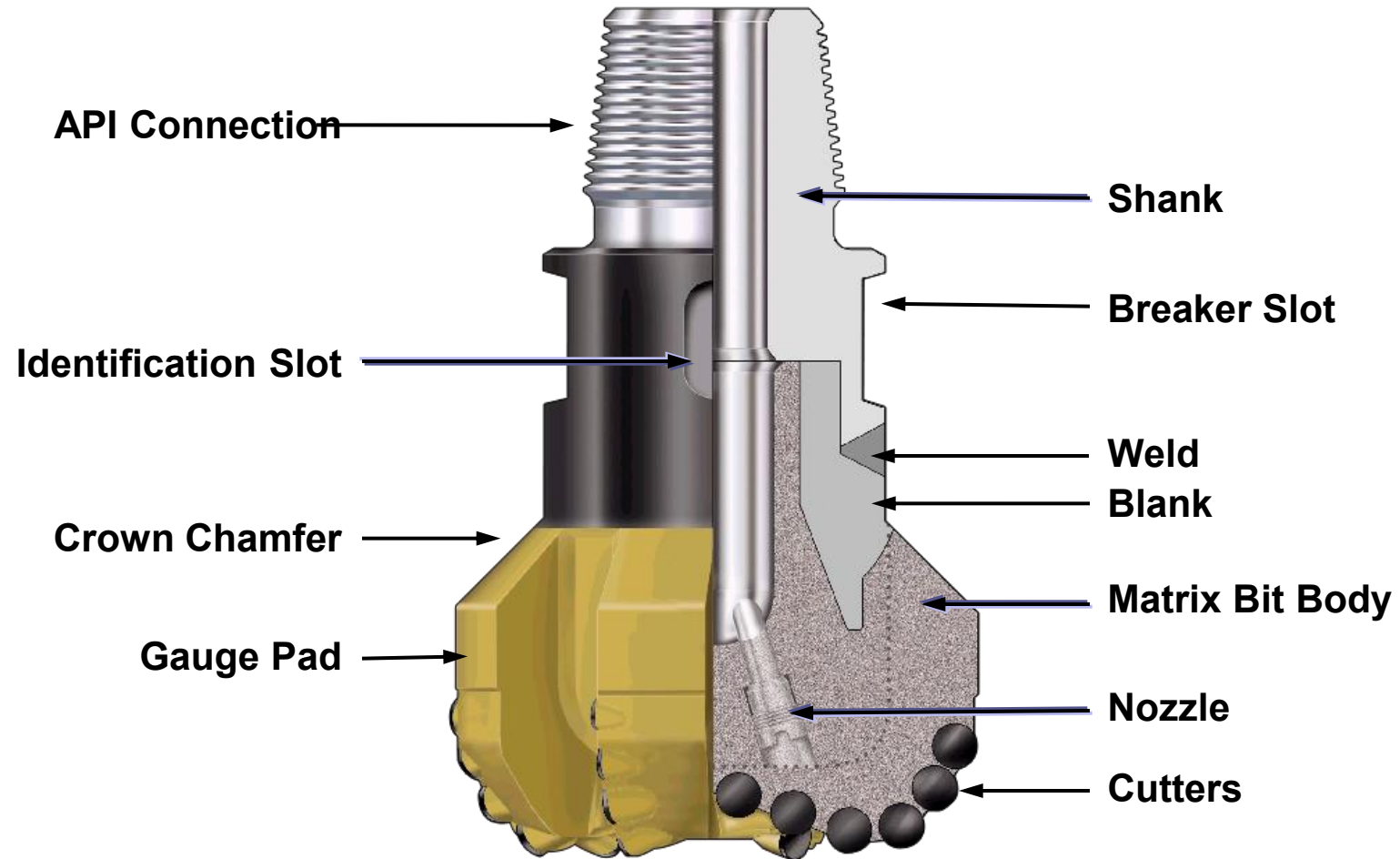


Hybrid Bit

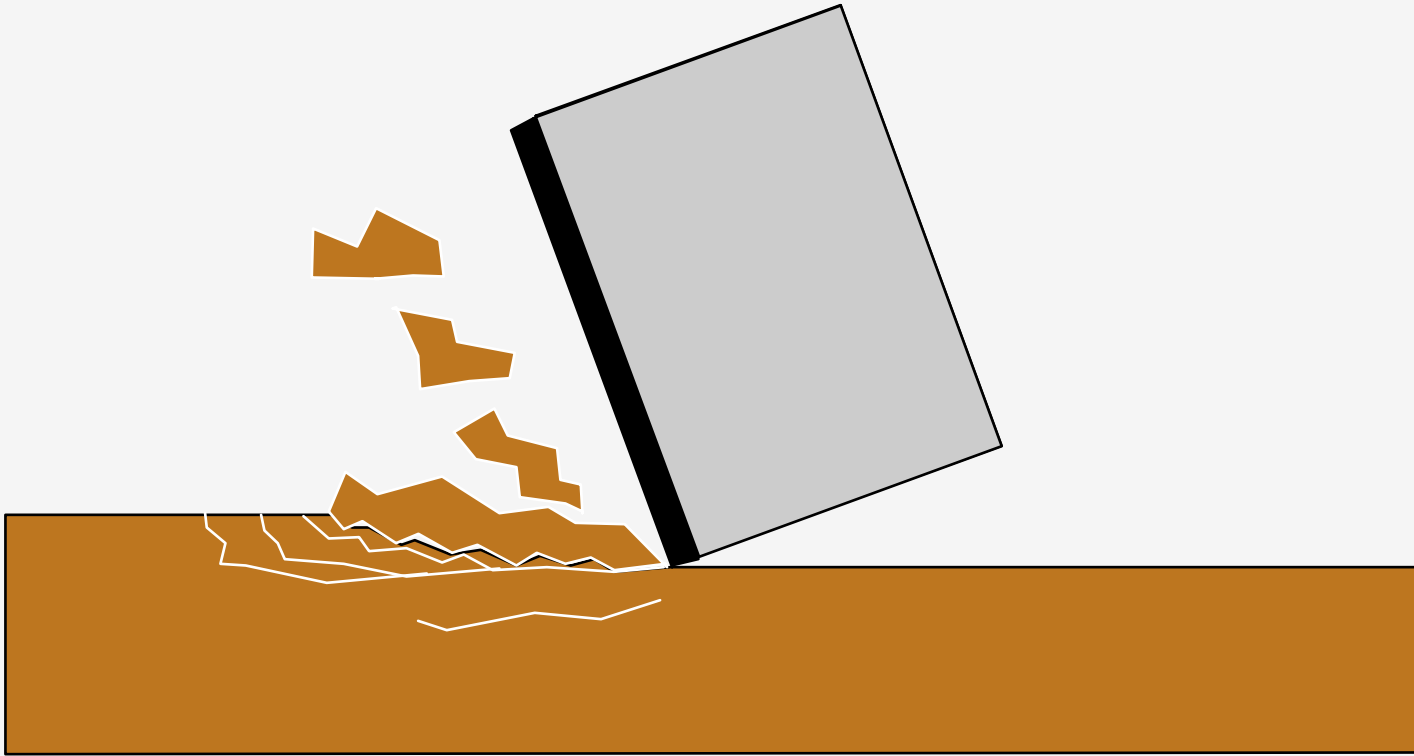
Tricone Bit Structure



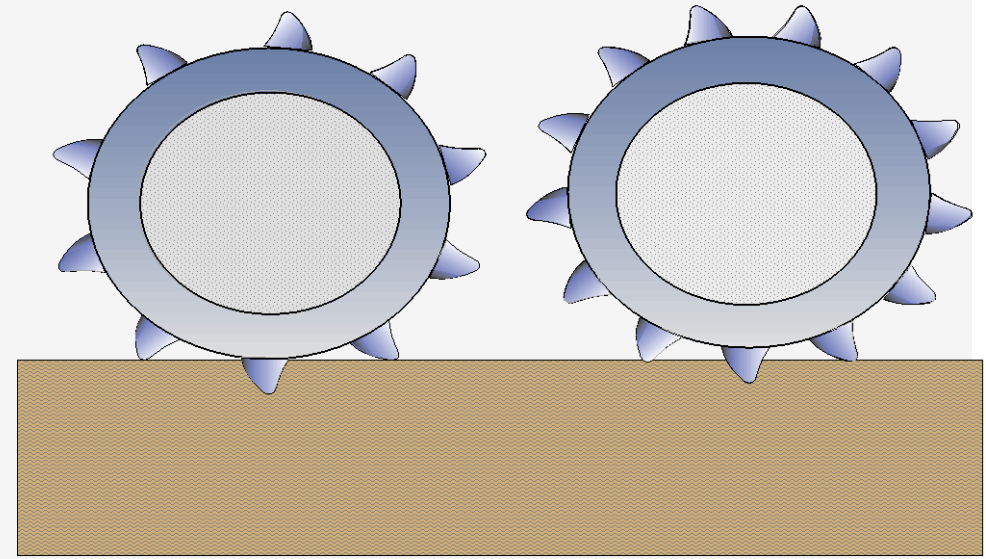
PDC Bit Structure



Cutting Method: PDC vs Rock Bit



Shearing the rock is the most efficient cutting method with most rock types.



Crushing the rock

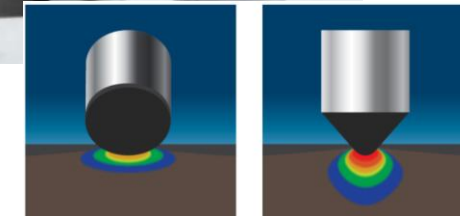
3. Well Construction – Best Performance Bit for Geothermal Drilling

26" Bit



- Tungsten Carbide Insert bit, IADC 415, 435, 515
- **Premium Bearing feature**
- **50-100% Diamond Gage Protection**

17-1/2", 12-1/4", 9-7/8", 7-7/8" Bit



FuseTek

StingBlade
Exe Blade

Hybrid Bit, Kymera

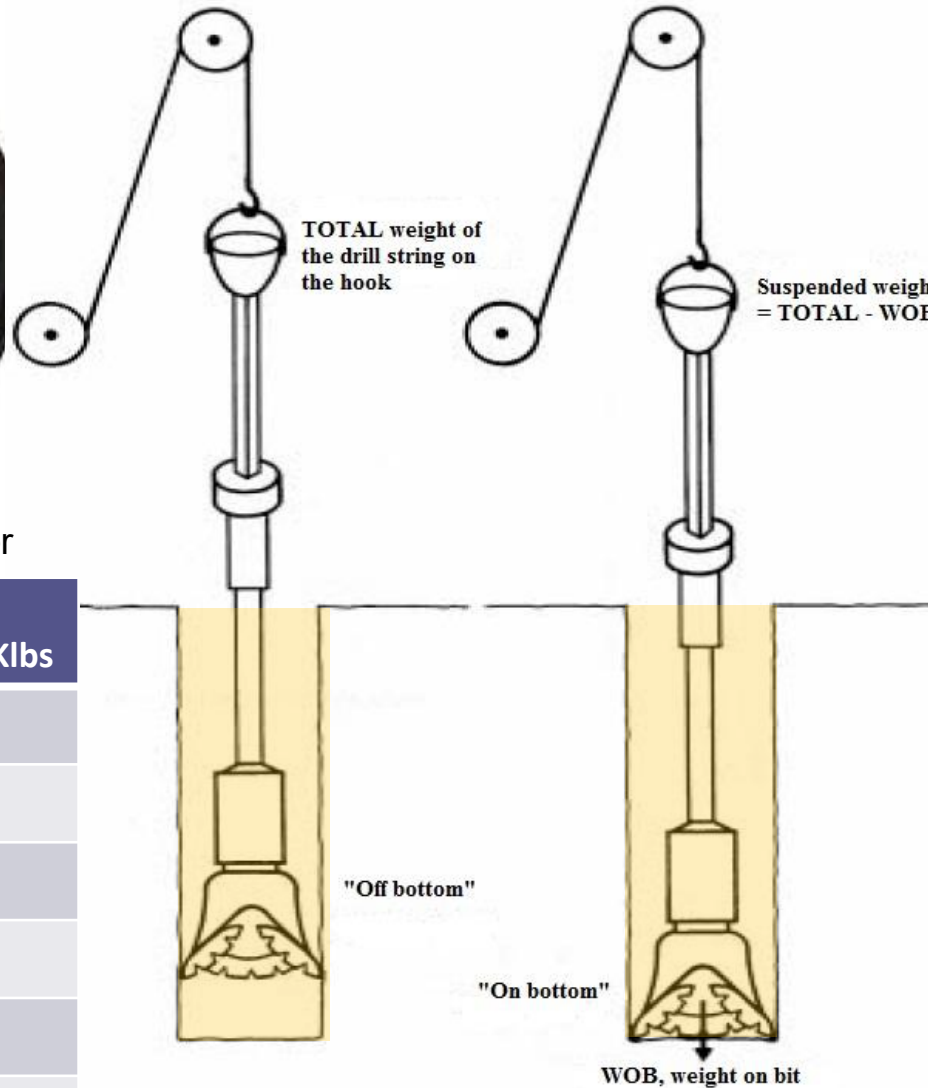
3. Well Construction – Weight On Feet = Bit



3. Well Construction – Weight On Bits (WOB)



Martin Decker Weight Indicator

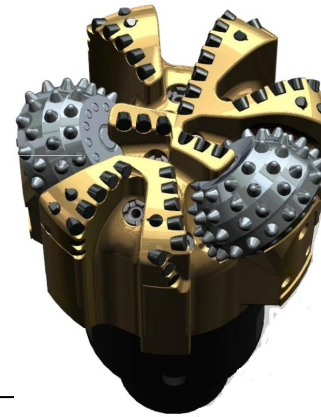
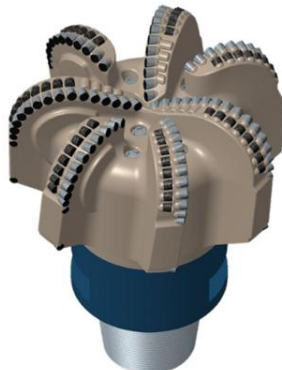
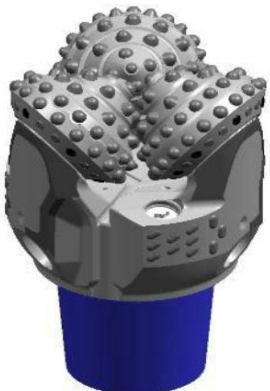


Condition	Weight Indicator, Klbs	WOB, Klbs
Off Bottom	240	0
On Bottom	230	10
On Bottom	220	20
On Bottom	210	30
On Bottom	200	40

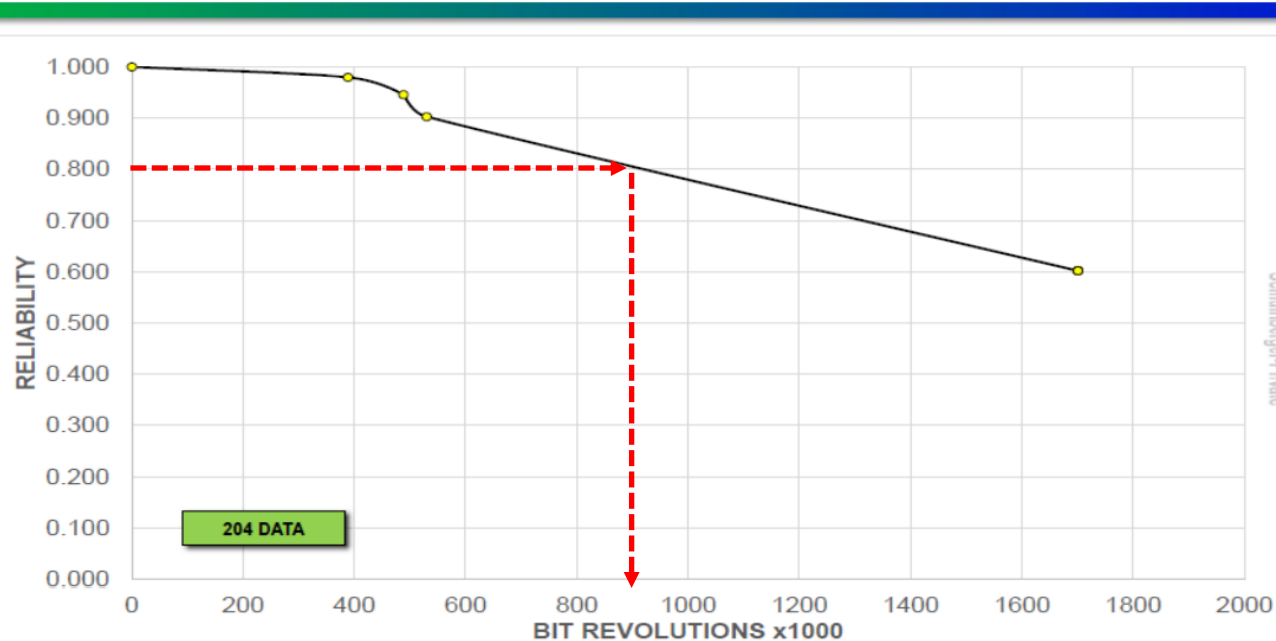
Weight Description	Weight Indicator, Klbs
Travelling Block	20
TDS (Top Drive System)	30
5" Drill Pipe	110
5" HWDP	25
6-1/2" DC	5
8-1/2" DC	50

Drilling Bit Program for Geothermal Well

Section	Bit Type	Remarks	Smith	Baker Hughes	NOV Reed
26"	TCI	TCI 26" for 500 m footage	TCI 415	TCI 435	-
17.5"	TCI or PDC	• Smith -> best TCI footage 590 m (WW) • Baker -> best TCI footage 638 m (SMP) ➤ If depth interval $\leq$ 600 m, then using TCI ➤ If depth interval $>$ 600, then using PDC/Hybrid	TCI 435/515 PDC (Stingblade) 616	TCI 435/515 Hybrid (Kymera 636)	PDC (FTKC76)
12.25"	PDC	Longest section	PDC (Stingblade 716)	Hybrid (Kymera 525)	PDC (FTKC76)
9.875"	TCI or PDC	• Smith -> Average TCI footage 186 m • Baker -> Average TCI footage 192 m (South Sumatera) ➤ If depth interval $\leq$ 200 m, then using TCI ➤ If depth interval $>$ 200 m, then using PDC/Hybrid	TCI 537/637 PDC (Stingblade 716)	TCI 537 Hybrid (Kymera 525)	PDC (FTKC76)
7.875" (Contingency)	TCI	Contingency Section	TCI 537/637	-	-
14-3/4"	Milltooth	Clean Out Receptacle	Milltooth 115	-	-



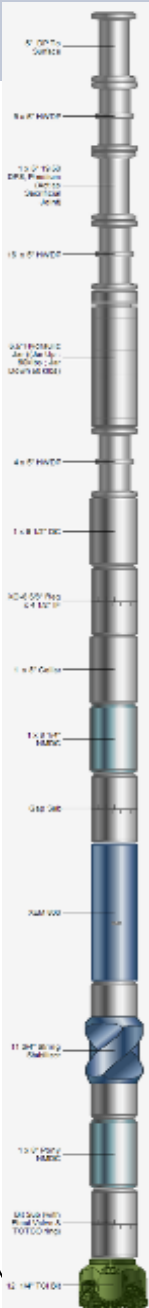
26" G04 / GS04 Series
Seal/Bearing Reliability (IADC 415)



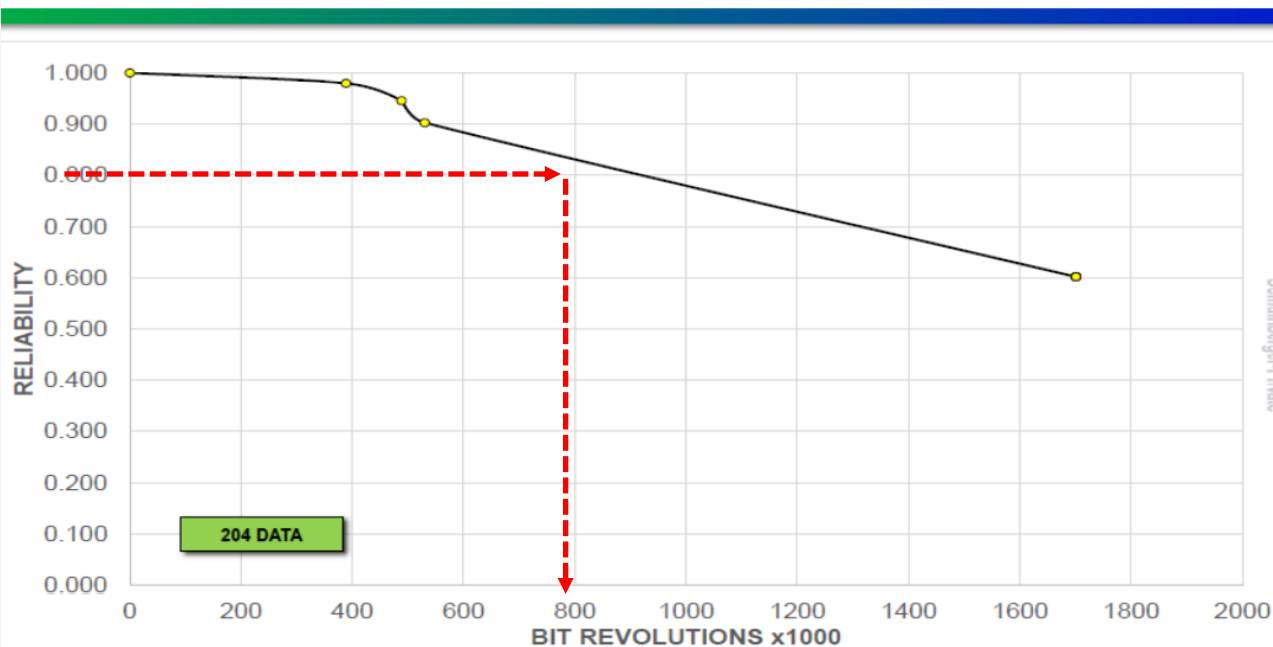
ROTARY DRILLING (WITHOUT MUD MOTOR):

- 80% Reliability, Bit revolution = 900.000 revolution
- RPM = Rotation Per Minute
- If Drill string is rotated at 100 RPM
- Bit Life = 80% Reliability, Bit revolution / Bit RPM
 - = 900.000 Revolution : 100 RPM
 - = 9000 Minutes
 - = 150 Hours
- Drilling Footage = ROP X Bit Life
 - = 10 ft per hour X 150 hours
 - = 1500 ft

6. Drilling Bit Life – DIRECTIONAL DRILLING USING MUD MOTOR



26" G04 / GS04 Series
Seal/Bearing Reliability (IADC 415)



- Rotary Drill with 60 rpm, Flow rate 1200 gpm. Use 9-5/8" Mud Motor. See Mud Motor performance
- How long Bit Life ???
- **How far the bit could drill formation, if average drill speed = 15 ft per hour**

9 5/8" (244 mm) 6-7 5.0 Stage

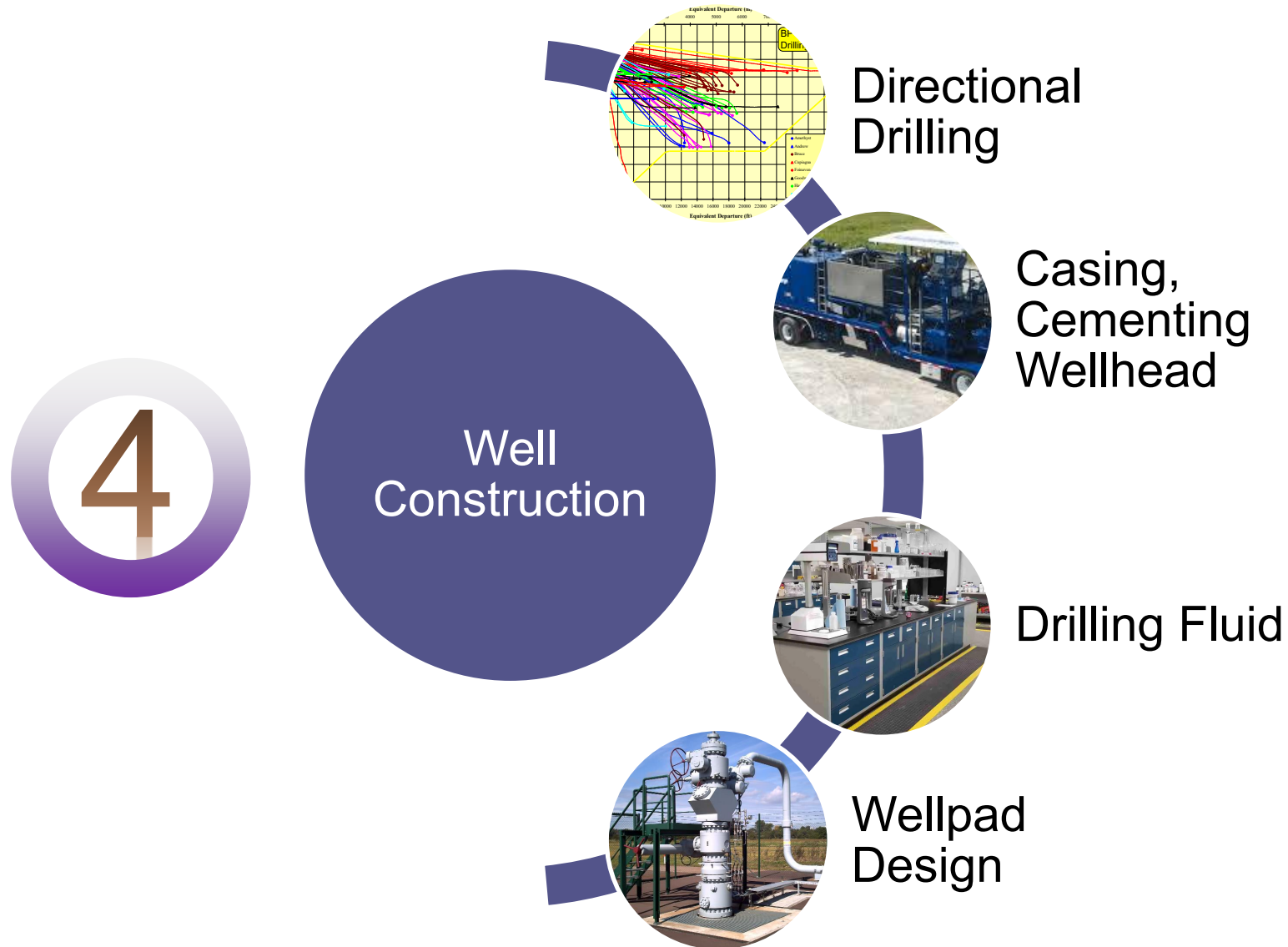
COUGAR
DRILLING SOLUTIONS

Maximum Pump Rate	1,200 gpm	4,542 Lpm
Revolutions per unit volume	0.130 rev/gal	0.034 rev/L
Pressure at full Load	775 psi	5,343 kPa
Torque at full load	13,302 ft lb	18,035 N m

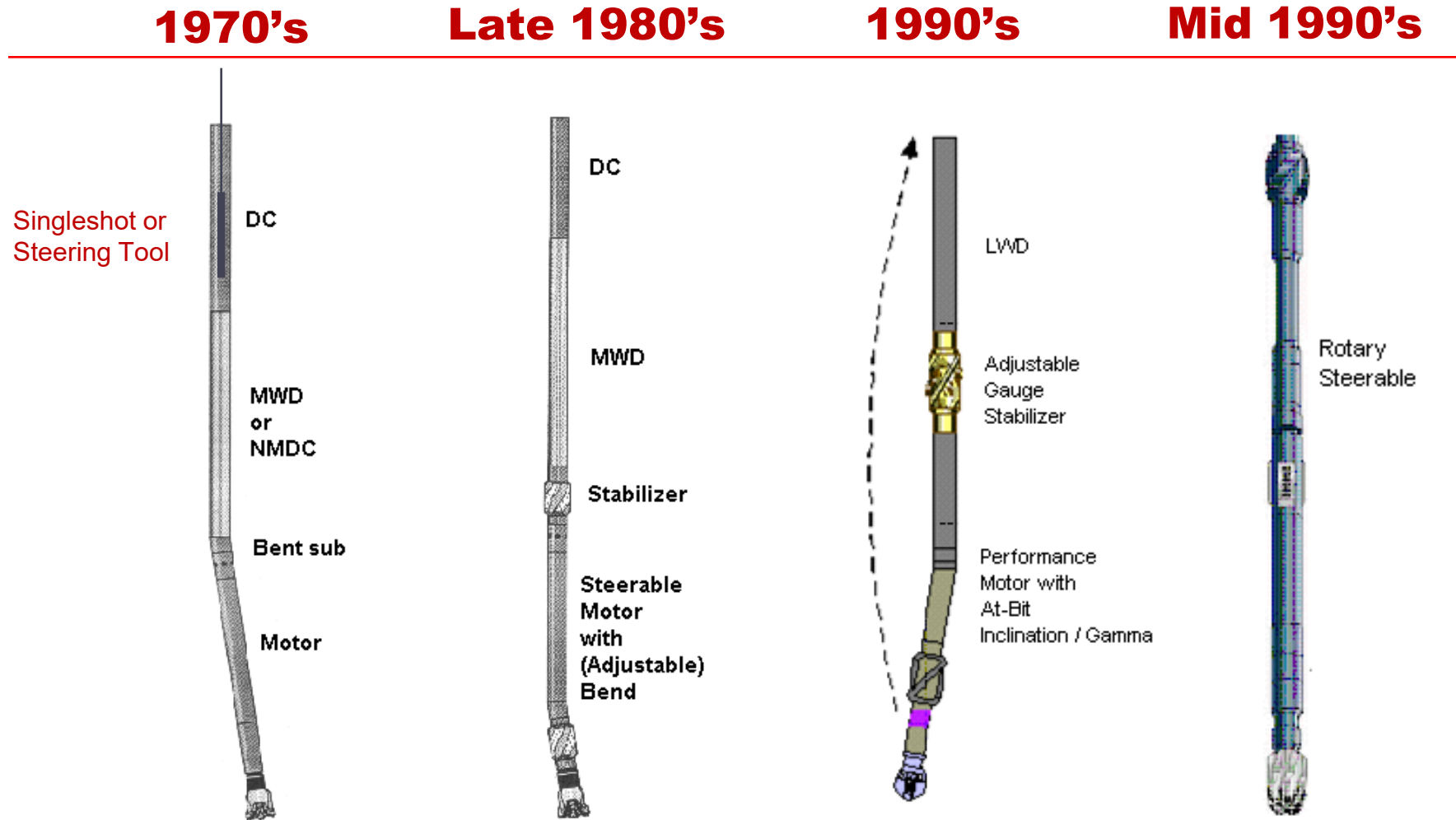
- RPM = Rotation Per Minute
- 80% Reliability, Bit revolution = 900.000 revolution
- Drill string is rotated at 60 RPM
- Down Hole Motor rotation
= 1200 gpm x 0.130 rev/gallon
= 156 RPM
- Total Bit RPM = Drill String RPM + Down Hole Motor
RPM = 60 RPM + 156 RPM = 216 RPM
- Bit Life = 80% Reliability, Bit revolution / Total Bit RPM
= 900.000 Revolution : 216 RPM
= 4166 Minutes
= 69 Hours
- Drilling Footage = ROP X Bit Life
= 20 ft per hour X 69 hours
= 1380 ft

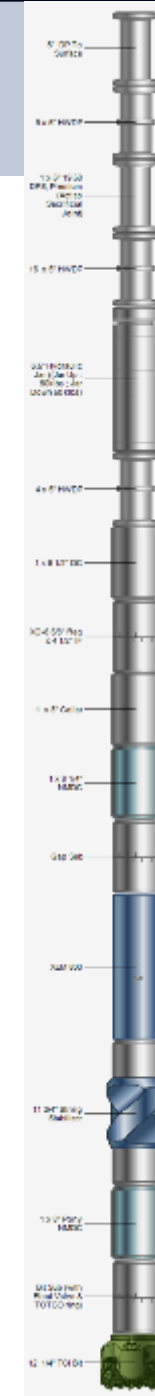
3

Well Construction



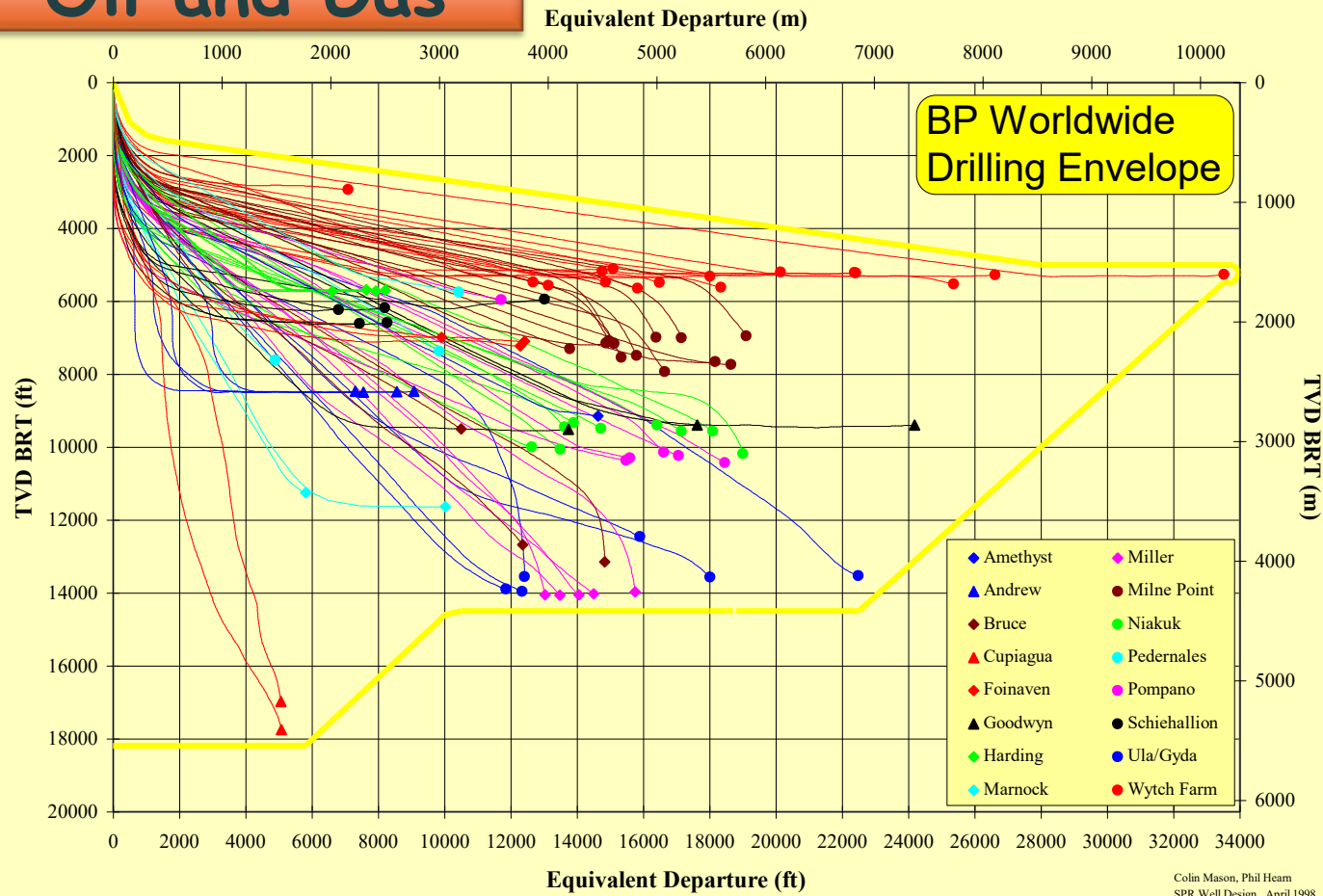
3. Well Construction – Directional Well



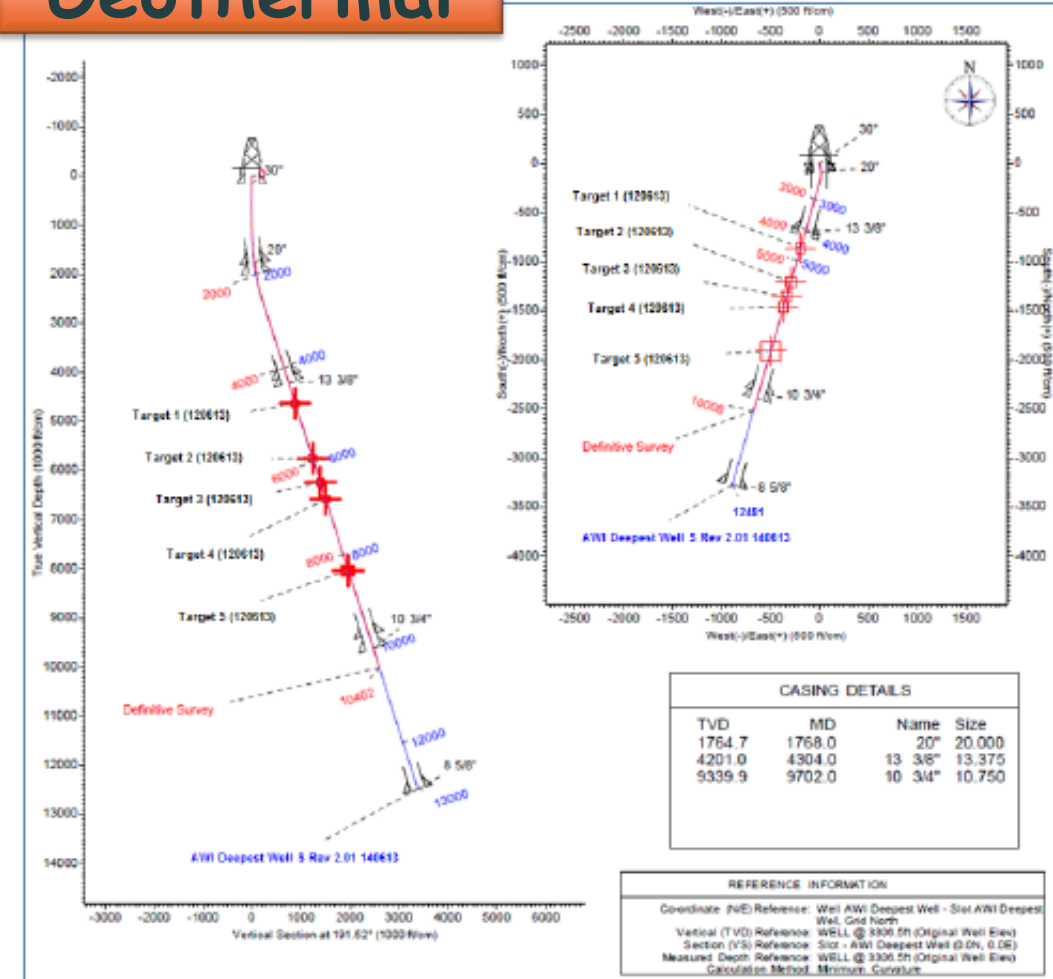


3. Well Construction – Directional Well

Oil and Gas

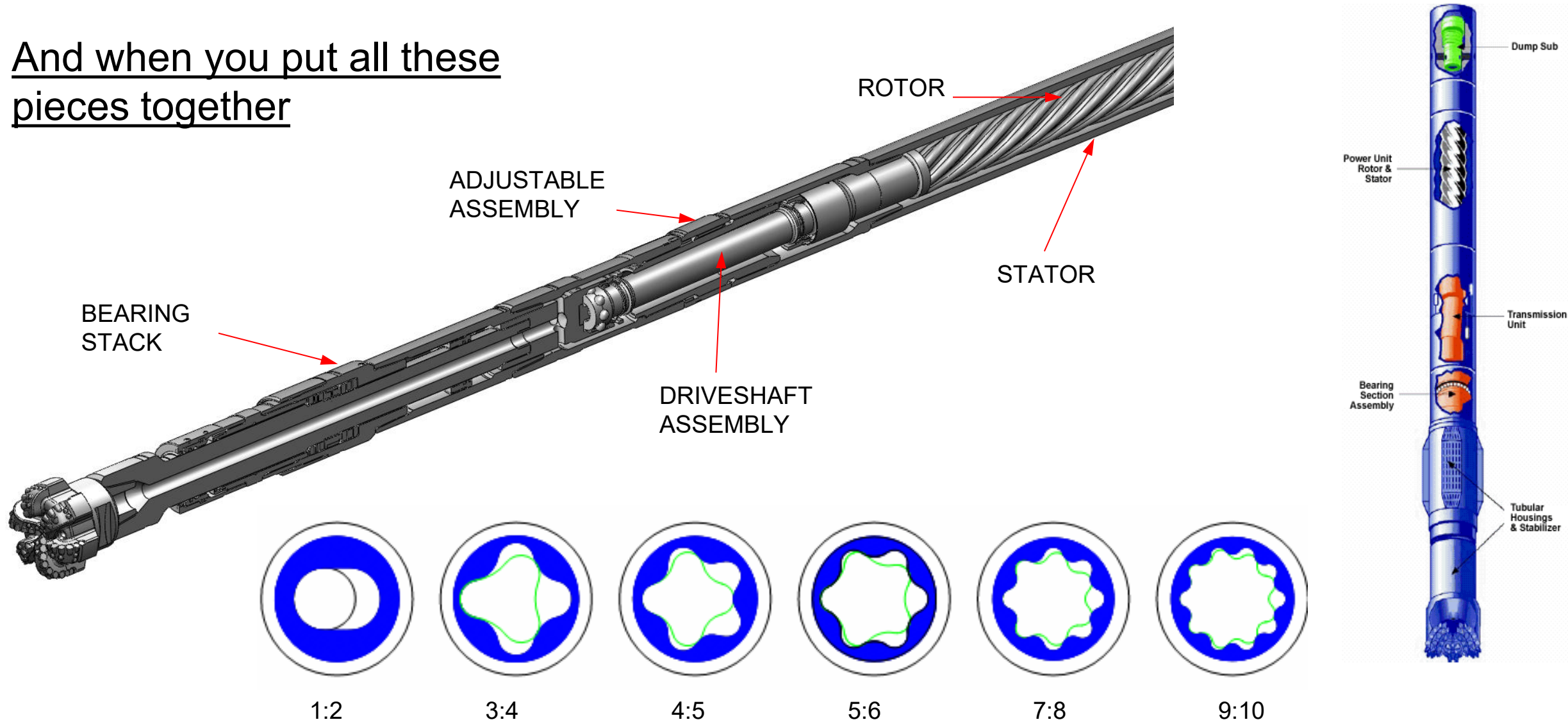


Geothermal

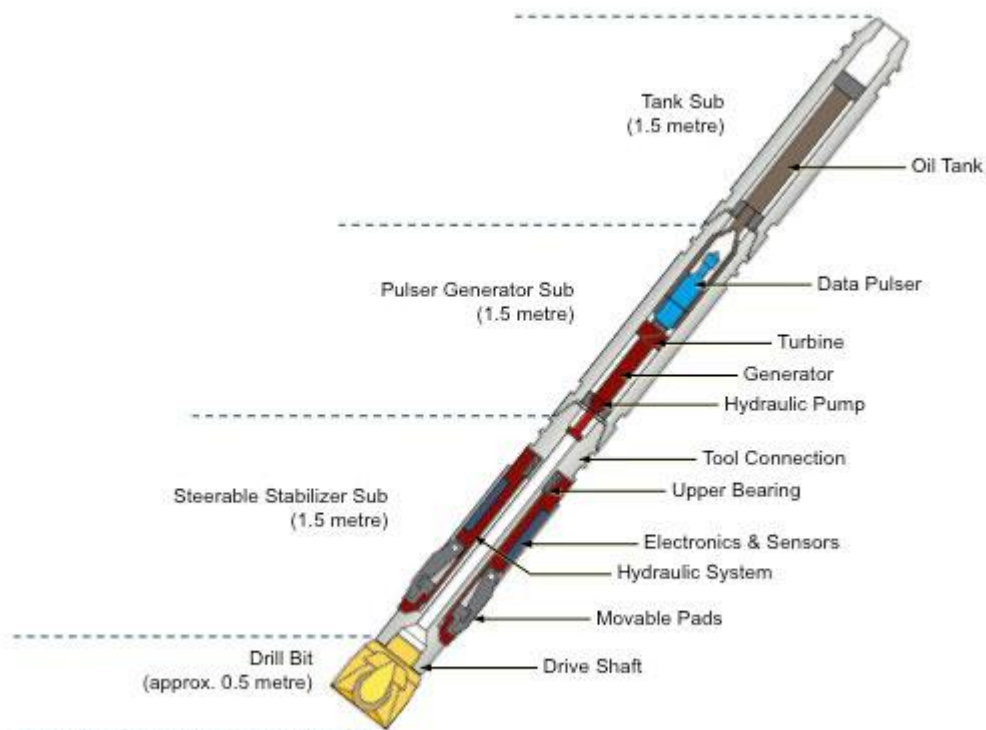


3. Directional Well - Drilling Mud Motor Technology

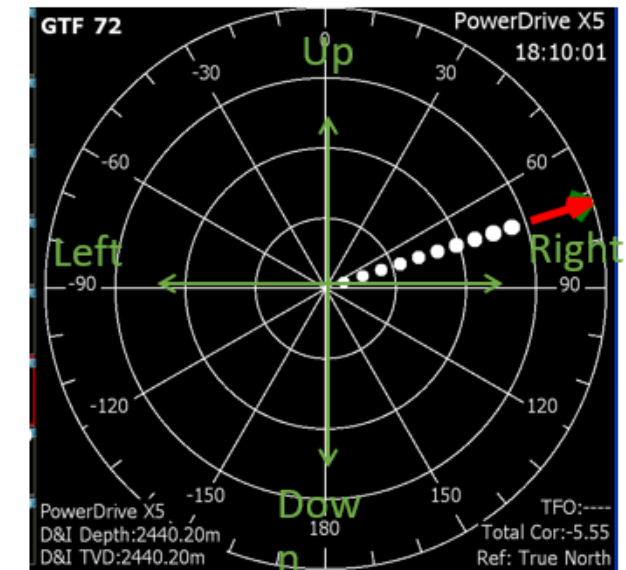
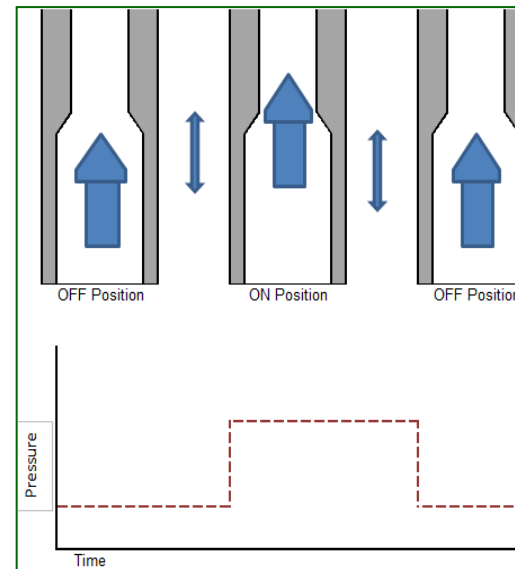
And when you put all these pieces together



3. Well Construction – Drilling Navigation Tool - MWD

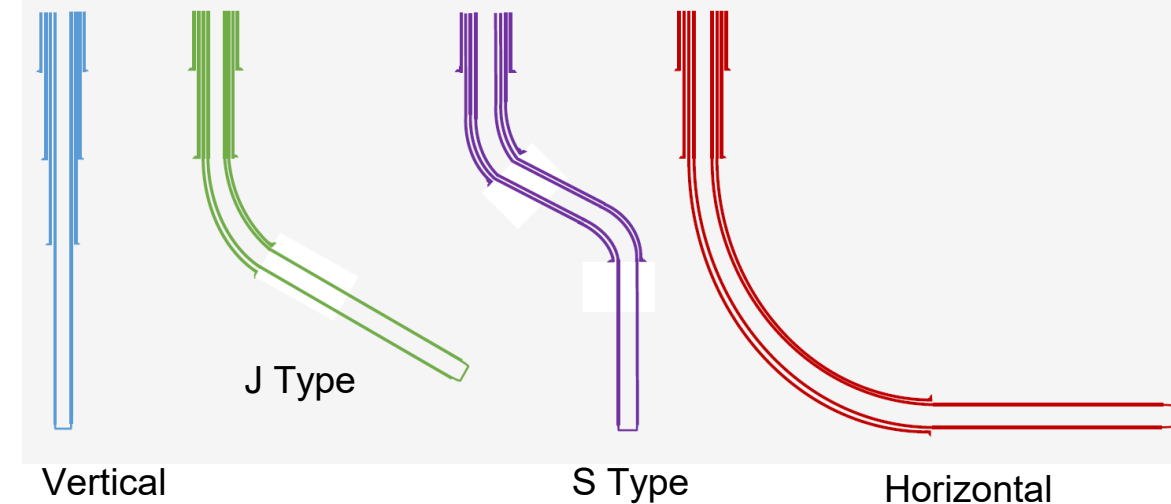
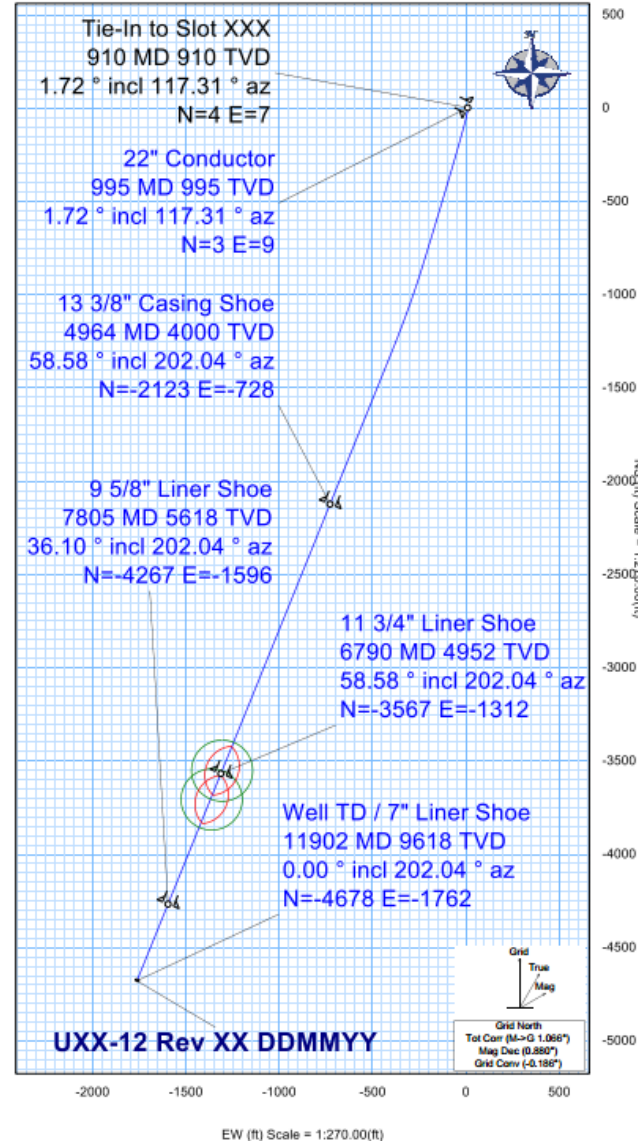
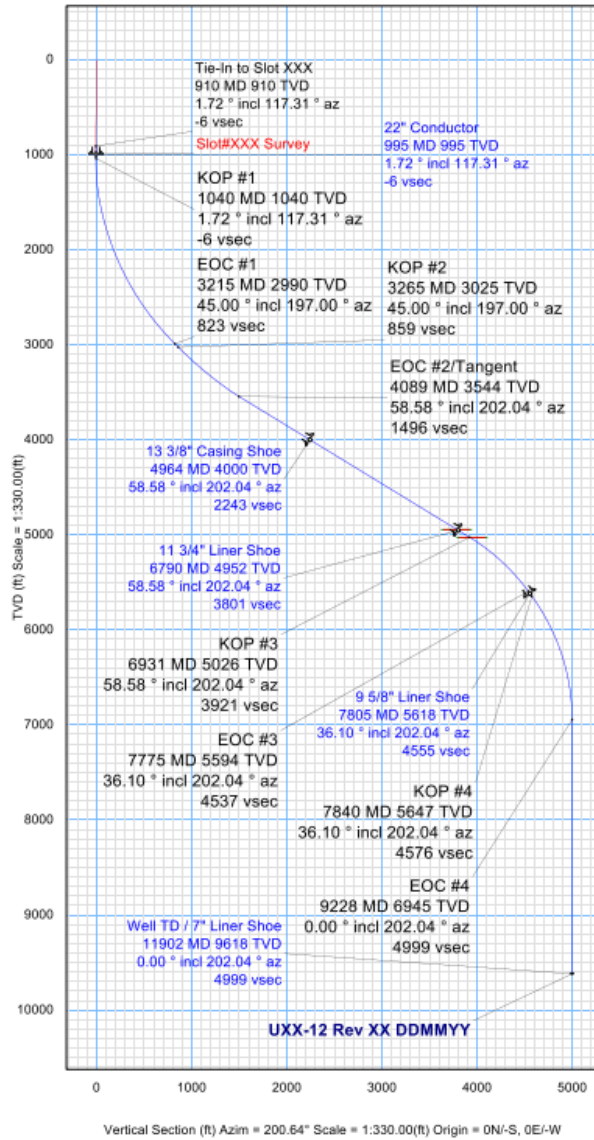


- Pulsa-pulsa ini akan dihantarkan melalui media lumpur dalam bentuk bilangan Biner
- Dengan program komputer, pulsa2 ini dikonvesikan dari bilangan biner menjadi decimal
- Hasil survey berupa: sudut lubang, azimuth lubang, temperatur.

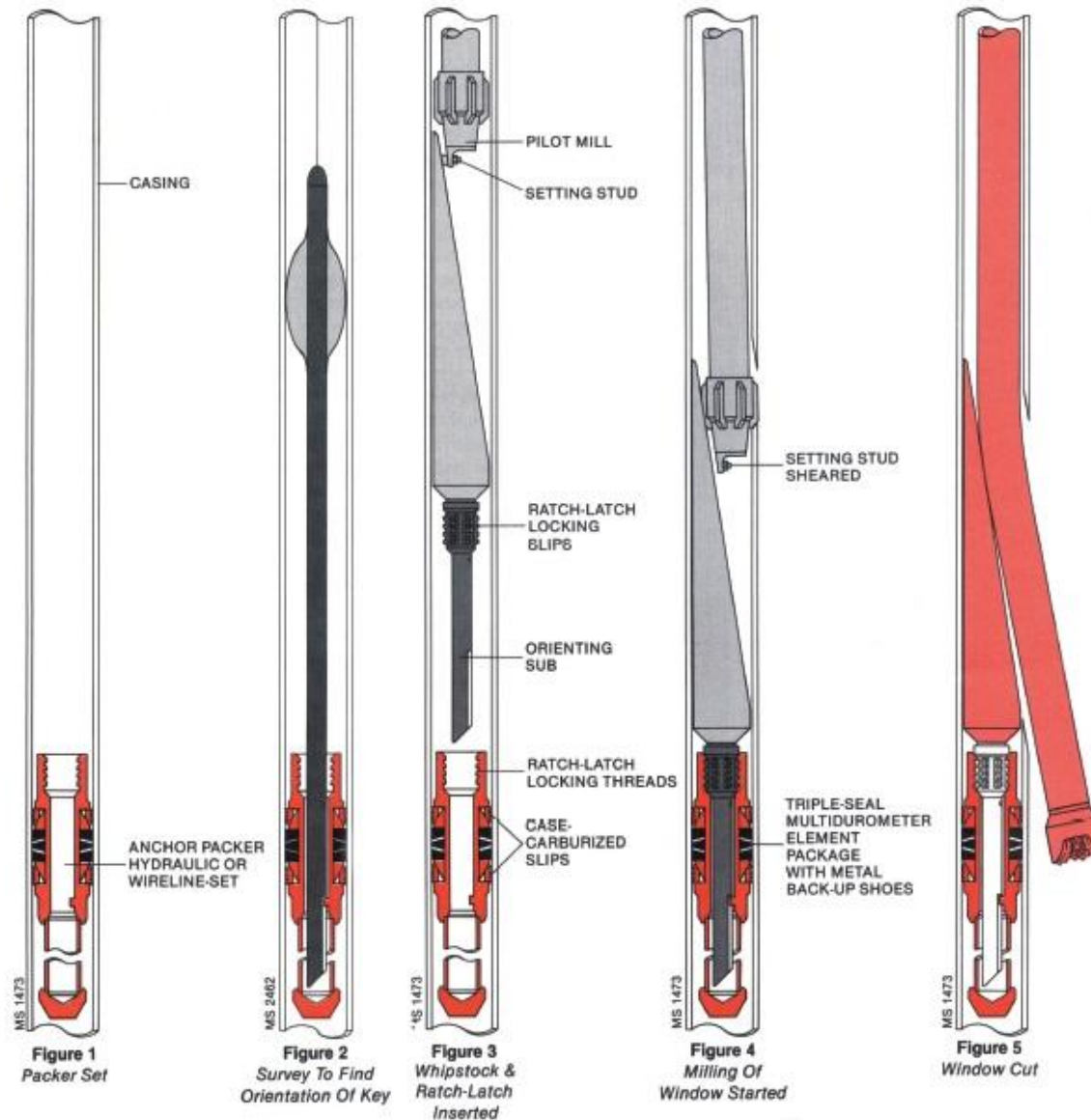


Decimal:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Binary:	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

3. Well Construction – Drilling Bits



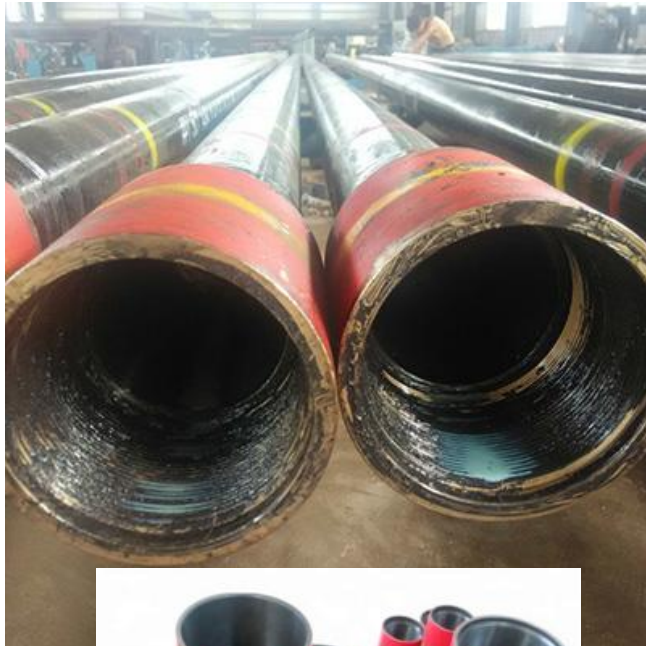
3. Well Construction – Drilling Side Track Hole



- 12-1/4" OD Window Mill
- 12-1/4" OD Water Melon Mill
- 12-1/4" OD Water Melon Mill
- XO
- 6-5/8" HWDP
- XO
- 3 x 8" DC
- XO
- 5 x 5" HWDP
- 6-1/2" Hydraulic Jars
- 16 x 5" HWDP
- 5" Drill Pipe to surface



3. Well Construction – Casing Design



- Box
- End

- Pin
- End



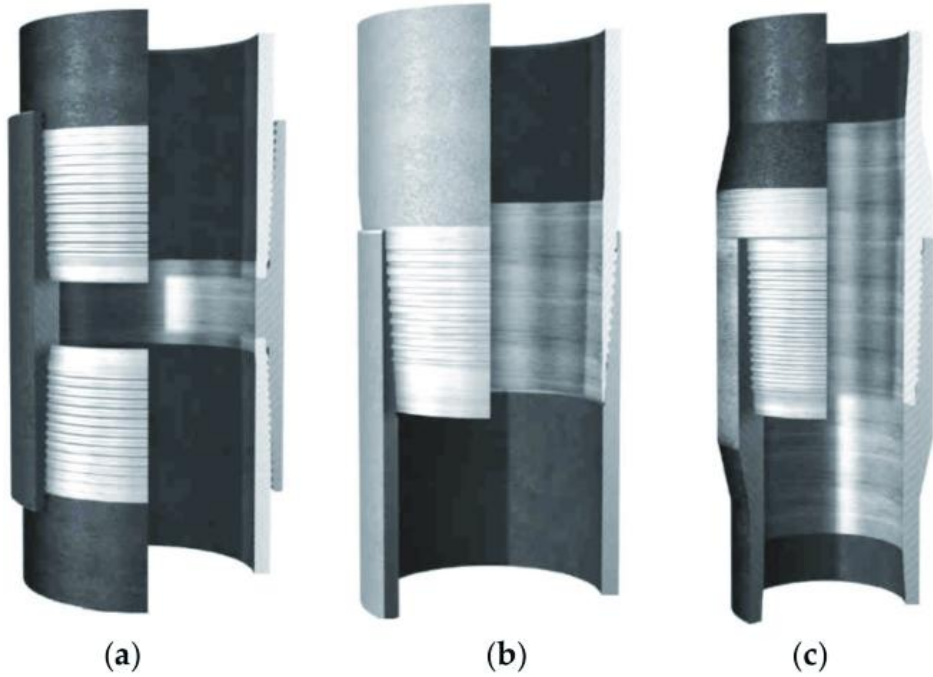
- Casing Collar
(Coupling)



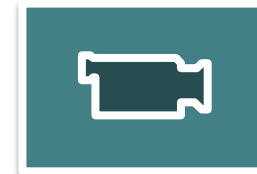
- Casing Protector



Casing Handling Equipment



Casing Thread Type

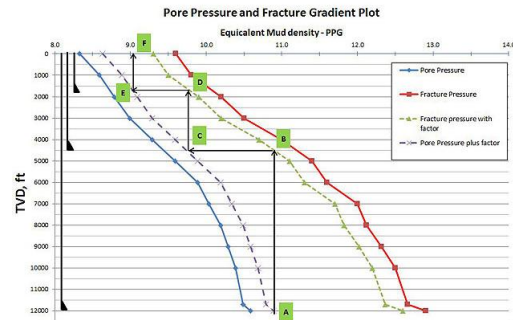


3. Well Construction – Casing Design



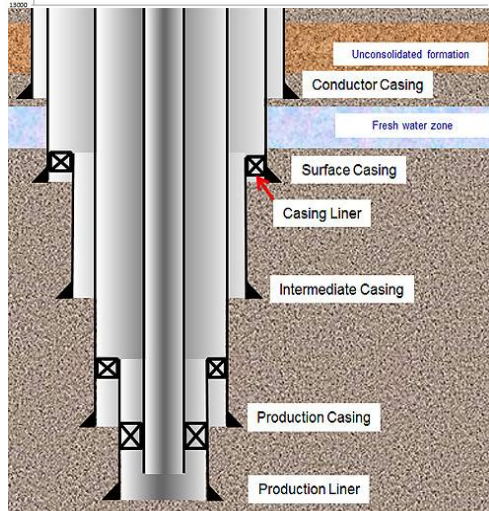
Casing Design

- Setting Casing Depth
- Size of Casing
- Casing Wall Thickness and Grade
- Casing Thread



Pressure

- Formation Pressure (Gradient)
- Fracture Pressure (Gradient)



Mud, Cement, Pressure Test Well Services

- Mud Weight, ppg or SG
- Cement Slurry Weight, ppg
- Well Stimulation Pressure
- Casing Pressure Test
- Well Control

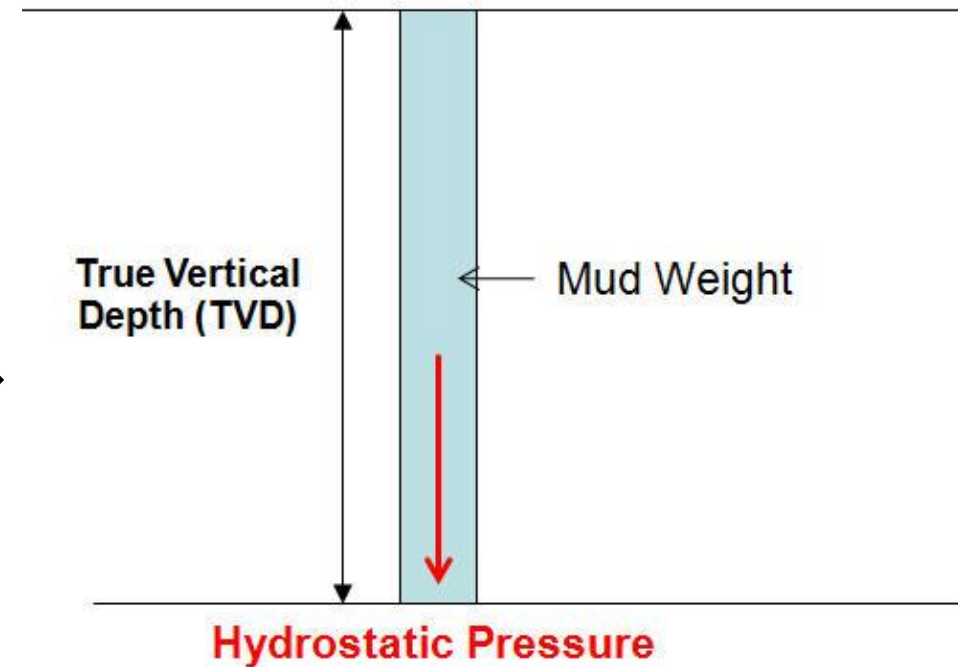
3. Hydrostatic Pressure Formula

Description	Formula	Remark
Hydrostatic	Pressure (Psi) = 0.052 X MW (ppg) X TVD (ft) MW (ppg) = $\frac{\text{Pressure (Psi)}}{0.052 \times \text{TVD (ft)}}$	Ppg = pound per gallon Psi = Pound per square inch $P = \rho gh$

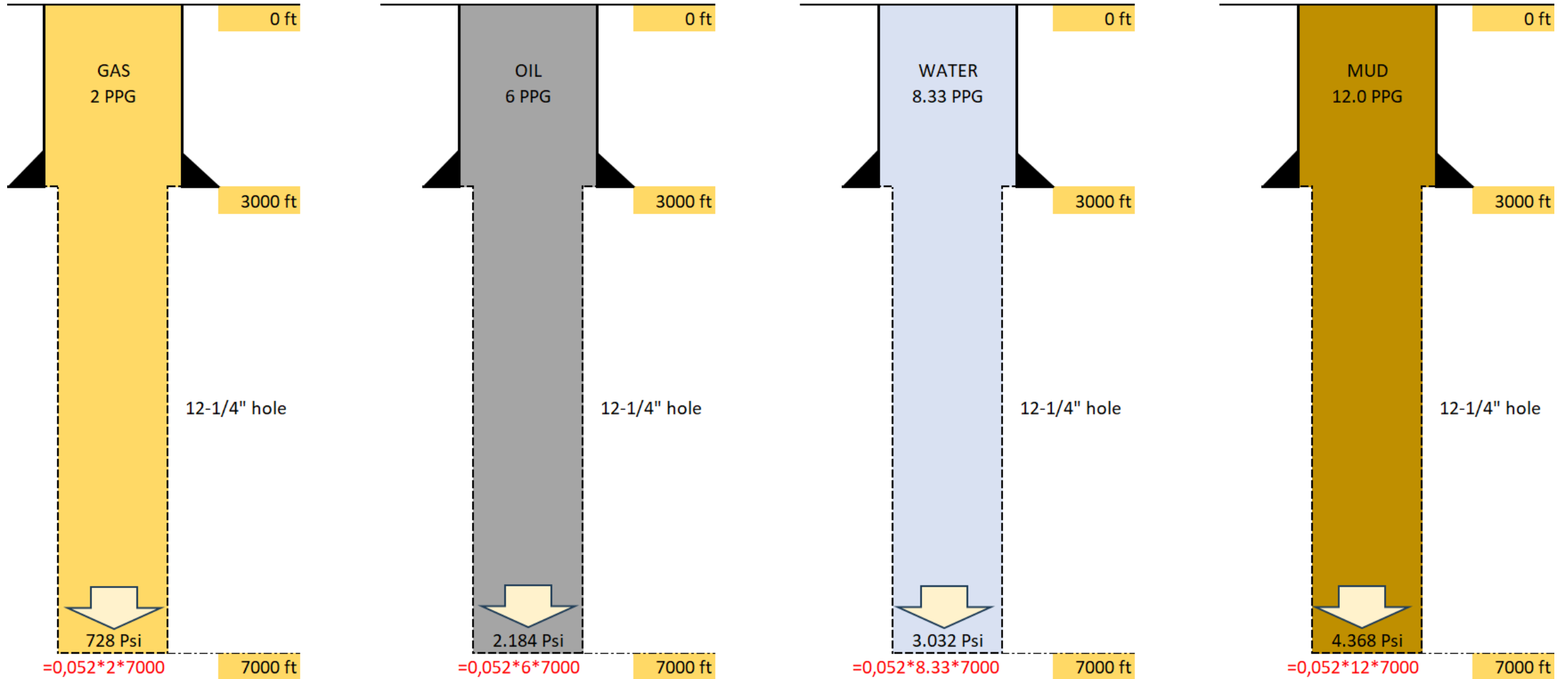


Mud Weight:

- Use Mud Balance → Ppg

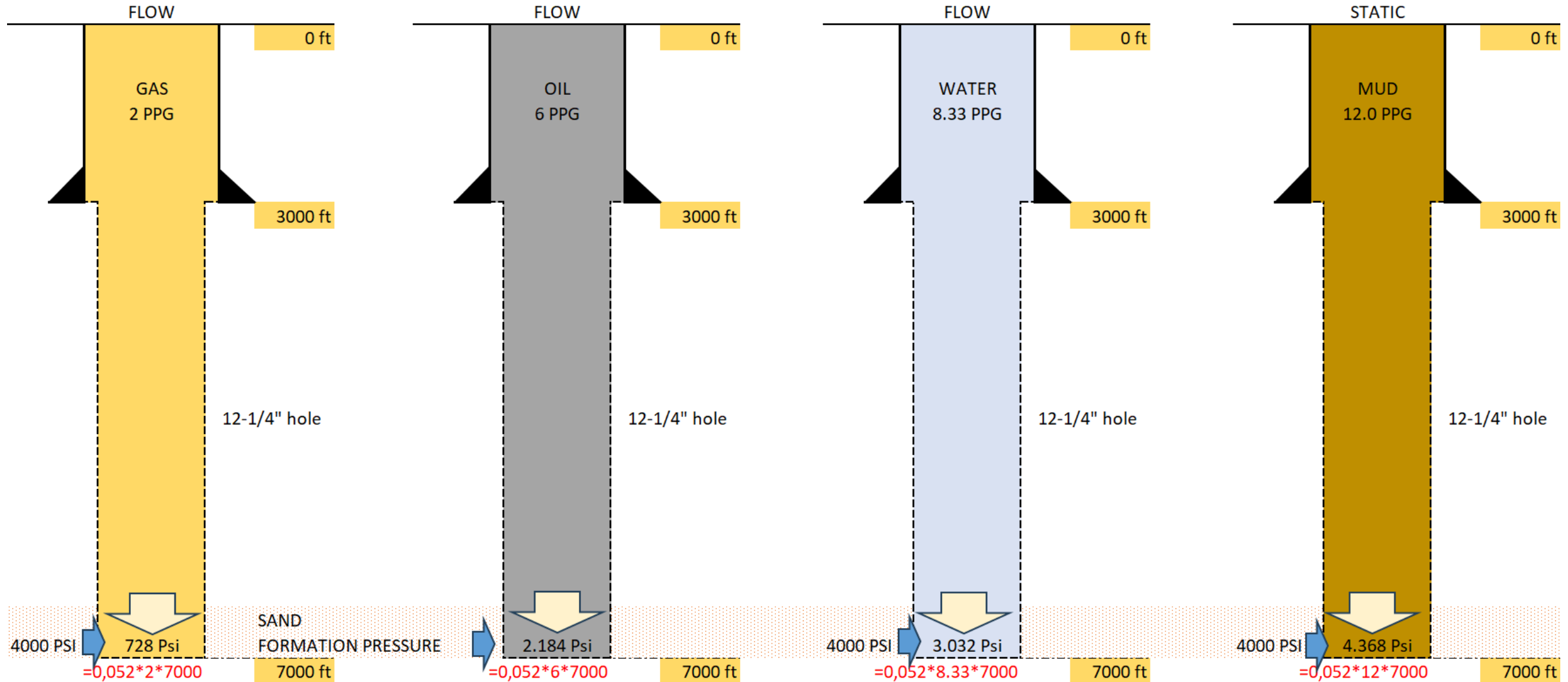


3. Hydrostatic Pressure Formula



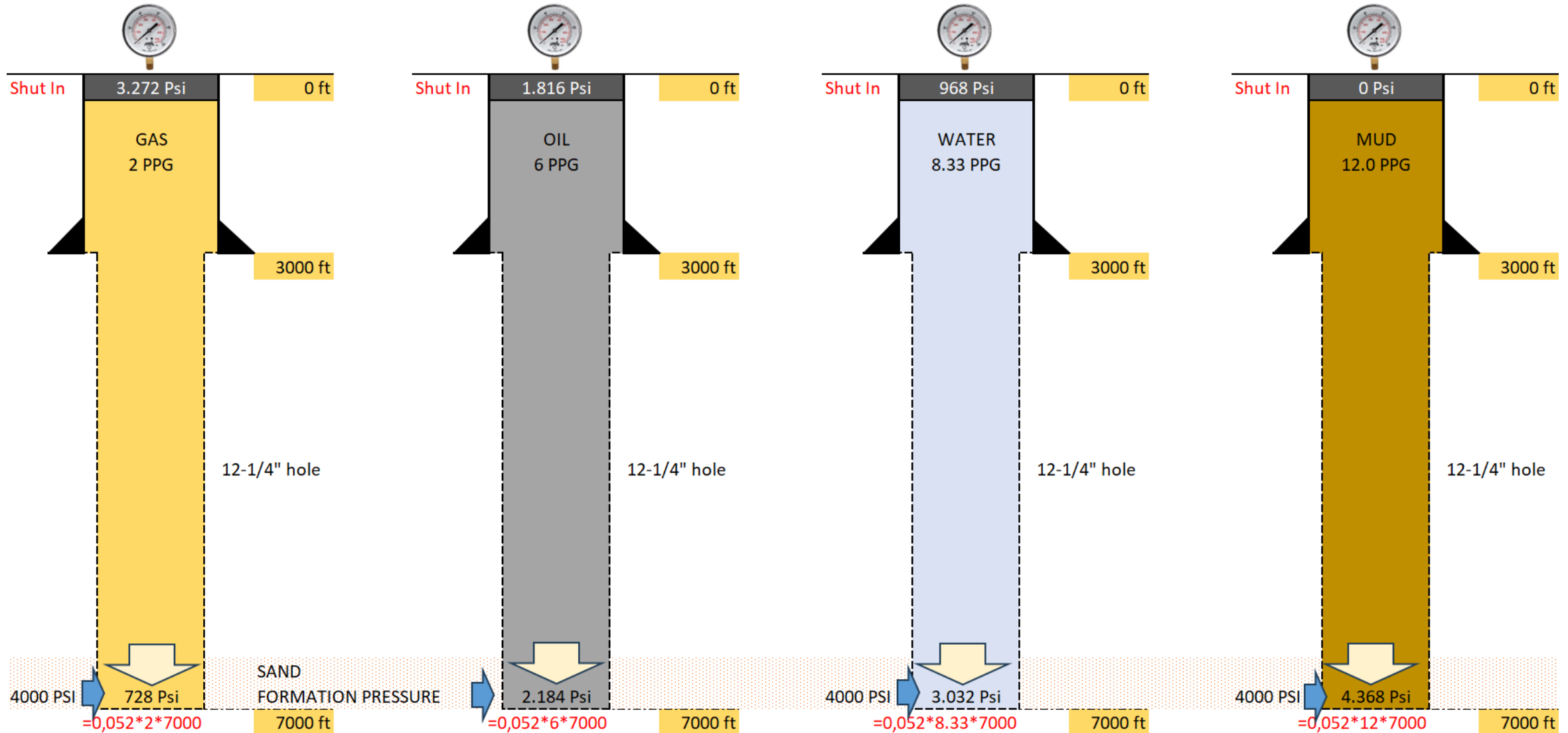
3. Hydrostatic Pressure Formula – KICK OR STATIC

$$\text{Pressure (Psi)} = 0.052 \times \text{MW (ppg)} \times \text{TVD (ft)}$$



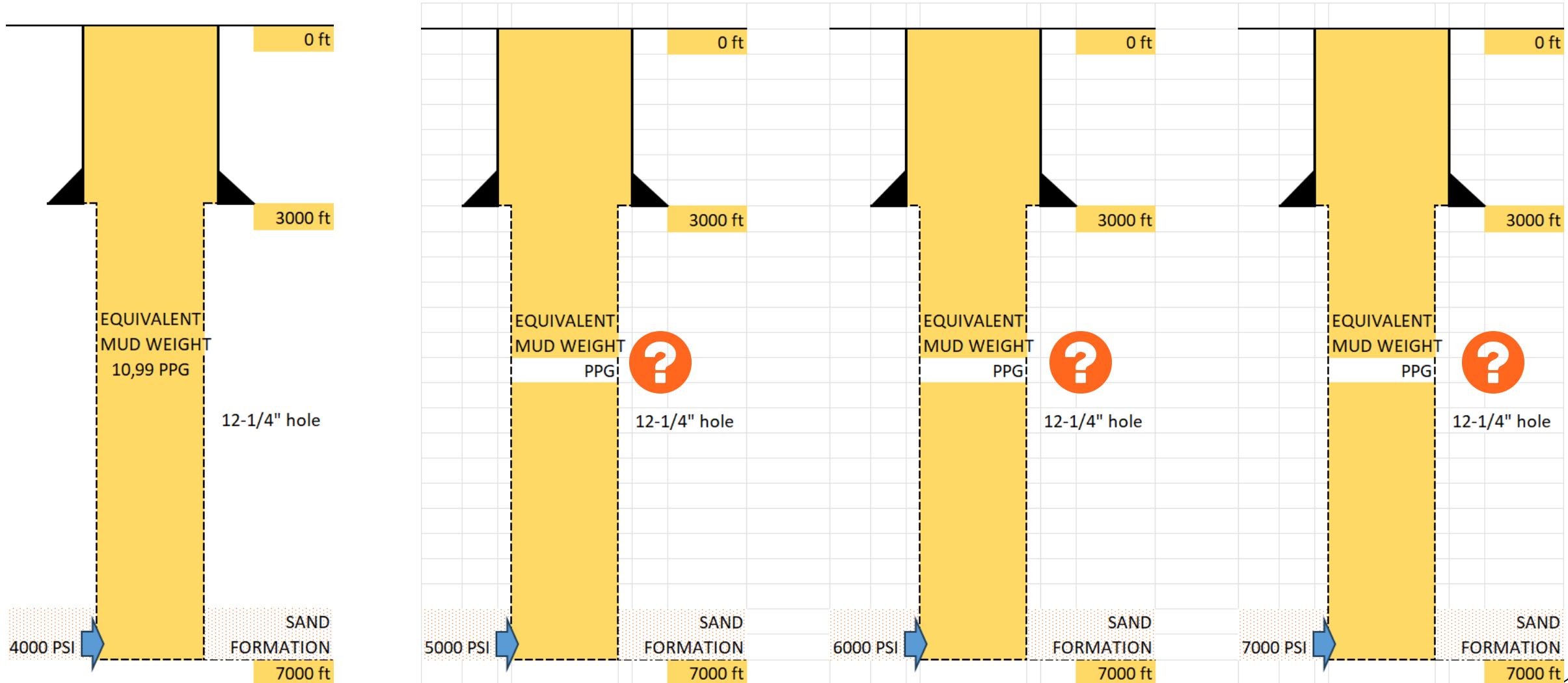
3. Hydrostatic Pressure Formula – SHUT IN PRESSURE

$$\text{Pressure (Psi)} = 0.052 \times \text{MW (ppg)} \times \text{TVD (ft)}$$

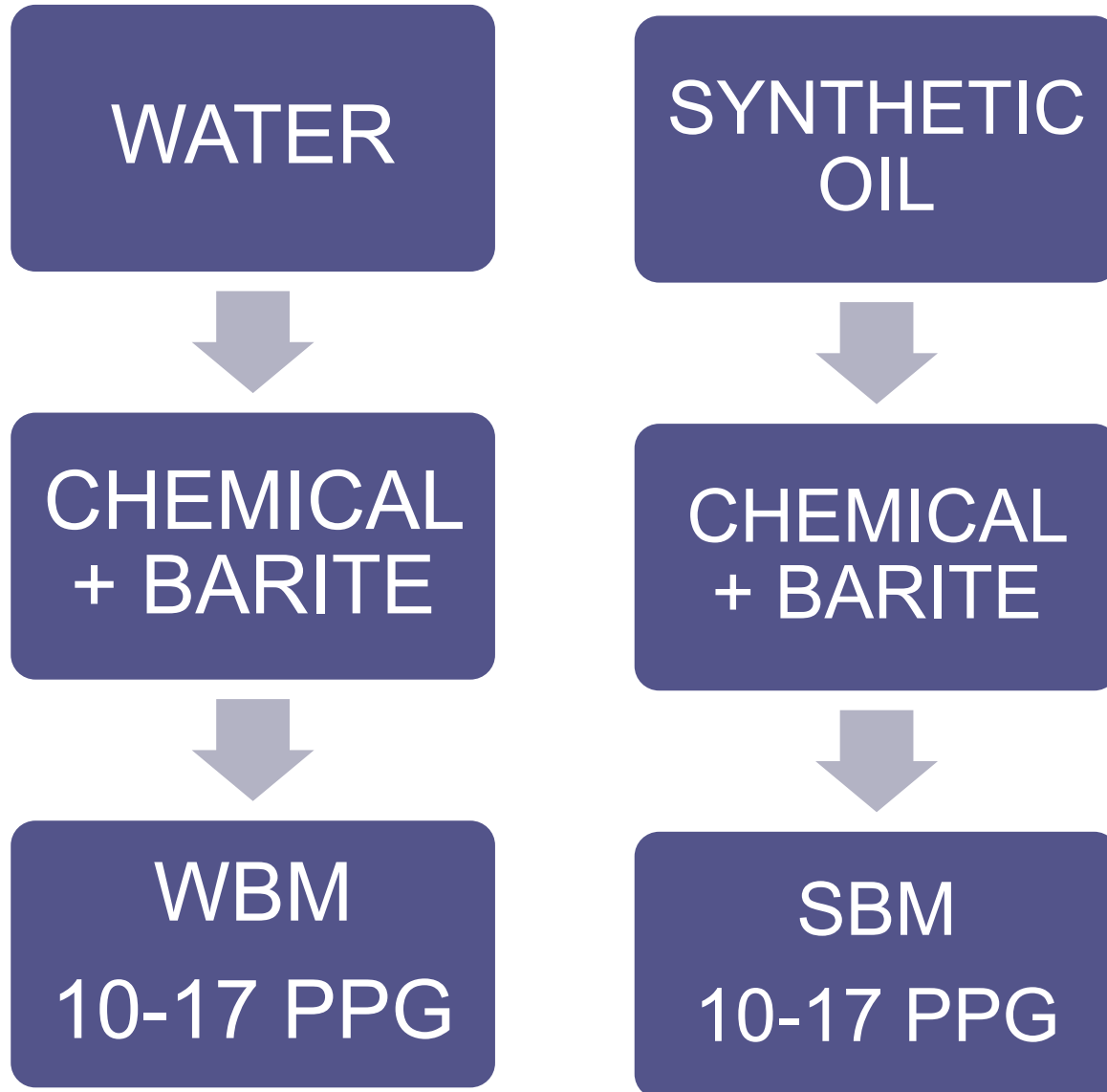


3. Hydrostatic Pressure Formula – EQUIVALENT MUD WEIGHT

$$\text{MW (ppg)} = \frac{\text{Pressure (Psi)}}{0.052 \times \text{TVD (ft)}} = \frac{4000 \text{ Psi}}{0.052 \times 7000} = \mathbf{10.99 \text{ PPG Equivalent}}$$



3. DRILLING FLUID, HOW TO MAKE HEAVY?



Mud & Cement Formulas

**Barite (100 lb sx) Per 100 bbl
Required for Weight-Up**

$$\text{Sacks per 100 bbl} = 1,470 \times \frac{\text{KWM}_{\text{ppg}} - \text{OMW}_{\text{ppg}}}{35 - \text{KWM}_{\text{ppg}}}$$

**Hematite (100 lb sx) Per 100 bbl
Required for Weight-Up**

$$\text{Sacks per 100 bbl} = 1,680 \times \frac{\text{KWM}_{\text{ppg}} - \text{OMW}_{\text{ppg}}}{40 - \text{KWM}_{\text{ppg}}}$$



KCL Mud during Drilling
1000 BBLS
9.5 PPG

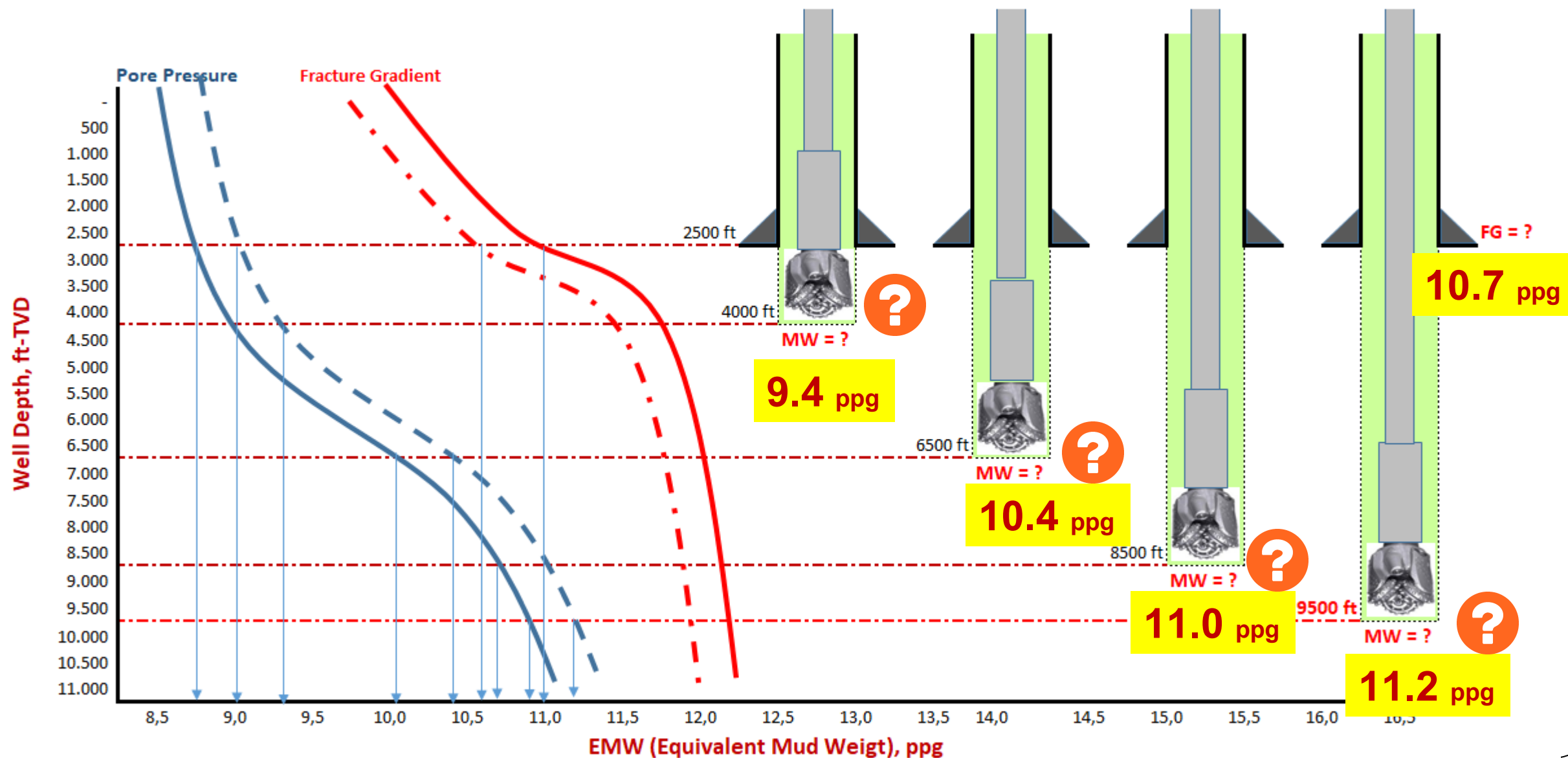


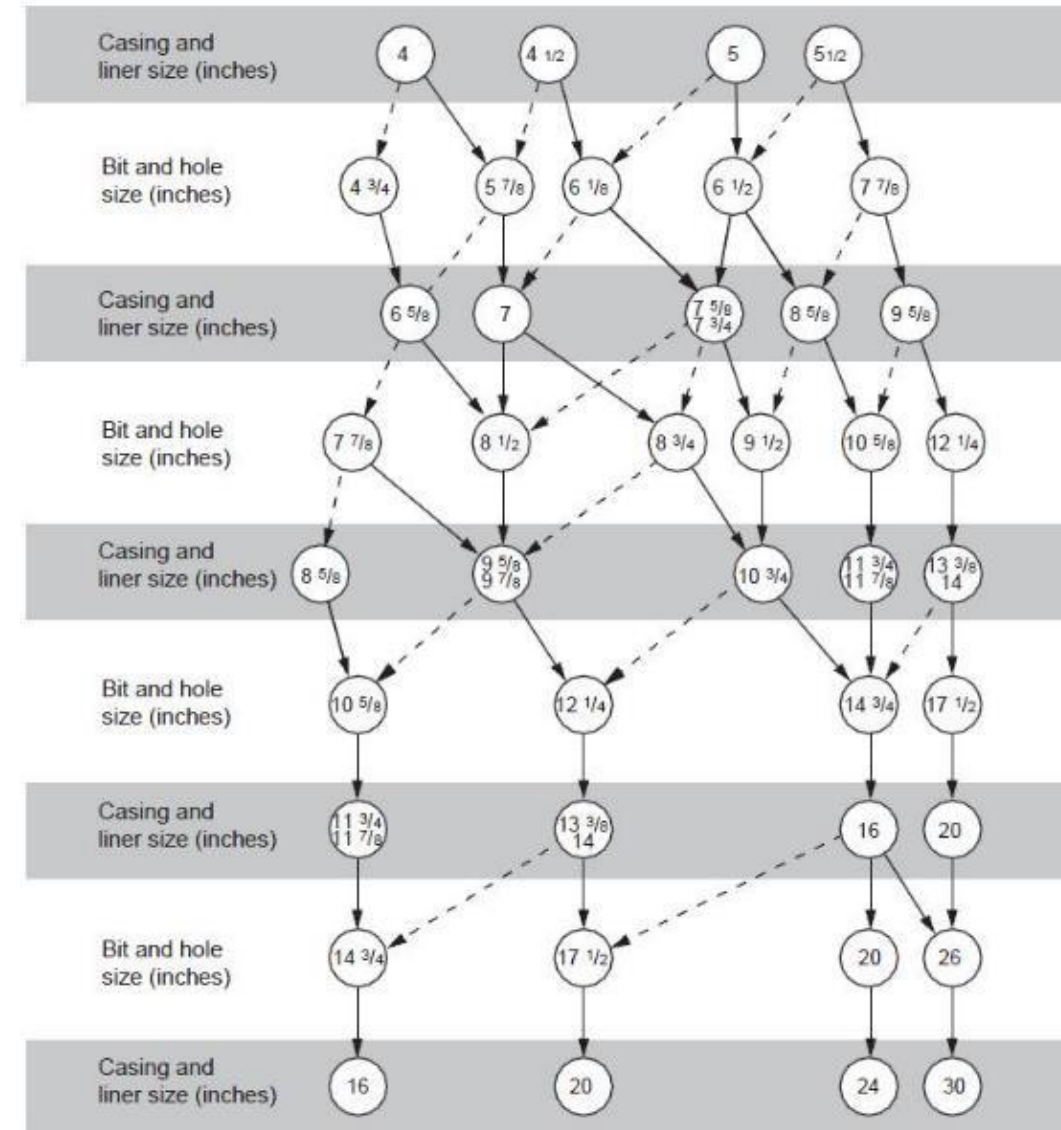
KCL Mud for Killing Well
1000 BBLS
11.5 PPG

- Sacks per 100 bbls = $1,470 \times (11.5 - 9.5) / (35 - 11.5) = 125$ sack
- Sacks for 1000 bbls = $1000 / 100 \times 125$ sack = 1250 sack.

3. Casing Depth Selection

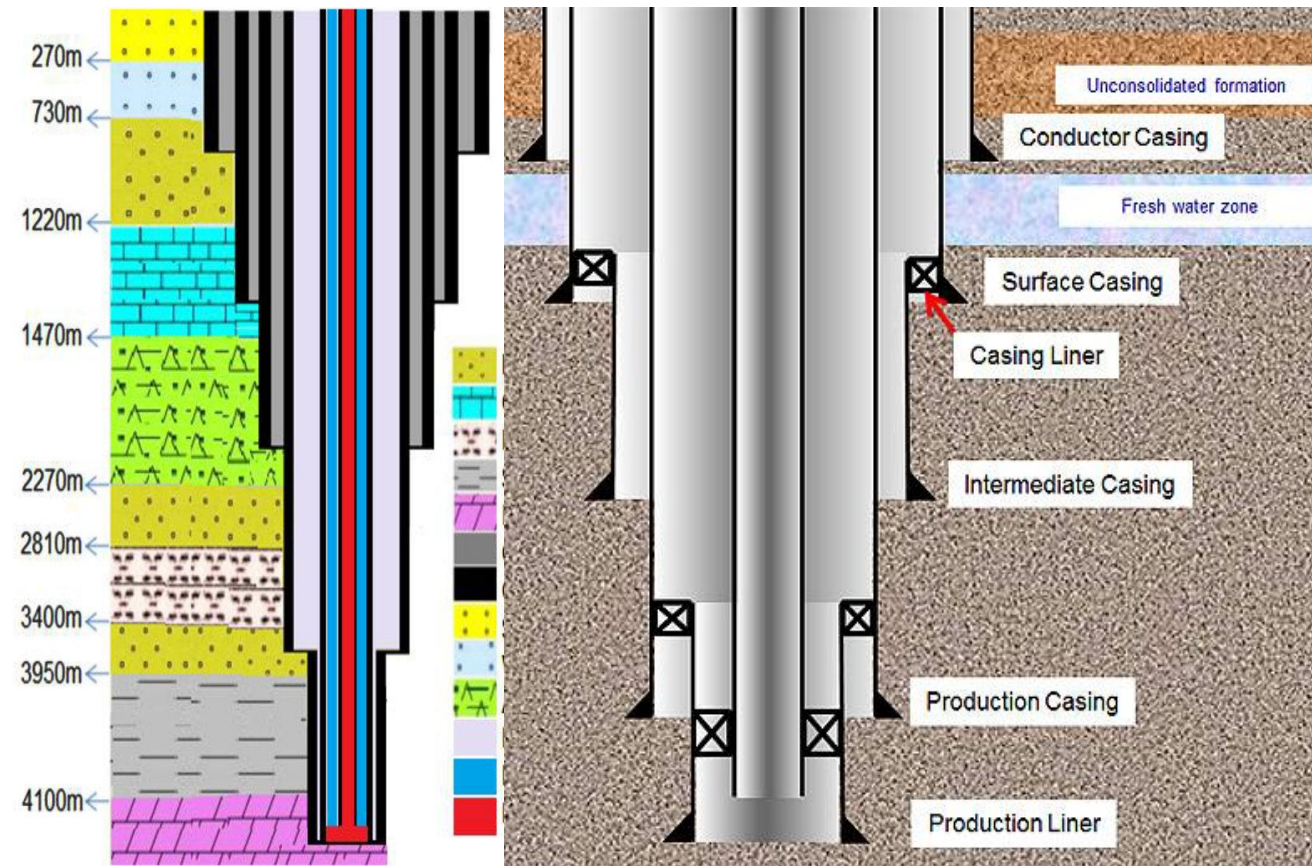
Pore Pressure, Fracture Pressure and Maximum Drilling Depth



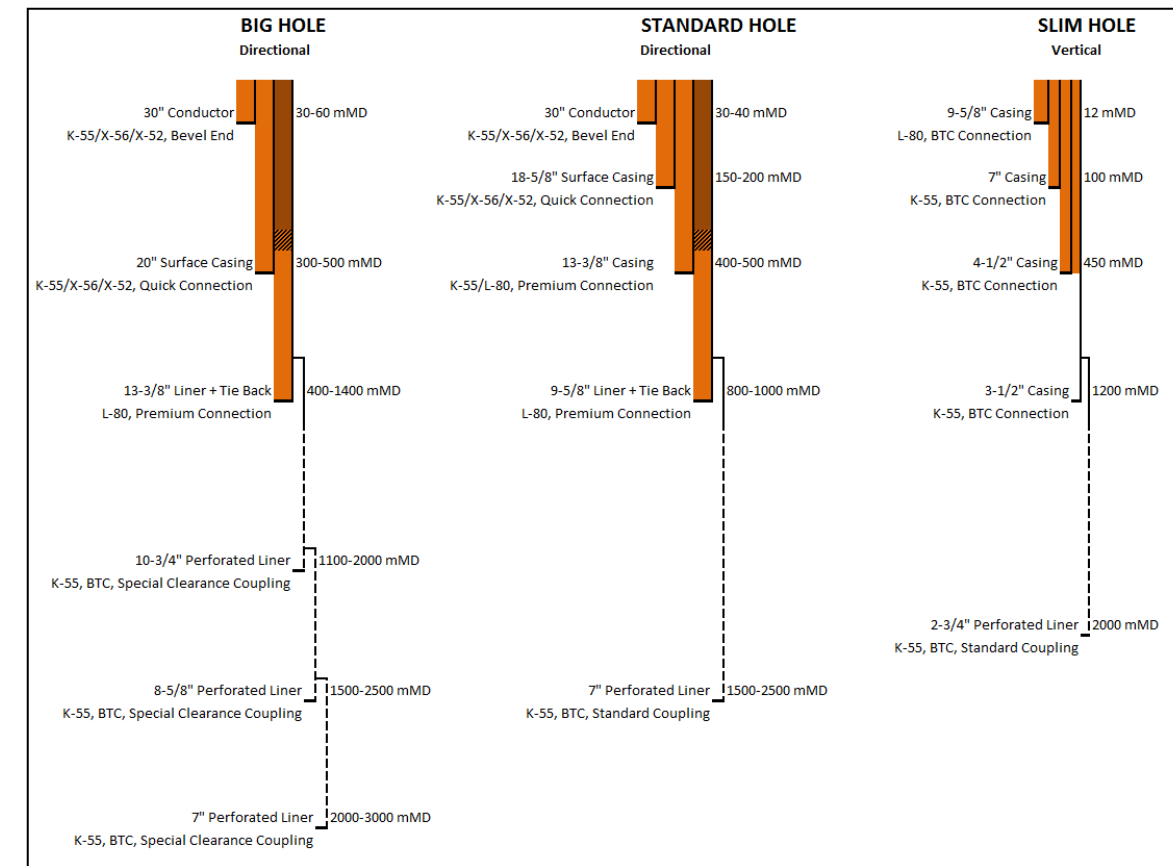


3. Casing Design Oil and Gas vs Geothermal Well

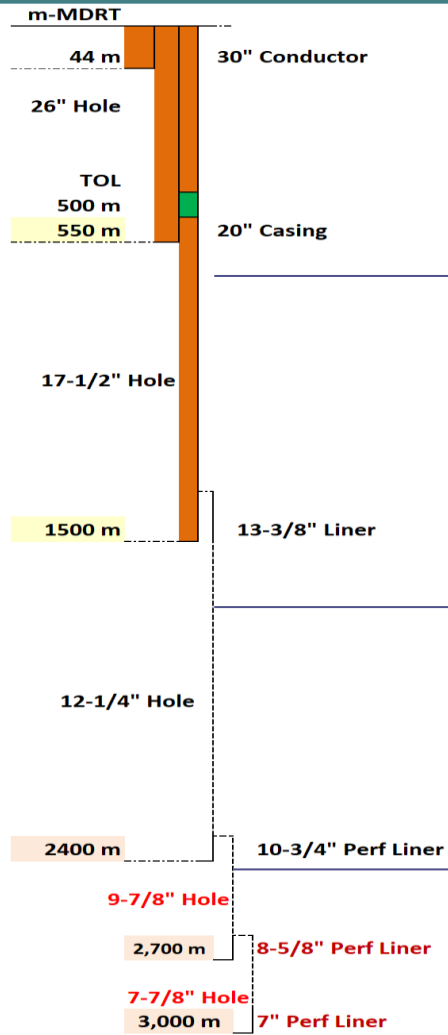
Oil and Gas



Geothermal



3. Casing Design For Geothermal



30" Conductor

- R-3, Plain end, Non-Seamless (Spiral / SAW / ERW)
- 0.5 – 0.75" Wall thickness.
- Bore pile and cemented

20" Casing

- 133 ppf, K-55 or X-56, R-3, Thread: 3 TPI or less (**Do not use BTC or equivalent**)
- Non-Seamless (SAW / ERW)

13-3/8" Casing Liner

- Seamless
- 68 ppf, L-80, R-3, **Premium Thread, TWCCP (?)**
- **Do NOT use K-55 casing**
- **Do NOT use BTC Thread**

13-3/8" Casing Tie Back

- Seamless
- 68 & 72 ppf, L-80, R-3, **Premium Thread, TWCCP (?)**
- **Do NOT use K-55 casing**
- **Do NOT use BTC Thread**

10-3/4" Perforated Liner

- Seamless
- 40 ppf, K-55 or L-80, R-3, **BTC SCC (Special Clearance Coupling)**

8-5/8" Perforated Liner

- Seamless
- 24 ppf, K-55, R-3, **BTC SCC (Special Clearance Coupling)**

7" Perforated Liner

- Seamless
- 23-26 ppf, K-55, R-3, **BTC SCC (Special Clearance Coupling)**

3. How to Purchase Casing ?

1. Unit Cost → **IDR per Joint** OR IDR per Meter
2. Delivery Point → Batam, Indonesia Water OR Wellsite
3. Mill or Green Pipe Sources → USA, Europe, Rusia, Japan, OR China
4. Delivery Term → DDU OR DDP
5. Payment Term → Down Payment ?, 30 days or 45 days
6. Late Delivery Penalty?
7. Handling cost at Batam? At Wellsite ?
8. Casing Storage cost at Batam?
9. Casing grouping ? Market Survey?
10. Casing Quantity Tolerance ? -3% + 0% OR -3% + 3%
11. Local Content ??

3. Well Construction – Cementing Design

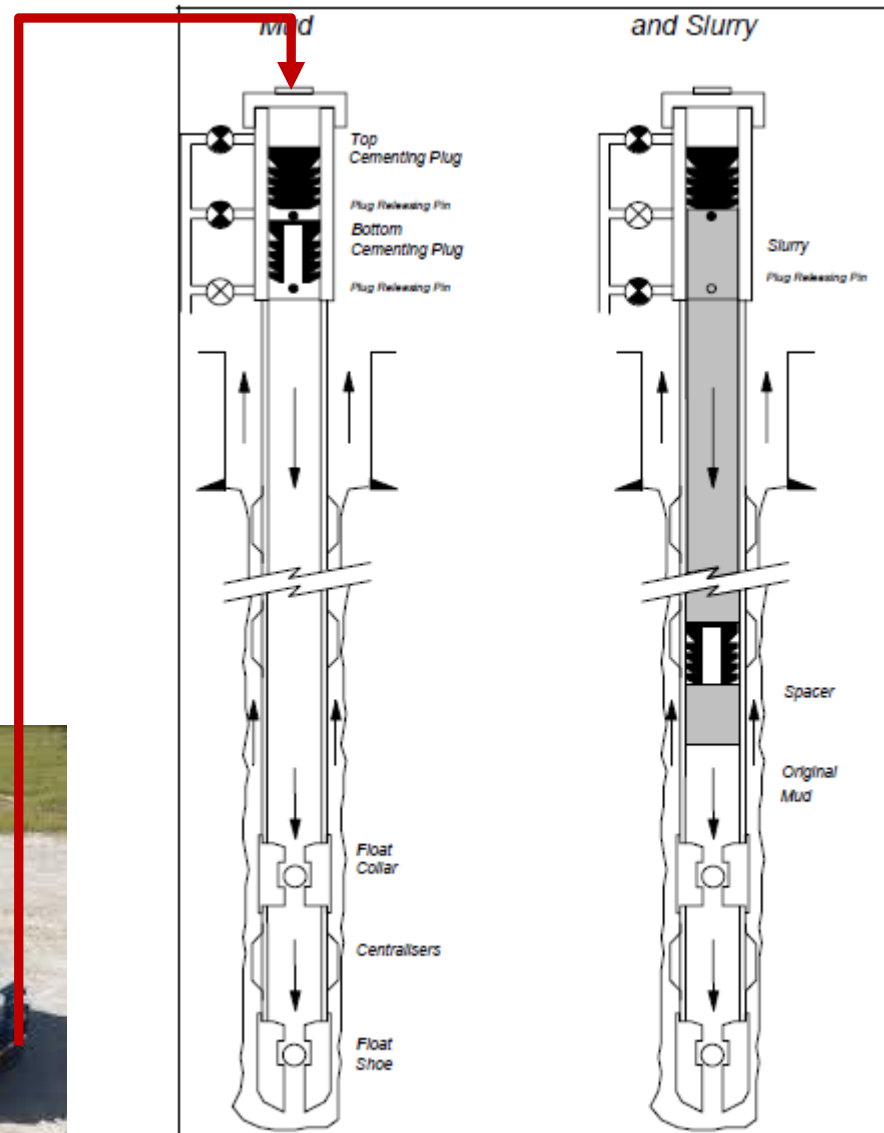


Figure 16-3- Typical One-Stage Primary Cement Job

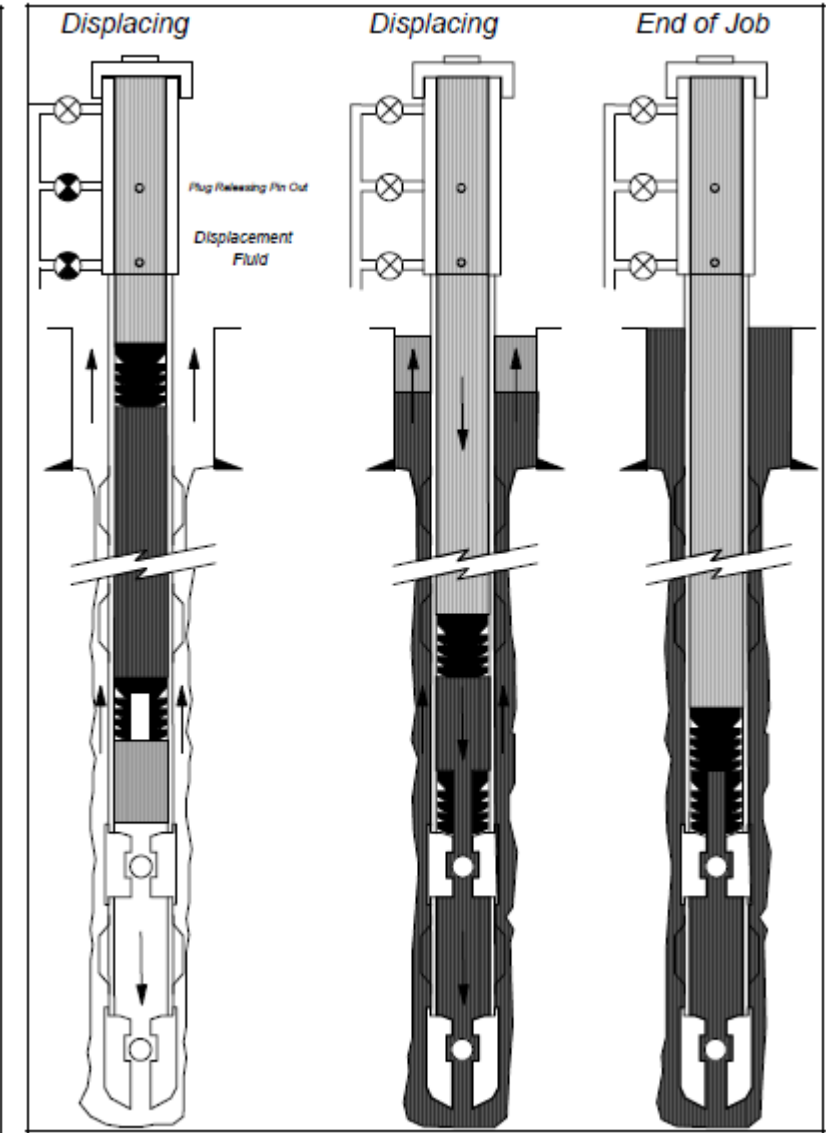
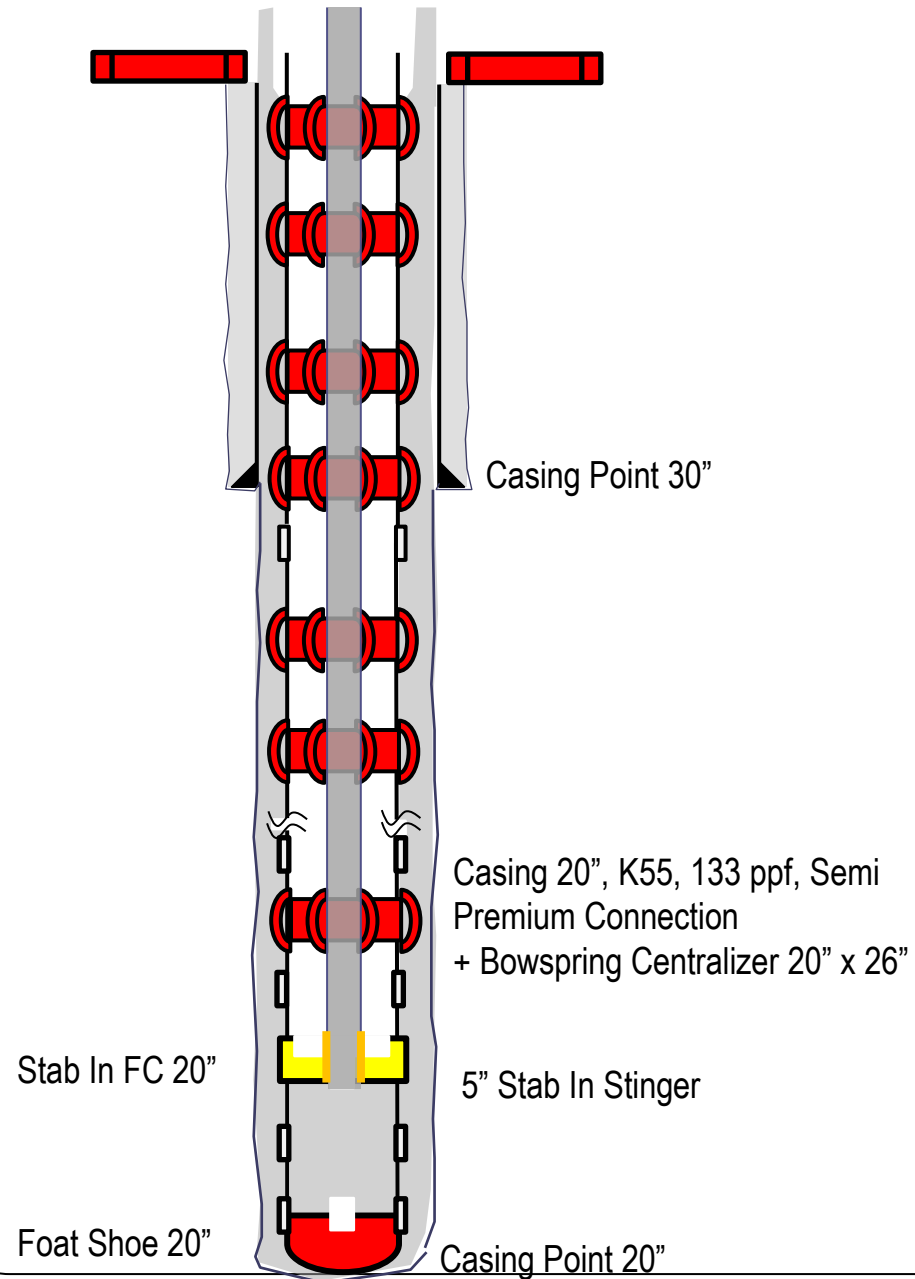


Figure 16-4- Typical One-Stage Cement Job (Cont)

26" Section – 20" Casing Cementing



PROCEDURES :

1. RIH Casing 20"
2. RIH Stinger
3. Pump spacer & lead-tail **slurry**
4. Displace string & POOH Stinger
5. WOC



CEMENTING ADDITIVES :

1. Class G Cement
2. Silica Flour
3. Retarders: Slow down the setting time of cement, Accelerators: Speed up cement hydration for faster strength development.
4. Plasticizers and Superplasticizers: Improve workability without increasing water content, enabling high-strength, dense concrete.
5. Dispersants: Reduce viscosity and improve fluid flow in cement slurries.
6. Fluid Loss Control Agents: Prevent excessive water loss into surrounding formations
7. Defomers and Anti-Gas Migration Additives: Control gas migration during cement setting, ensuring zonal isolation in wells.



3. Cementing Laboratory

HALLIBURTON

Indonesia, Jakarta

Lab Results- Lead

Job Information

Request/Sherry	2658046/1	Rig Name	PRA-15	Date	17/NOV/2020
Submitted By	Uda Hardyanti	Job Type	Surface Casing	Bulk Plant	Jakarta
Customer	SMGP	Location	Sorik Masapi	Well	SMP P-110

Well Information

Casing/Liner Size	20 in	Depth MD	1477 ft	BHST	60°C / 140°F
Hole Size	26 in	Depth TVD	1477 ft	BHCT	60°C / 140°F
Pressure	950 psi				

Cement Information - Lead Design

Conc	UOM	Cement/Additive	MP	Sample Type	Sample Date	Lot No.	Cement Properties	
100	% BWOC	Holcim Class G		Chemicals	22.07.20	CL11119-CM2020	Slurry Density	14.5 lbm/gal
14.01	gal/sack	Fresh Water		Chemicals	25.04.20	AEK ZUNIK	Slurry Yield	3.404 ft ³ /sack
0.01	ggs	D-Air 2	PH	Chemicals	06.04.20	BN : J160701-2	Water Requirement	14.007 gal/sack
2	ggs	Silicate Liquid	PH	Chemicals	04.08.20	SM072015	Total Mix Fluid	16.447 gal/sack
40	% BWOC	SSA-1 (Silica Fume) - PB	PB	Lab	11.05.20	J2004005	Water Source	Fresh Water
0.28	ggs	HALAD-344L	PH	Chemicals	06.04.20	BN : 0708		
0.1	ggs	SCR-100L NS	PH	Chemicals	06.04.20	BN : 0294		
100	% BWOC	Enhancer 923	PB	Lab	14.10.20	N/A		
0.05	ggs	CFR-3L	PH	Chemicals	06.04.20	BN : 0242		

Operation Test Results Request ID 2658046/1

Thickening Time, Request Test ID:37838341

Temp (degF)	Pressure (psi)	Start BC	70 Bc (hh:mm)
140	950	8.6	3:59

UCA Comp. Strength, Request Test ID:37838342

End Temp (degF)	Pressure (psi)	500 psi (hh:mm)	12 hr CS (psi)	24 hr CS (psi)
140	3000	8.33	1048	2745

API Rheology, Request Test ID:37838343

Temp (degF)	300	200	100	60	30	6	3
80 (up)	53	38	26	17	10	5	2
80 (down)	52	37	25	16	8	4	2
80 (avg.)	53	38	26	17	9	5	2
PV (cP) & YP (lbs/100ft ²):	49.46	4.83	(Least-squares method)				
PV (cP) & YP (lbs/100ft ²):	40.5	12	(Traditional method (300 & 100 rpm based))				
Generalized Herschel-Bulkley 4: $YP(lbs/100ft^2)=0.62$ $Muln(cP)=41.04$ $m=1$ $n=0.73$							

API Rheology, Request Test ID:37838344

Temp (degF)	300	200	100	60	30	6	3	Cond Time (min)
140 (up)	44	30	19	12	8	4	2	30
140 (down)	44	30	19	12	8	4	2	30
140 (avg.)	44	30	19	12	8	4	2	30
PV (cP) & YP (lbs/100ft ²):	40.83	3.58	(Least-squares method)					
PV (cP) & YP (lbs/100ft ²):	37.5	6.5	(Traditional method (300 & 100 rpm based))					
Generalized Herschel-Bulkley 4: YP(lbs/100ft ²)=1.29 Muln(cP)=31.4 m=0.52 n=0.52								

API Fluid Loss, Request Test ID:37838345

Test Temp (degF)	Test Pressure (psi)	Test Time (min)	Calculated FL (30 min)	Conditioning time (min)
140	1000	30	90	30

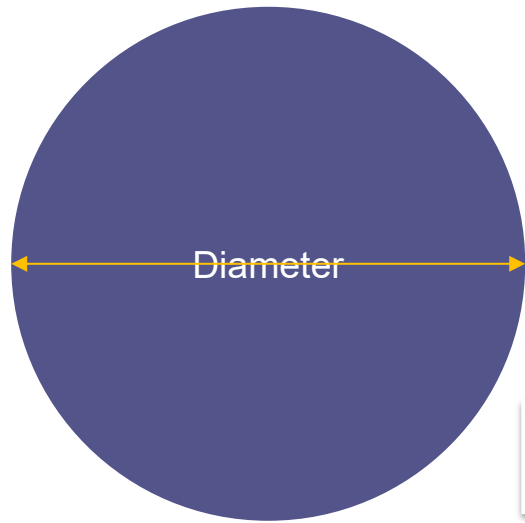
Free Fluid API 10B-2, Request Test ID:37838346

Con. Temp (degF)	Cond. Time (min)	Static time (min)	% Fluid
140	30	120	0

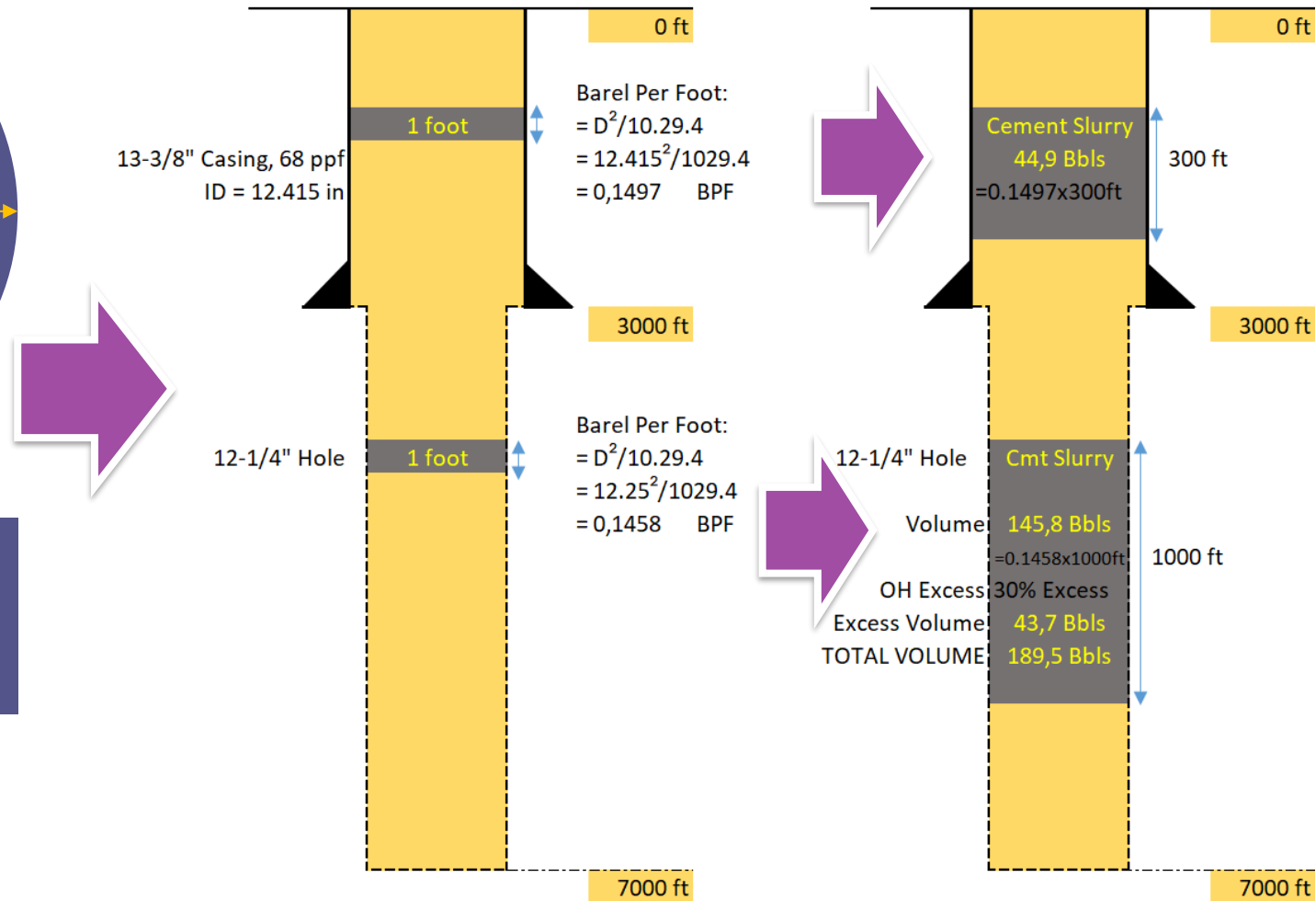
3. Casing Technical Data Sheet

O.D. (inch)	Nominal Weight T & C lbs/ft	Grade	Collapse Pressure (psi)	Internal Yield Pressure Minimum Yield (psi)				Joint Strength 1000 lbs			Body Yield 1000 lbs	Wall (inch)	I.D. (inch)	Drift Diameter (inch)	Displacement (bbl/ft)	Capacity (bbl/ft)
				PE	STC	LTC	BTC	STC	LTC	BTC						
20	94	H-40	520	1530	1530		1530	581		1041	1077	0.438	19.124	18.936	0.03329	0.35528
20	94	J-55	520	2110	2110	2110	2110	783	907	1402	1480	0.438	19.124	18.936	0.03329	0.35528
20	94	K-55	520	2110	2110	2110	2110	824	955	1479	1480	0.438	19.124	18.936	0.03329	0.35528
20	106.5	J-55	770	2410	2410	2410	2410	913	1056	1595	1685	0.5	19	18.812	0.03789	0.35069
20	106.5	K-55	770	2410	2410	2410	2410	960	1113	1683	1685	0.5	19	18.812	0.03789	0.35069
20	106.5	N-80	770	3500	3500	3500	3500	1307	1514	2281	2450	0.5	19	18.812	0.03789	0.35069
20	133	K-55	1500	3060	3060	3060	3060	1253	1453	2123	2125	0.635	18.73	18.542	0.04778	0.34079
20	133	L-80	1600	4450	4450	4450	4450	1692	1958	2849	3091	0.635	18.73	18.542	0.04778	0.34079
20	133	N-80	1600	4450	4450	4450	4450	1707	1976	2877	3091	0.635	18.73	18.542	0.04778	0.34079
20	169	K-55	2500	3910	3230	3430	3380	1402	1732	2689	2692	0.812	18.376	18.188	0.06054	0.32803
20	169	L-80	3020	5680	4690	4990	4920	2202	2549	3610	3916	0.812	18.376	18.188	0.06054	0.32803
20	169	N-80	3020	5680	4690	4990	4920	2221	2573	3645	3916	0.812	18.376	18.188	0.06054	0.32803
13 3/8	68	HCK-55	2850	3450	3450		3450	905		1496	1069	0.48	12.415	12.259	0.02405	0.14973
13 3/8	68	LS-65	2110	4080	4080		4080	791		1318	1264	0.48	12.415	12.259	0.02405	0.14973
13 3/8	68	L-80	2260	5020	5020		5020	952		1545	1556	0.48	12.415	12.259	0.02405	0.14973
13 3/8	68	HCL-80	2910	5020	5020		5020	1093		1732	1556	0.48	12.415	12.259	0.02405	0.14973
13 3/8	68	N-80	2260	5020	5020		5020	963		1585	1556	0.48	12.415	12.259	0.02405	0.14973
9 5/8	47	L-80	4760	6870		6870	6870		893	1122	1086	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	HCL-80	7100	6870		6870	6870		1027	1234	1086	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	N-80	4760	6870		6870	6870		905	1161	1086	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	HCN-80	7100	6870		6870	6870		1027	1234	1086	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	C-90	5000	7720		7720	7720		987	1210	1221	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	S-95	7100	8150		8150	8150		1053	1311	1289	0.472	8.681	8.525	0.01679	0.07321
9 5/8	47	T-95	5090	8150		8150	8150		1040	1273	1289	0.472	8.681	8.525	0.01679	0.07321

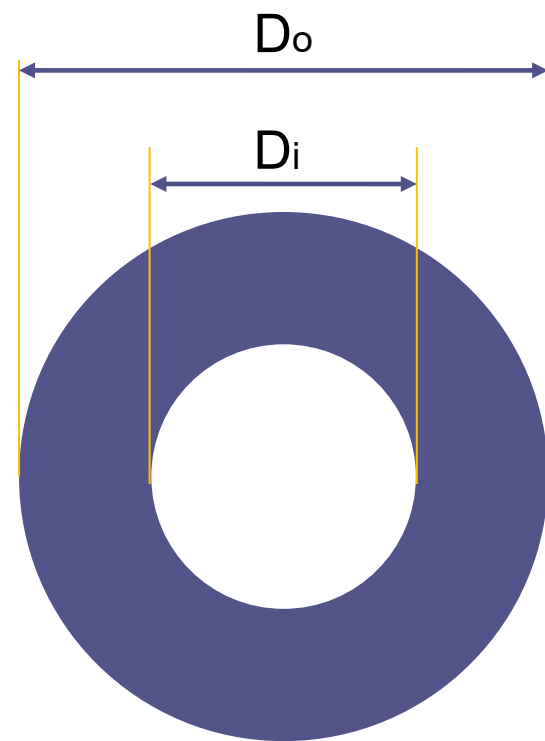
3. Cementing Volume Calculation



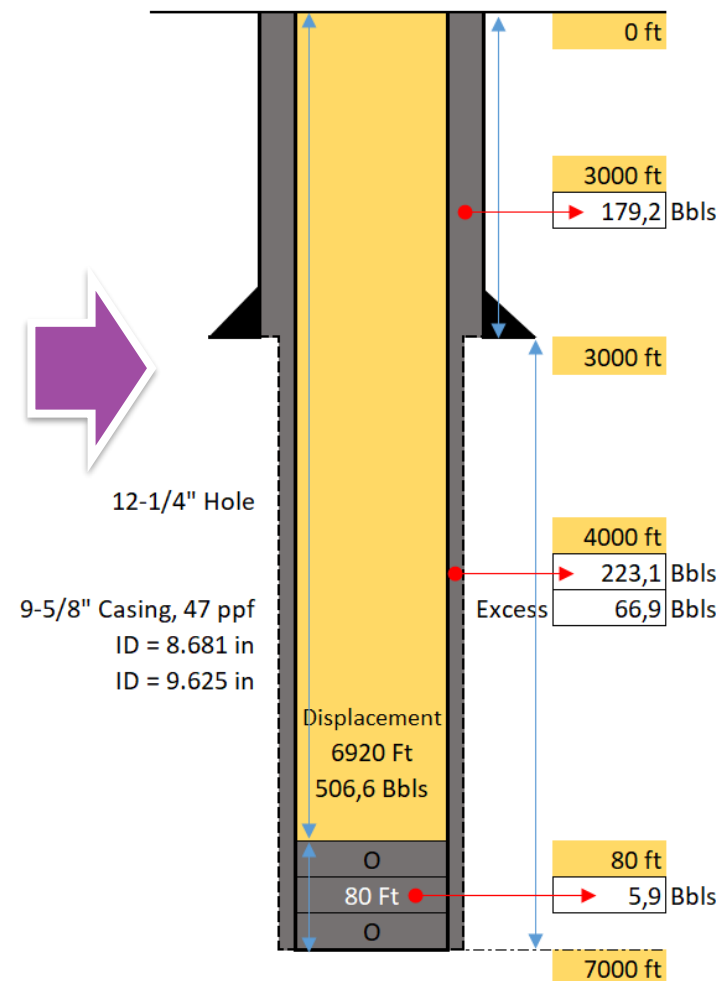
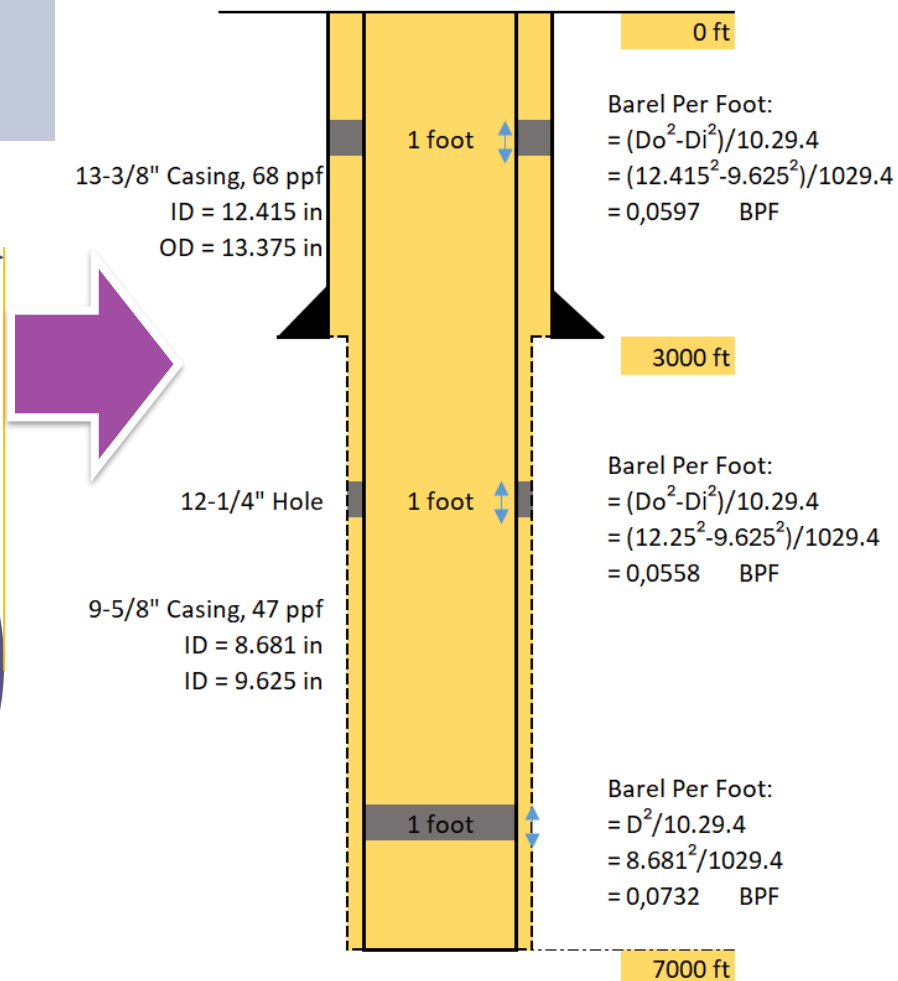
$$\text{BBF (Bbls per Foot)} = \frac{\text{Diameter}^2}{1029.4}$$



3. Cementing Volume Calculation

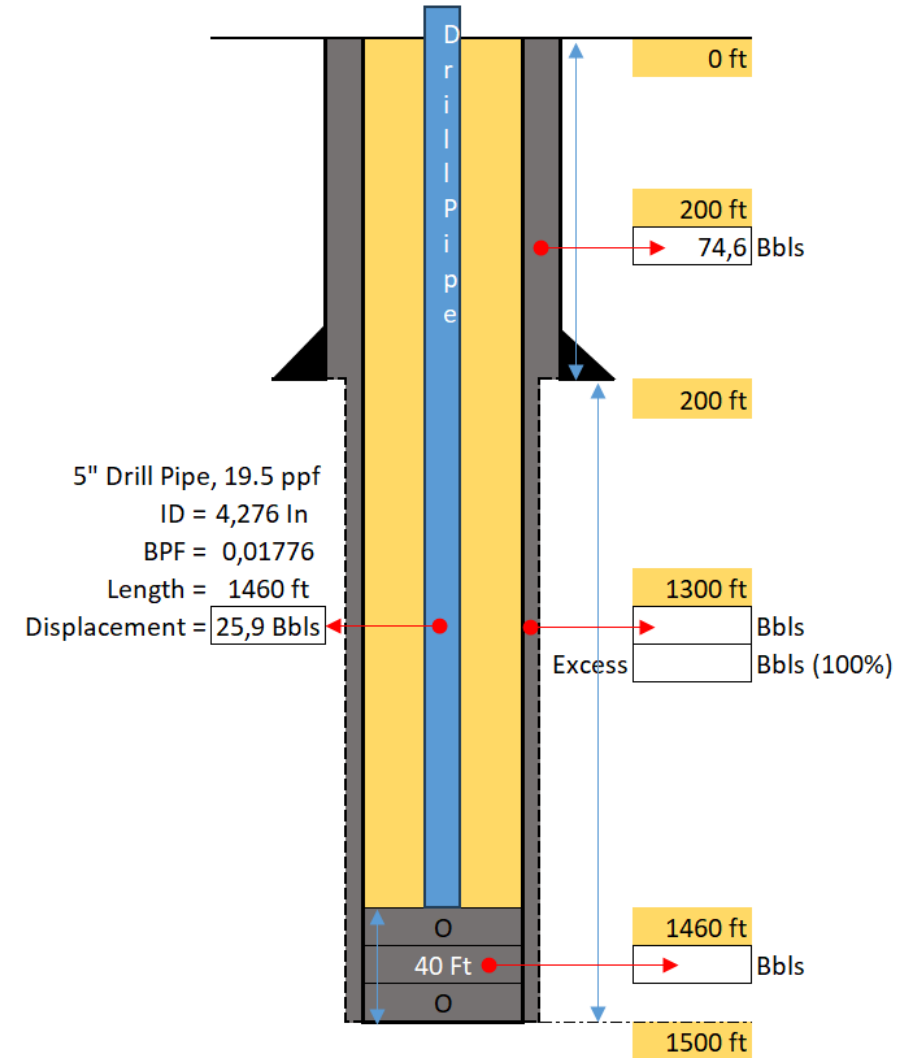
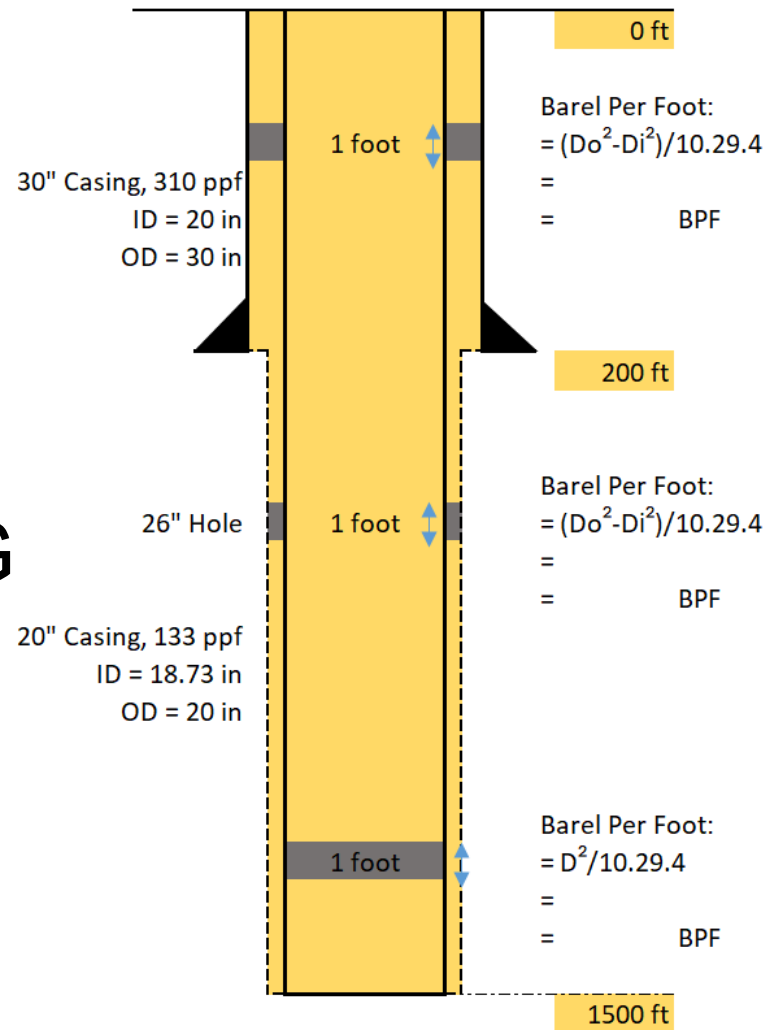


BBF (Bbls per Foot)
$$= \frac{D_o^2 - D_i^2}{1029.4}$$



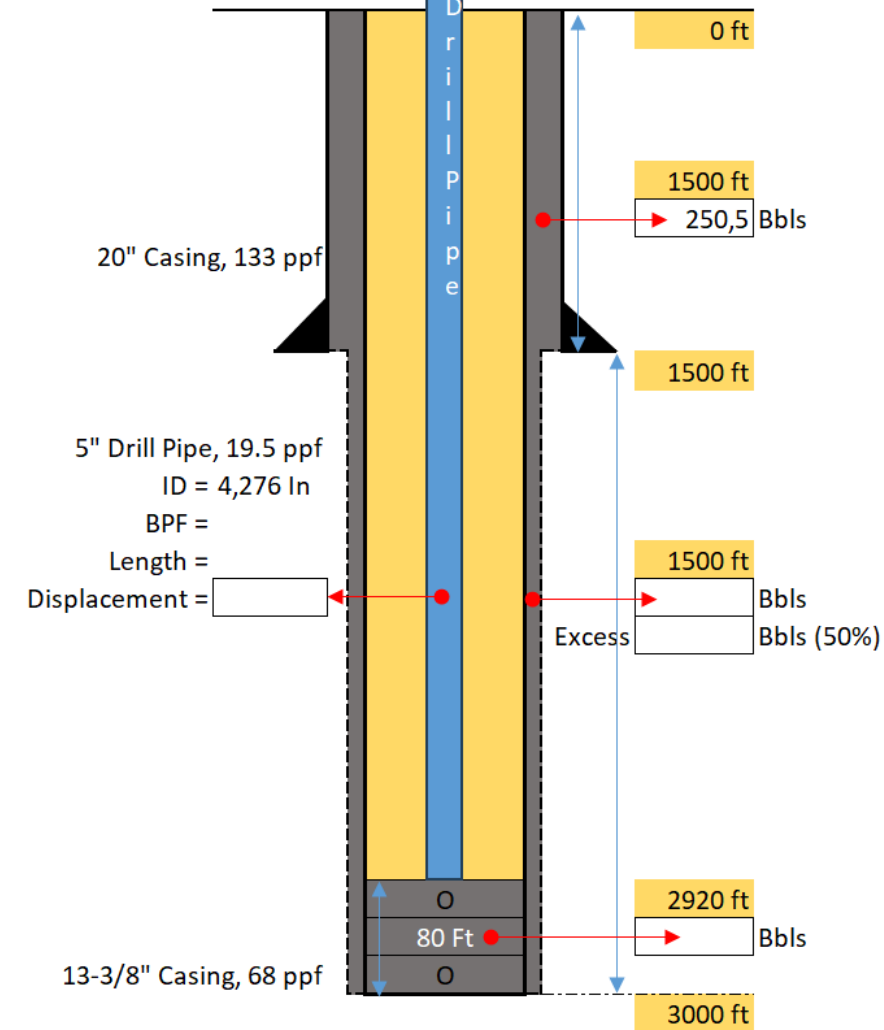
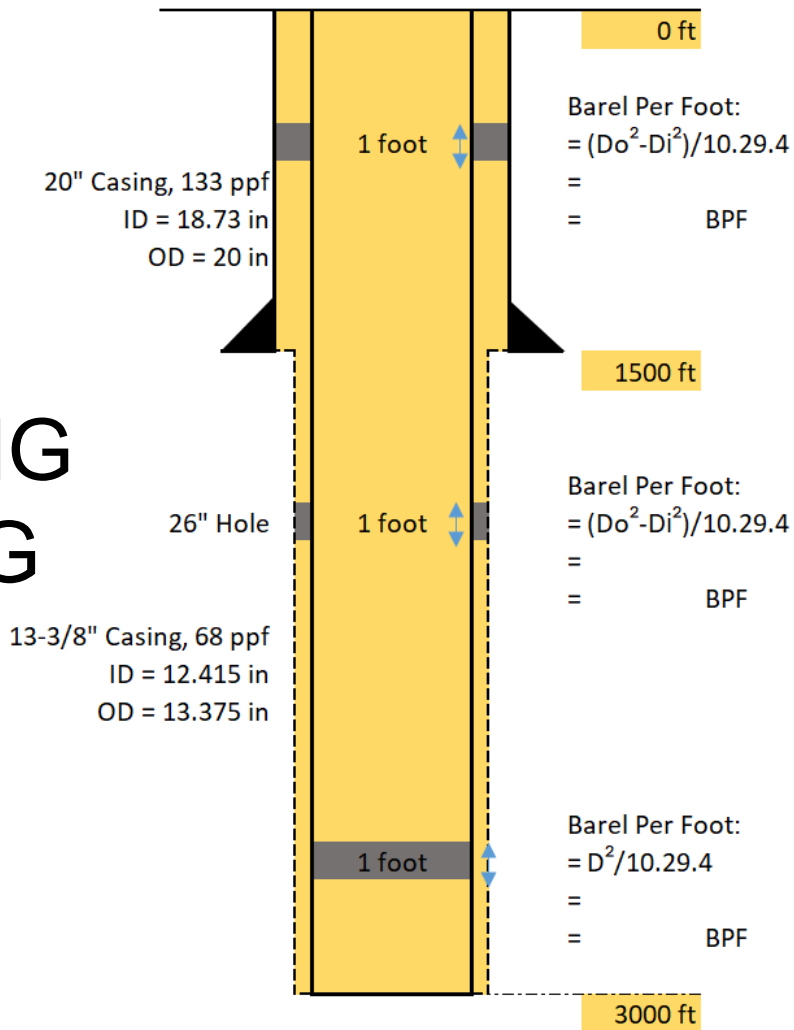
TOTAL CEMENT SLURRY	475,1 Bbls
CEMENT PUMPING RATE	4,0 BPM (Bbls / Minute)
CEMENT PUMPING TIME	118,8 Minutes
TOTAL DISPLACEMENT VOLUME	506,6 Bbls
DISPLACEMENT PUMPING RATE	4,0 BPM (Bbls / Minute)
DISPLACEMENT TIME	126,6 Minutes
SAFETY FACTOR	90,0 Minutes
THICKENING TIME REQ	335,4 Minutes
THICKENING TIME REQ	5,0 Hours
	35,4 Minutes

20" CASING CEMENTING VOLUME



TOTAL CEMENT SLURRY	785,3 Bbls
CEMENT PUMPING RATE	4,0 BPM (Bbls / Minite)
CEMENT PUMPING TIME	196,3 Minutes
TOTAL DISPLACEMENT VOLUME	25,9 Bbls
DISPLACEMENT PUMPING RATE	4,0 BPM (Bbls / Minite)
DISPLACEMENT TIME	6,5 Minutes
SAFETY FACTOR	90,0 Minutes
THICKENING TIME REQ	Minutes
THICKENING TIME REQ	4,0 Hours
	52,8 Minutes

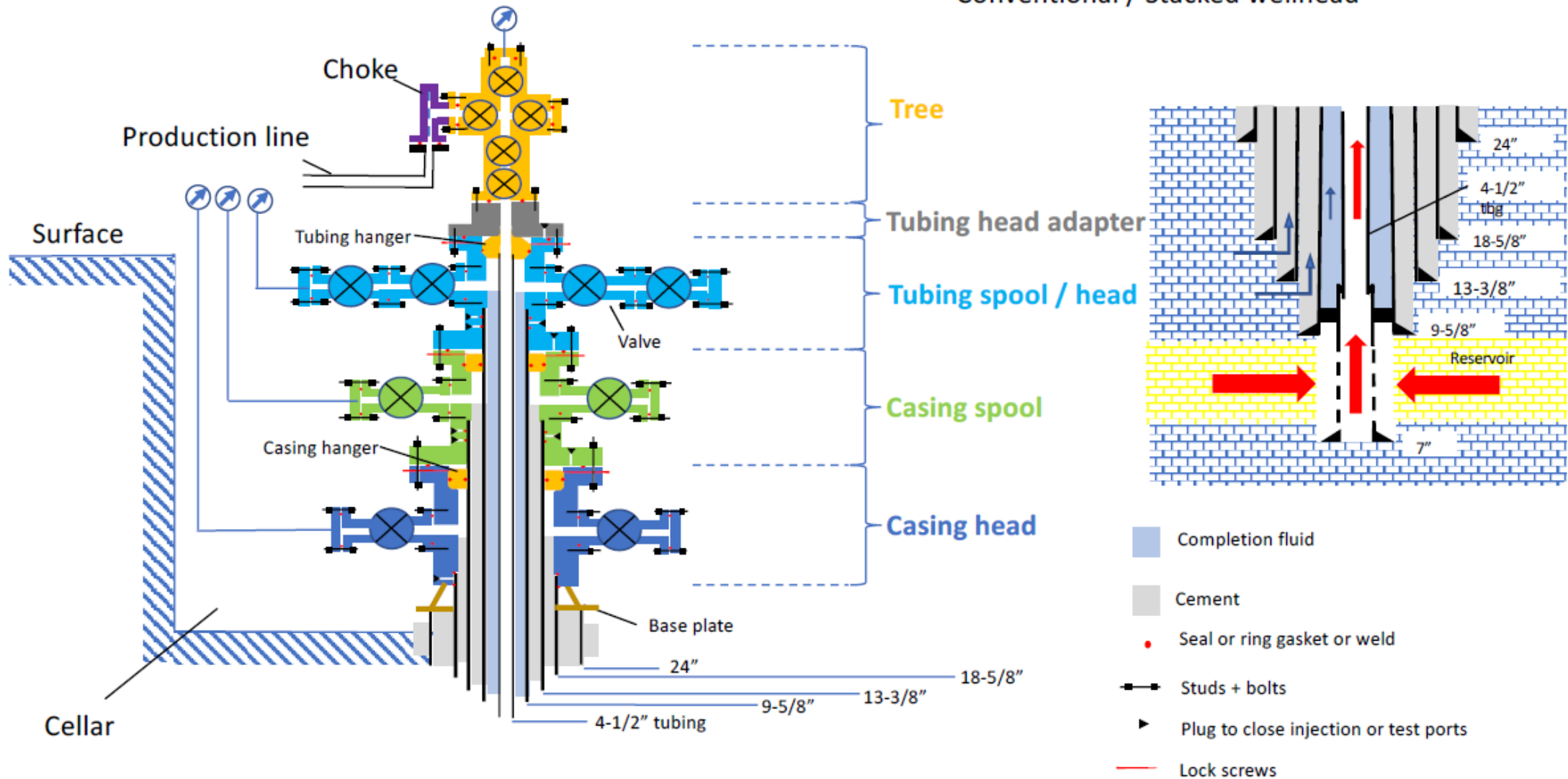
13³/₈" CASING CEMENTING VOLUME



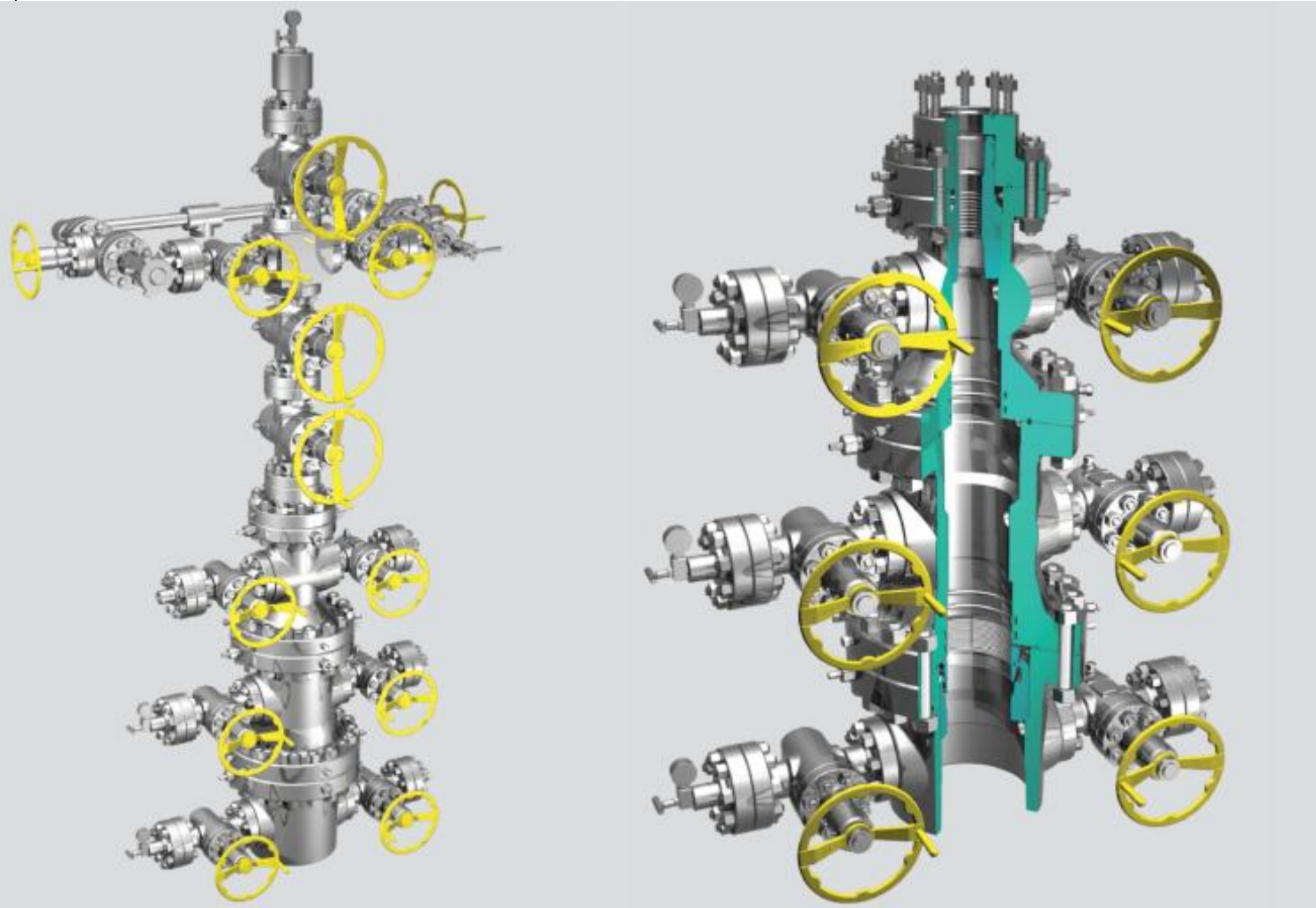
TOTAL CEMENT SLURRY		Bbls
CEMENT PUMPING RATE	4,0	BPM (Bbls / Minute)
CEMENT PUMPING TIME	216,4	Minutes
TOTAL DISPLACEMENT VOLUME		Bbls
DISPLACEMENT PUMPING RATE	4,0	BPM (Bbls / Minute)
DISPLACEMENT TIME	13,0	Minutes
SAFETY FACTOR	90,0	Minutes
THICKENING TIME REQ	319,4	Minutes
THICKENING TIME REQ	5,0	Hours
		19,4 Minutes

4. Drilling Equipment - Wellhead

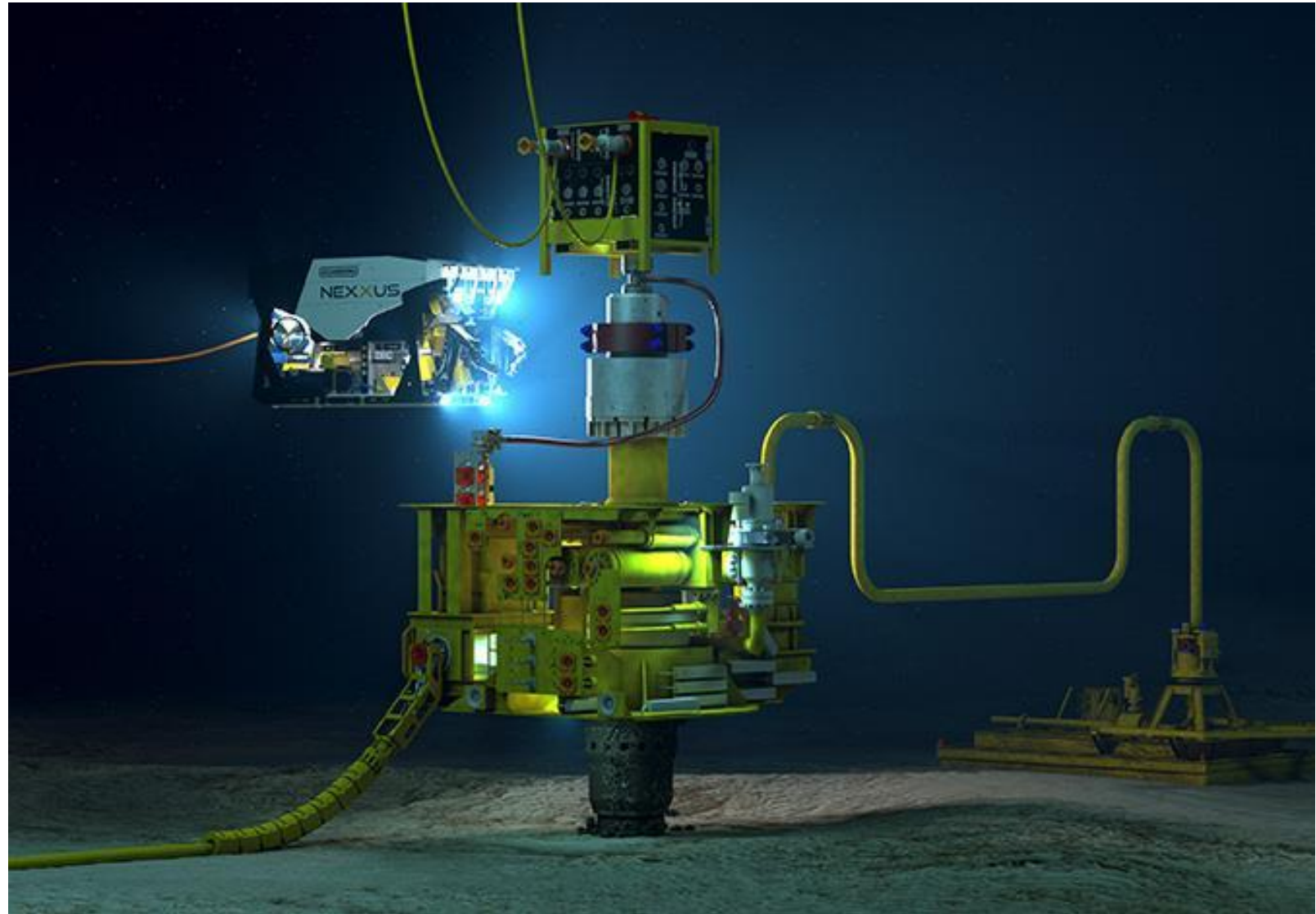
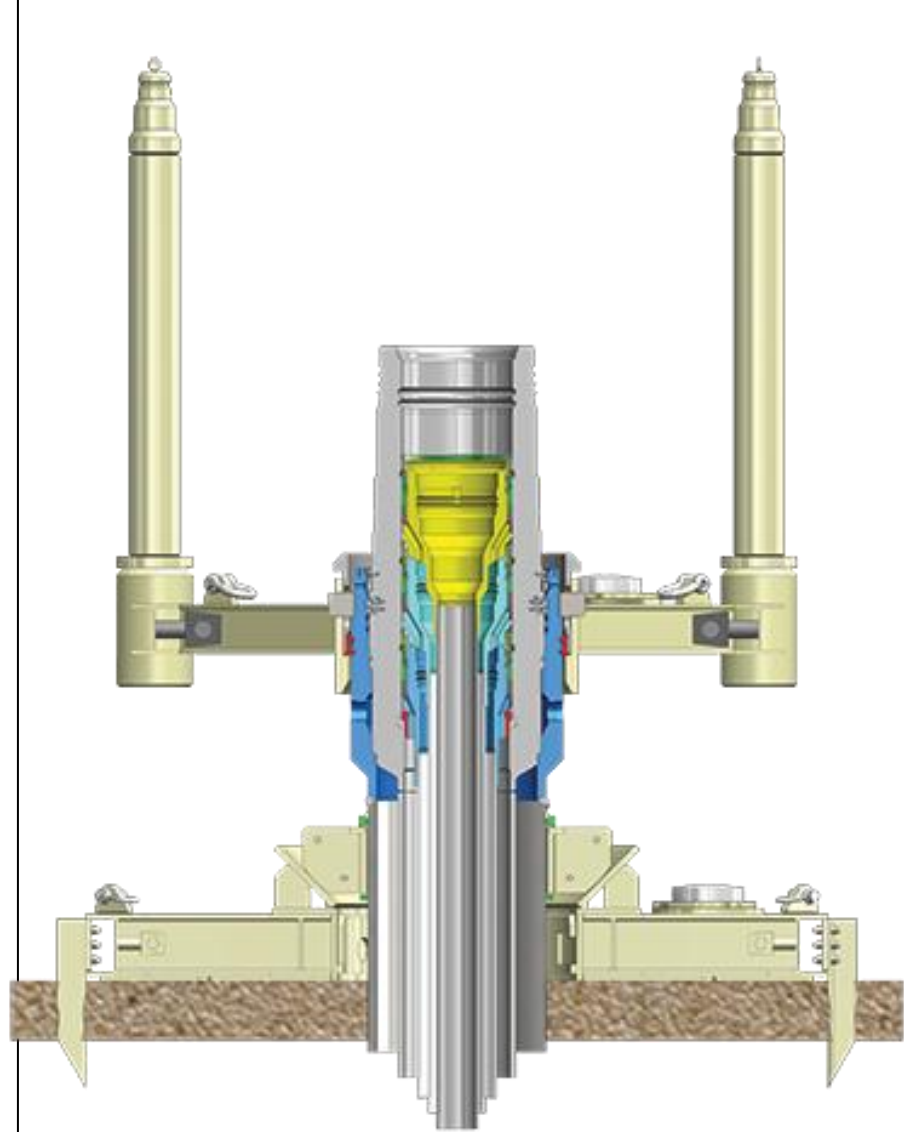
Conventional / Stacked wellhead



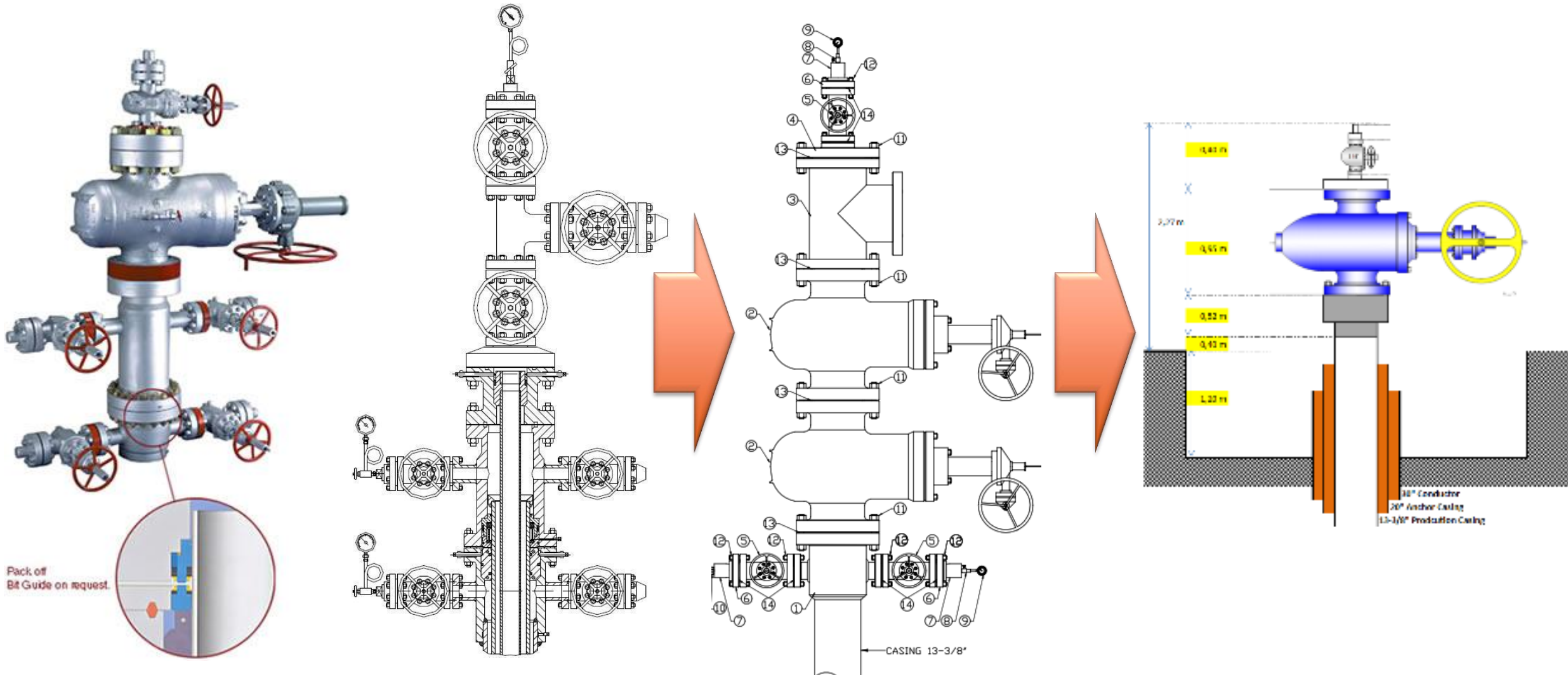
4. Drilling Equipment – Wellhead for Onshore Oil and Gas



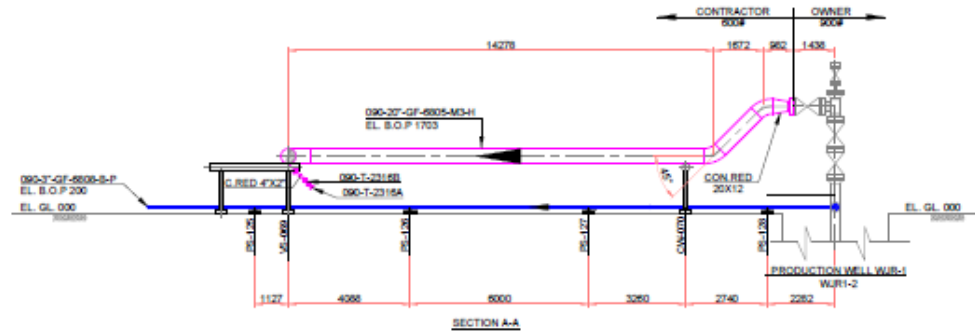
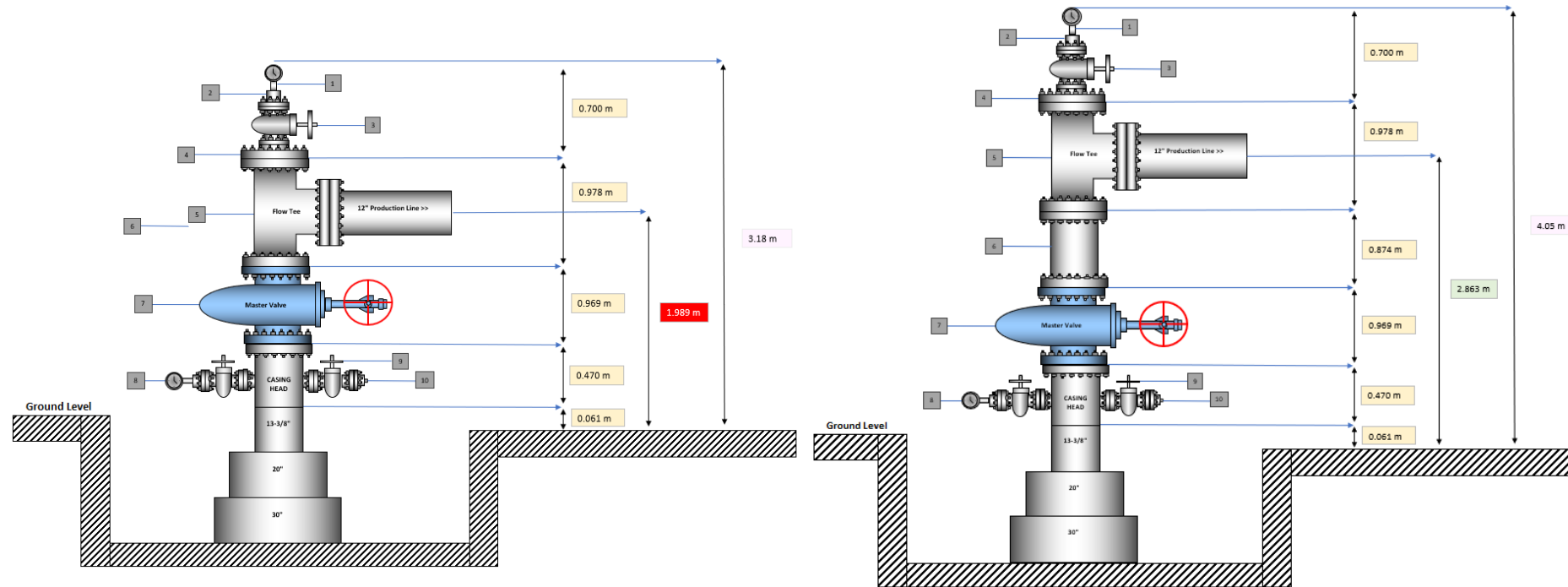
4. Drilling Equipment – Wellhead for Offshore Oil and Gas



4. Drilling Equipment – Wellhead for Geothermal



4. Wellhead and Master Valve at WJR1-2



We are fine tuning the design of wellhead configuration for WJR1-3 and WJR1-4:

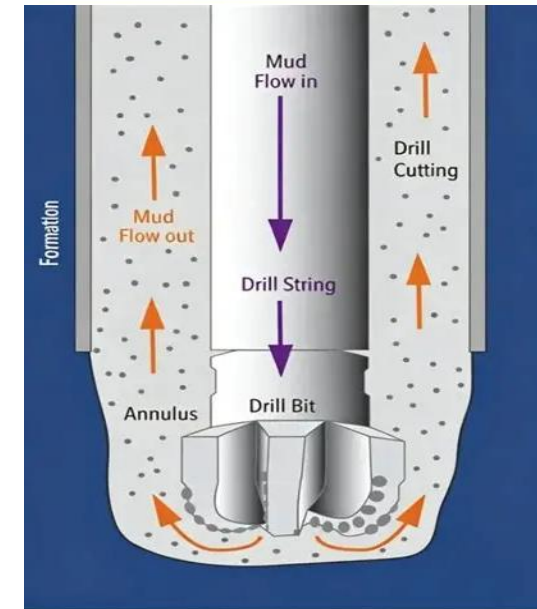
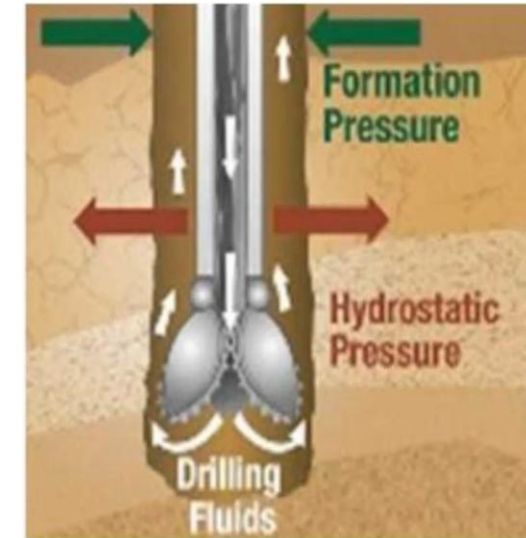
- Using 1 Master Valve 12" 900#
- Using extension/spacer spool to reduce pipe stress in the 12" production line.

DRILLING FLUID

WBM (Water Base Mud)

OBM (Oil Base Mud)

SBF (Synthetic Base Fluid)

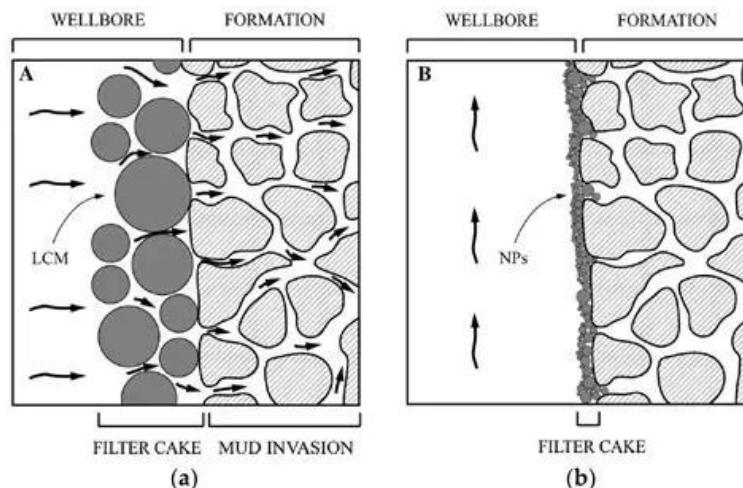


Well cutting removal by circulating drilling mud on bottom well

3. Drilling Fluid (MUD)

Drilling fluid, also known as drilling mud, is used to

- **Well Control:** Maintains hydrostatic pressure to prevent blowouts
- **Cuttings Transport:** Carries rock fragments to the surface for removal
- **Cooling and Lubrication:** Reduces wear on the drill bit and drill string
- **Wellbore Stability:** Prevents collapse or erosion of the borehole
- **Formation Protection:** Minimizes damage to productive zones and reactive clays



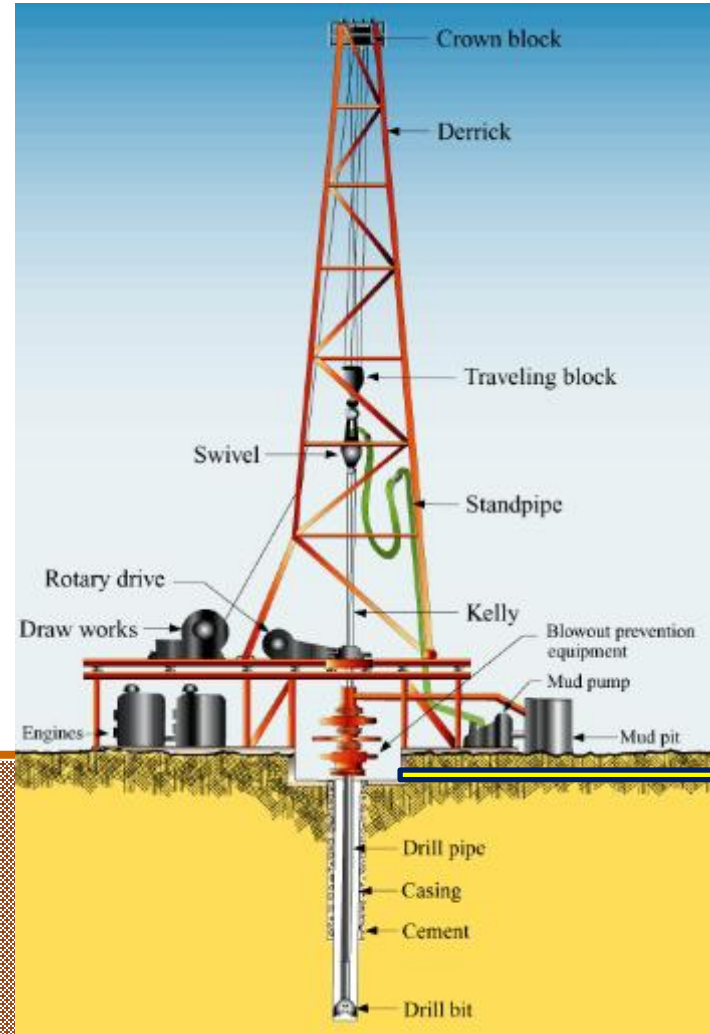
Drilling fluids are a mixture of liquid and solid components, including:

- **Clays (e.g., bentonite)** for viscosity and suspension
- **Weighting agents** (e.g., barium sulfate, calcium carbonate) to control density
- **Polymers** and thickeners (e.g., xanthan gum, guar gum, carboxymethyl cellulose) to adjust viscosity
- **Deflocculants** (e.g., lignosulfonates) to reduce viscosity
- **Corrosion inhibitors**, lubricants, and pH-control agents to protect equipment and formations
- **Specialized additives** optimize fluid loss control, suspension properties, and thermal stability, tailored to the specific wellbore conditions.

3. Mud Laboratory



3. Well Construction - Ideal Elevated Wellpad Design



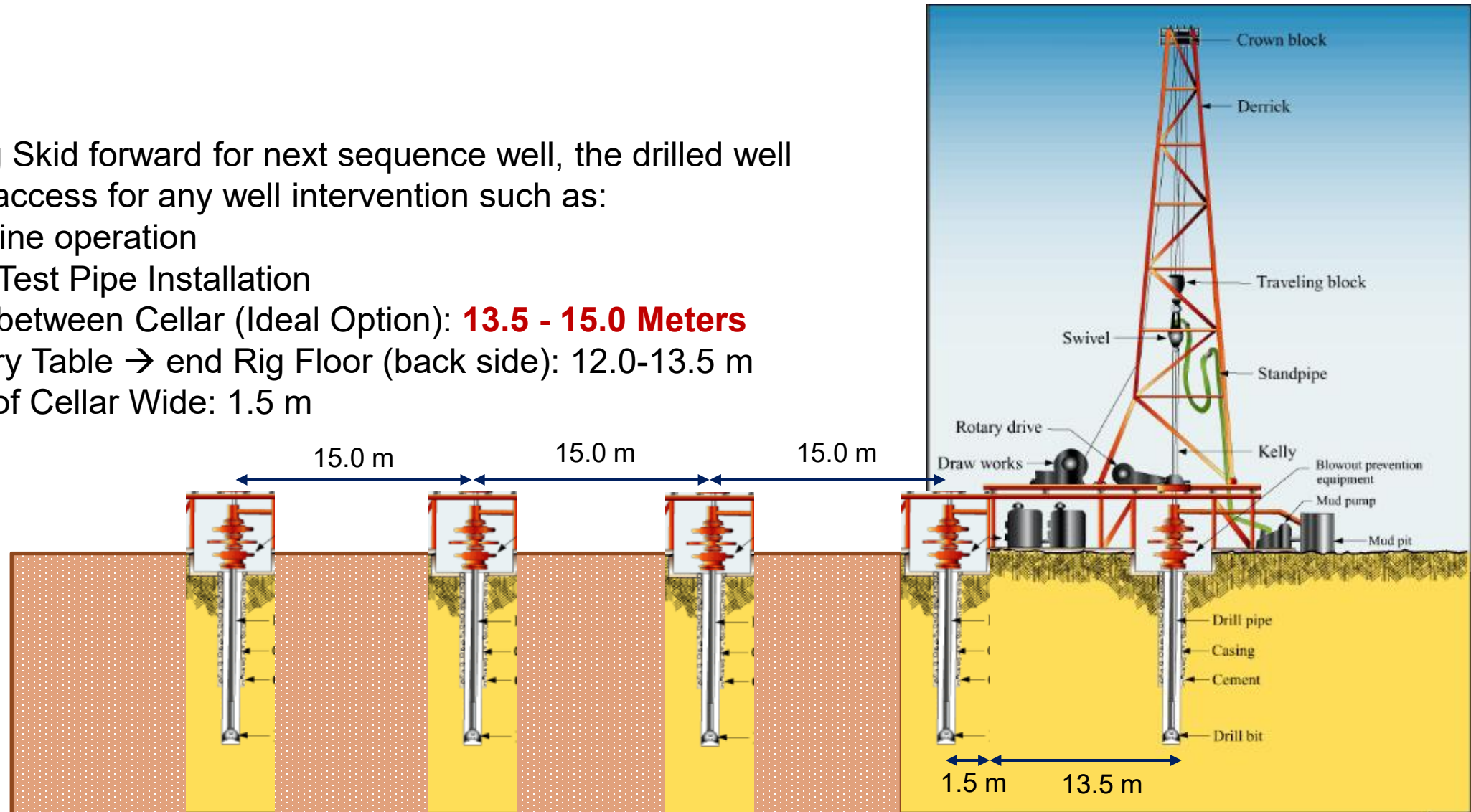
Drainage Line from Cellar to Mud Pond

Settling Pond

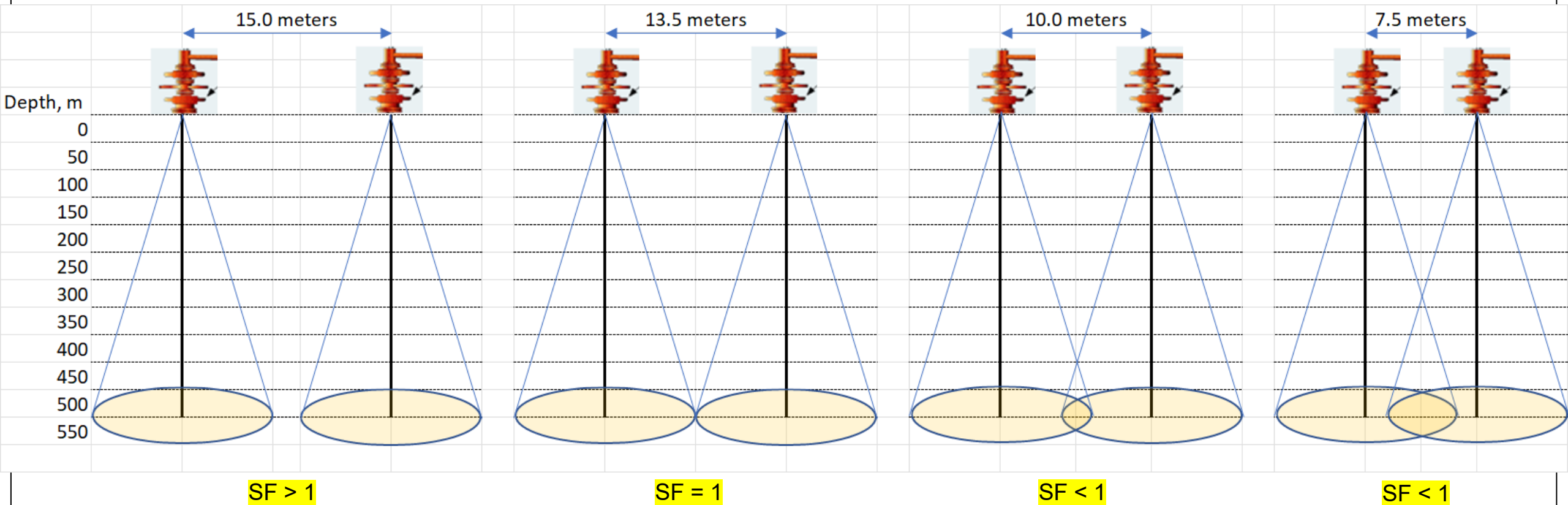
Water and Mud Pond

3. Well Construction – Ideal Distance Between Well

- ❑ When Rig Skid forward for next sequence well, the drilled well could be access for any well intervention such as:
 - Wireline operation
 - Well Test Pipe Installation
- ❑ Distance between Cellar (Ideal Option): **13.5 - 15.0 Meters**
 - Rotary Table → end Rig Floor (back side): 12.0-13.5 m
 - Half of Cellar Wide: 1.5 m



3. Well Construction – Ideal Distance Between Well



SF = Separation Factor

$SF = (\text{Ellipse Radius-1} + \text{Ellipse Radius-2}) / \text{Jarak lubang}$

3. Well Construction



Advantage:

- More simple → Less Construction Cost
- Always dry cellar
- No risk accumulated H₂S

5

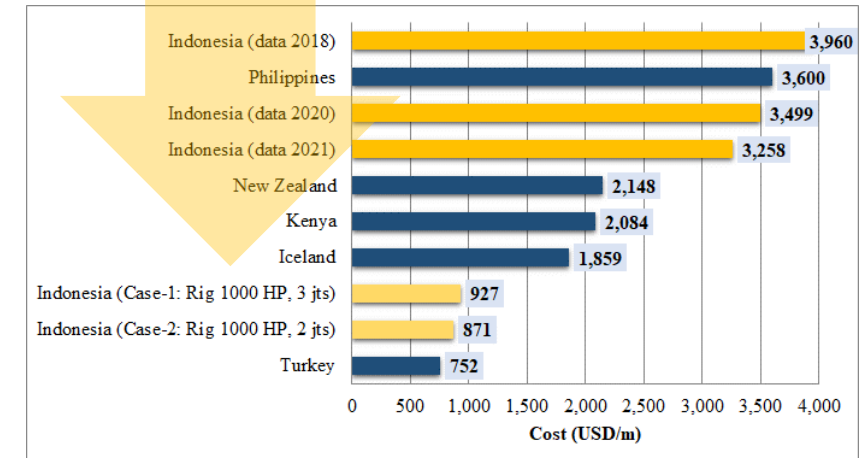
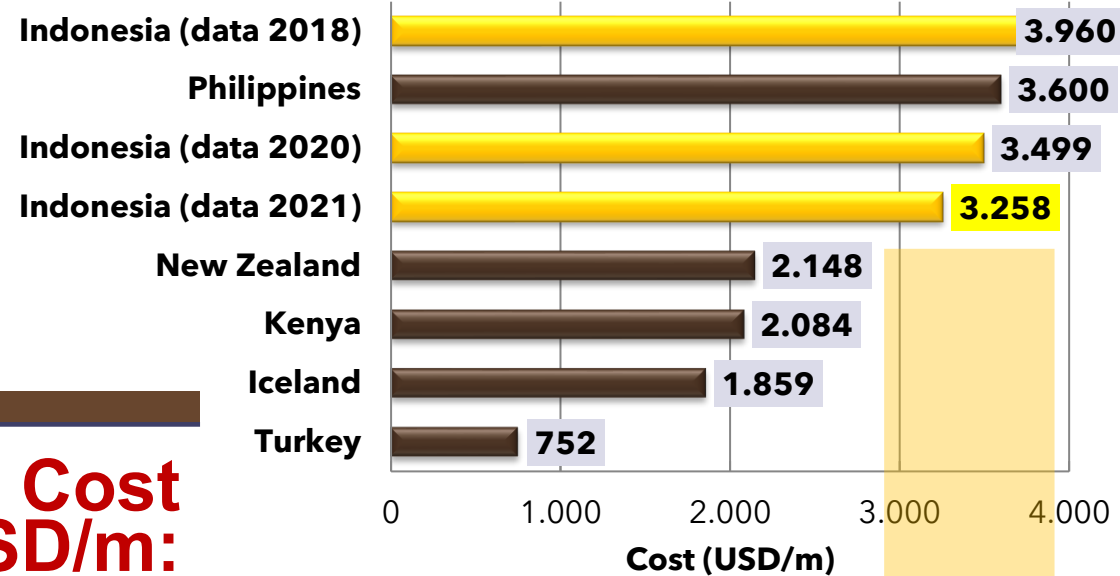
Drilling Cost & Procurement

Oil and Gas Well Cost:

- One well: USD 1 million up to 140 Million

Geothermal Well Cost USD/m:

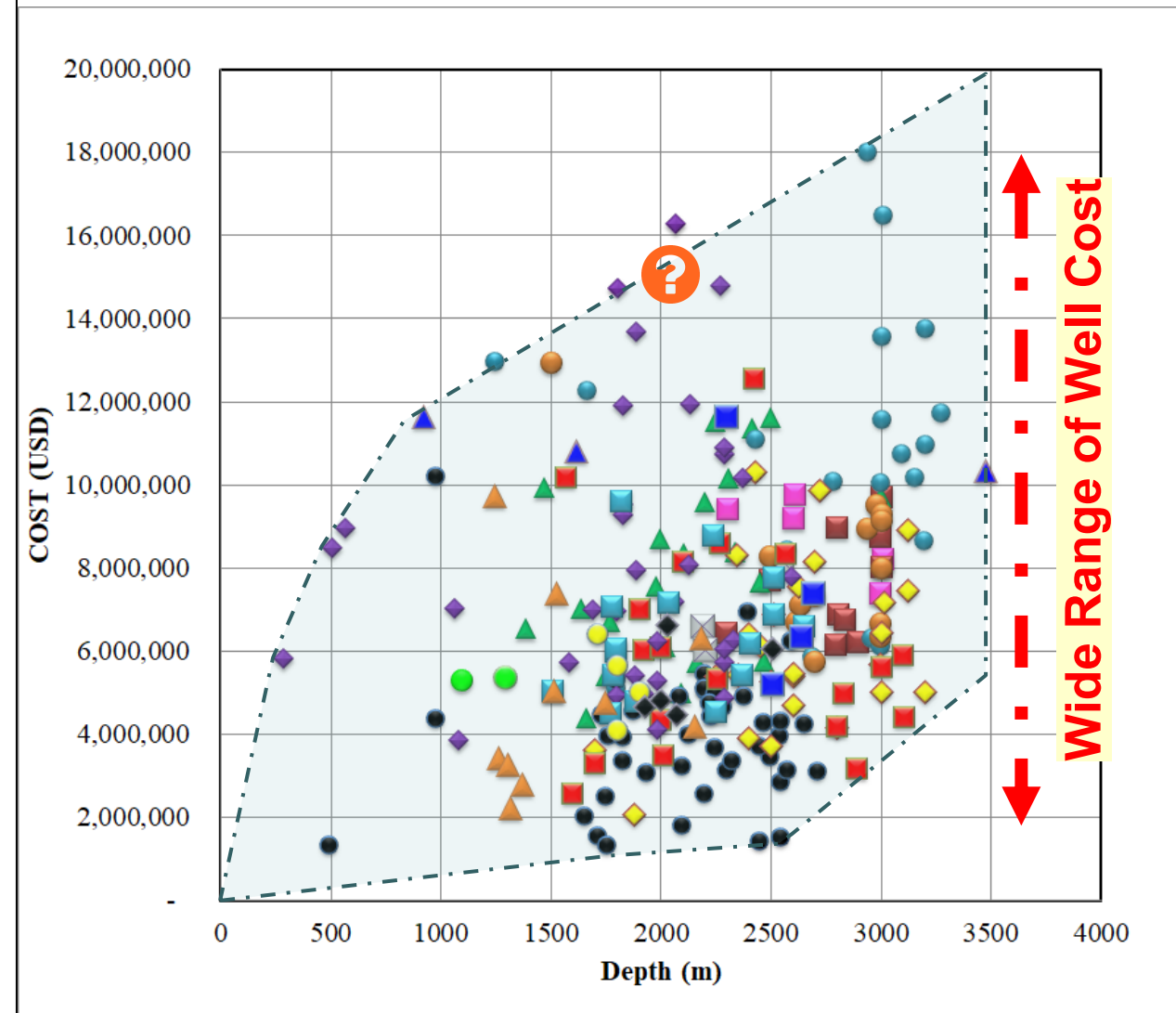
- Minimum: 2,400
- Average (year 2021): 3,258
- Maximum 7,400



SOL Geothermal Drilling Cost USD/m:



4. Geothermal Drilling Cost in Indonesia

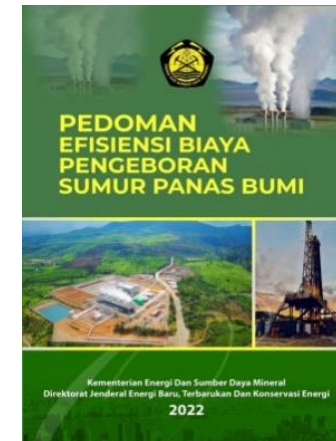
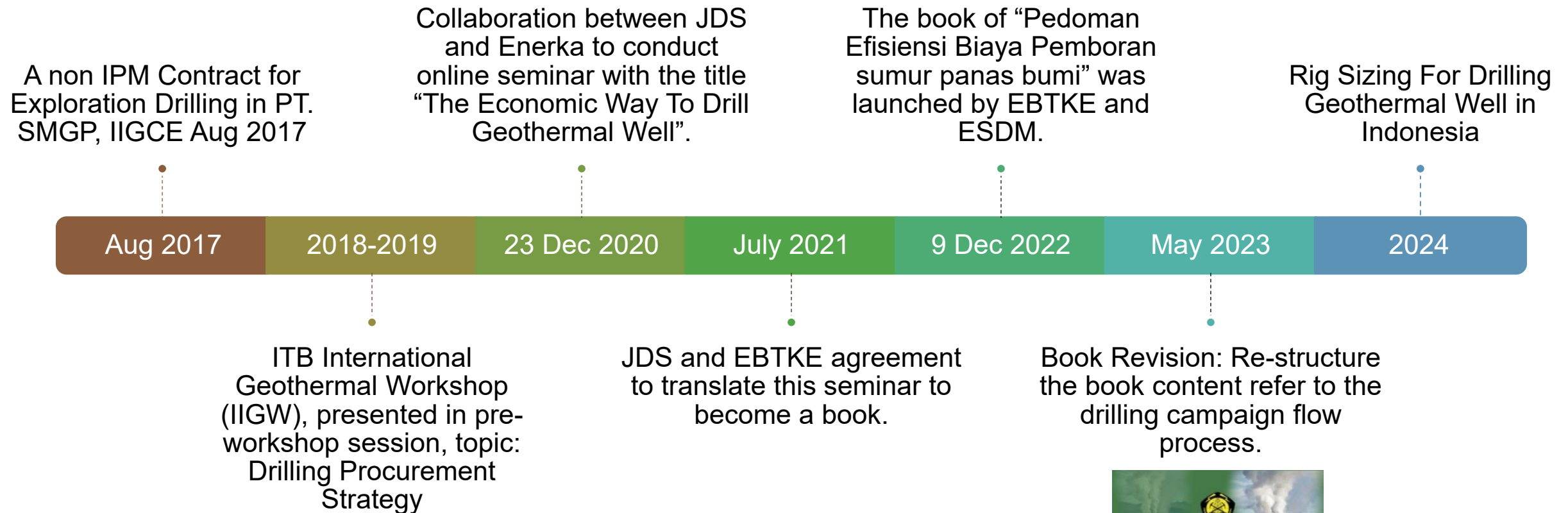


Expected < 1500 USD/m

- (2011-Q3 2021): Geothermal drilling cost taken from **242 wells**.
- Widespread of Geothermal drilling cost → **potential drilling cost saving**.
- Some area have achieved low drilling cost where **good lesson learn, and practices** have been identified and documented.
- Book, **“The Economic Way To Drill Geothermal Well”**, by JDS & EBTKE



5. The Economic Way To Drill Geothermal Well in Indonesia



5. Rank of Drilling Cost Efficiencies Method

1 Drilling Personnel Integrity

2 Procurement Strategy

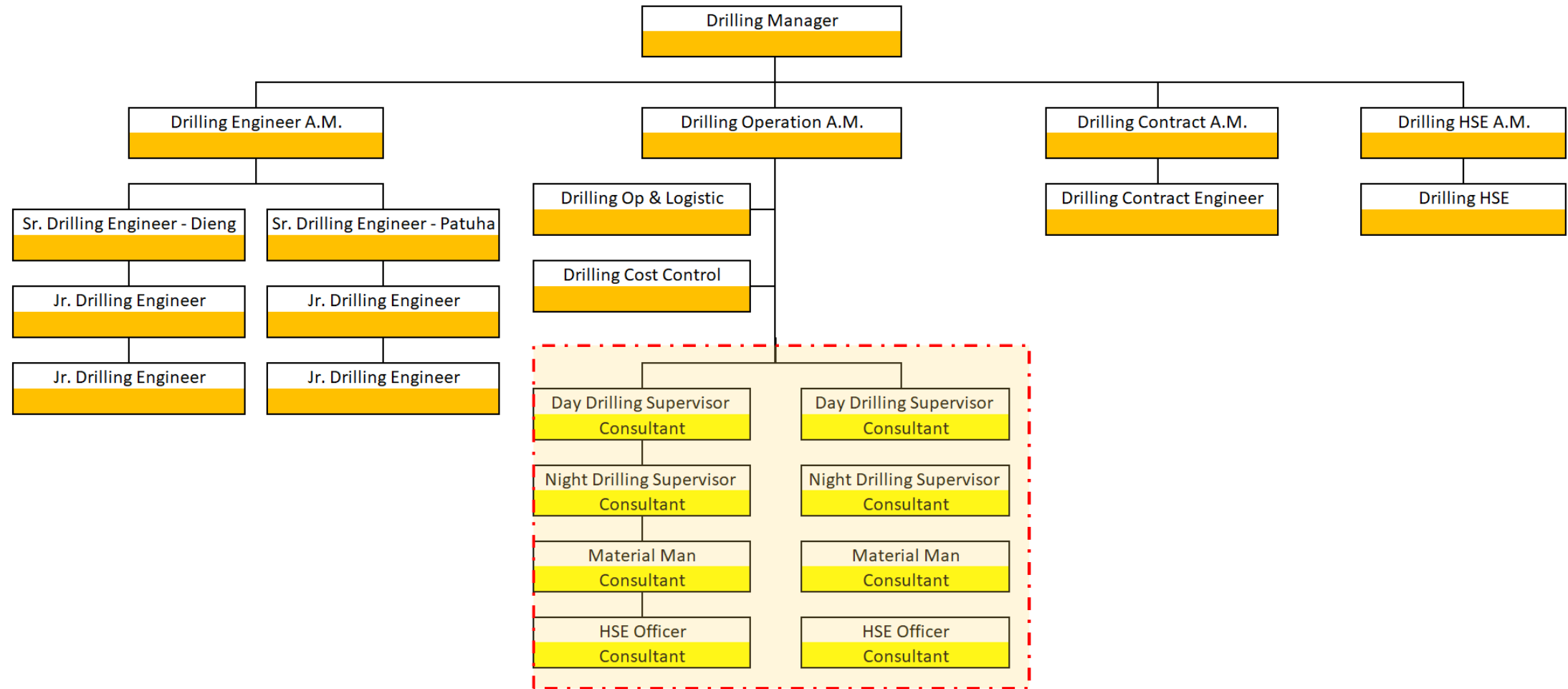
3 Rig Sizing

4 The Economic Way to drill Geothermal Well

5 Post Drilling Review and Lesson Learnt



5. Ideal Drilling Organization Chart



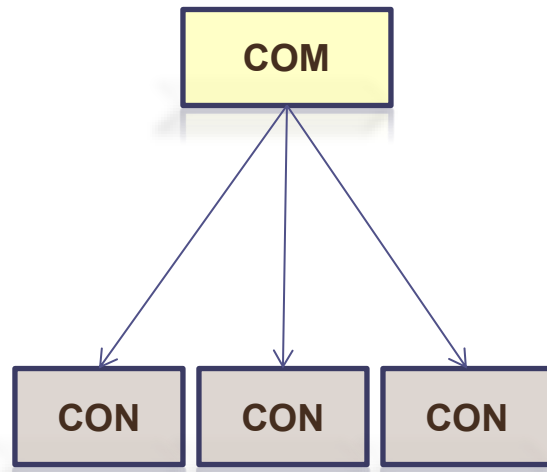


Purchase	Services	Support
1. Casing	1. Drilling Rig	1. Drilling Site Construction
2. Wellhead	2. Cement and Mud	2. Diesel Fuel
3. Liner	3. Directional Drilling	3. Water Pump Services
4. Adapter	4. Air Drilling	4. UKL / UPL
5. Drilling Bits	5. Mud Coller	5. Security Services
6. Casing	6. Wellhead Installation	6. Road Survey
Accessories	7. Drilling Personnel	7. Tubular Inspection
6. DSA and X/O	8. Heavy Equipment	8. Explosive Handling
	9. Drilling Waste Handling	9. Permits
	10. VSAT	10. Perforated Liner
	11. Drilling Report	11. Logistic Services
	12. Casing Handling	12. General Services
	13. H2S Service	
	14. Mud Logging	
	15. Drill String Inspection	
	16. Hard Bending	
	17. Solid Control Services	
	18. Fishing Tool Services	
	19. Coring Services	
	20. E-Line & Explosive	

5. Drilling Procurement Strategy

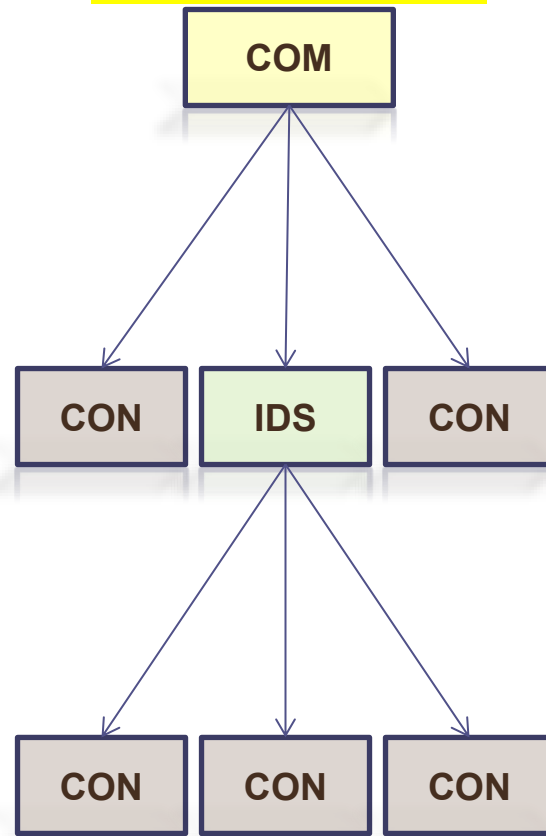
Discrete

35 Contract



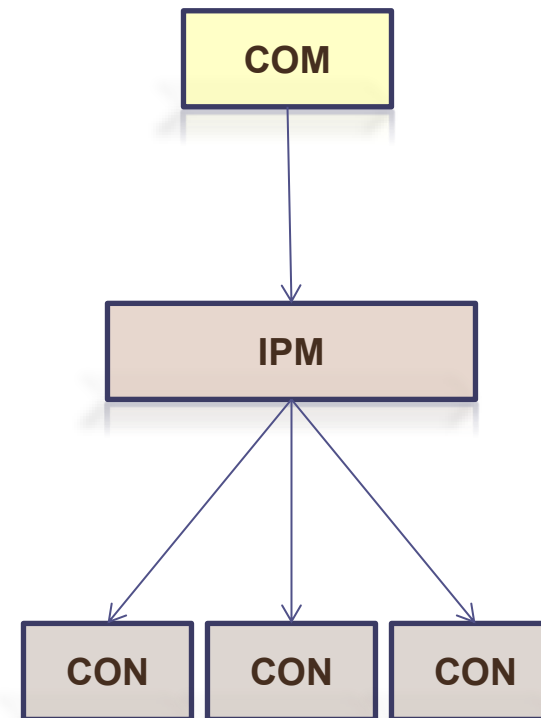
Bundled

5-10 Contract

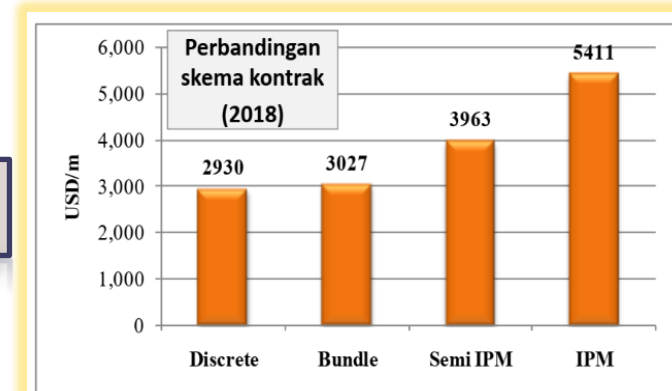


IPM / Turnkey

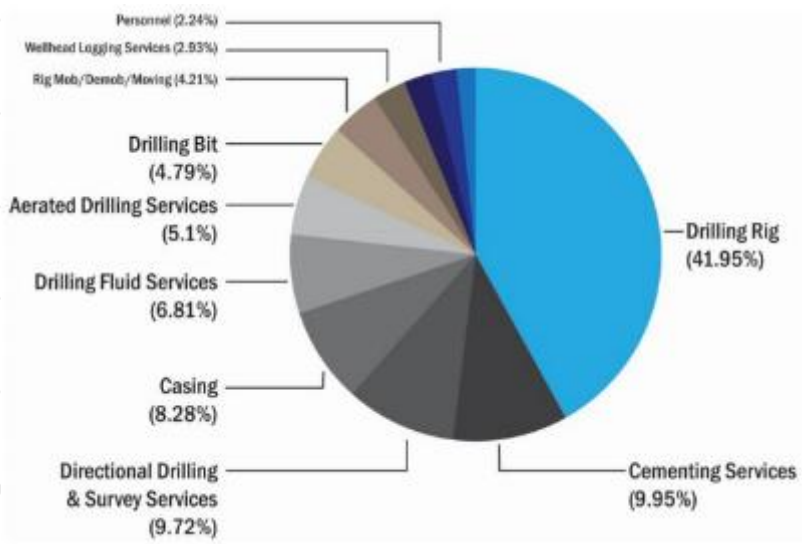
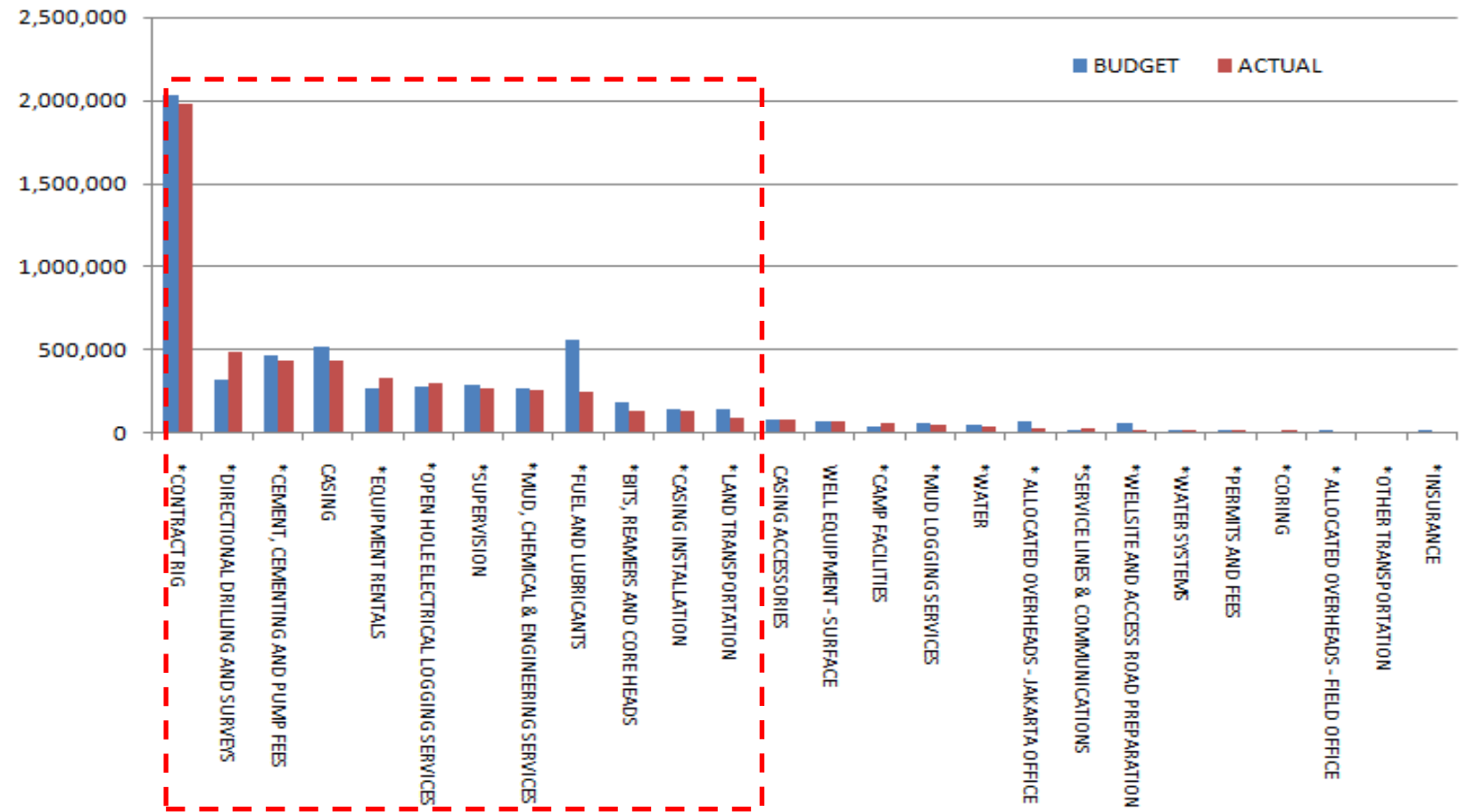
1-2 Contract



- **COM: Company**
- **CON: Contractor**
- **IDS: Integrated Drilling Services**



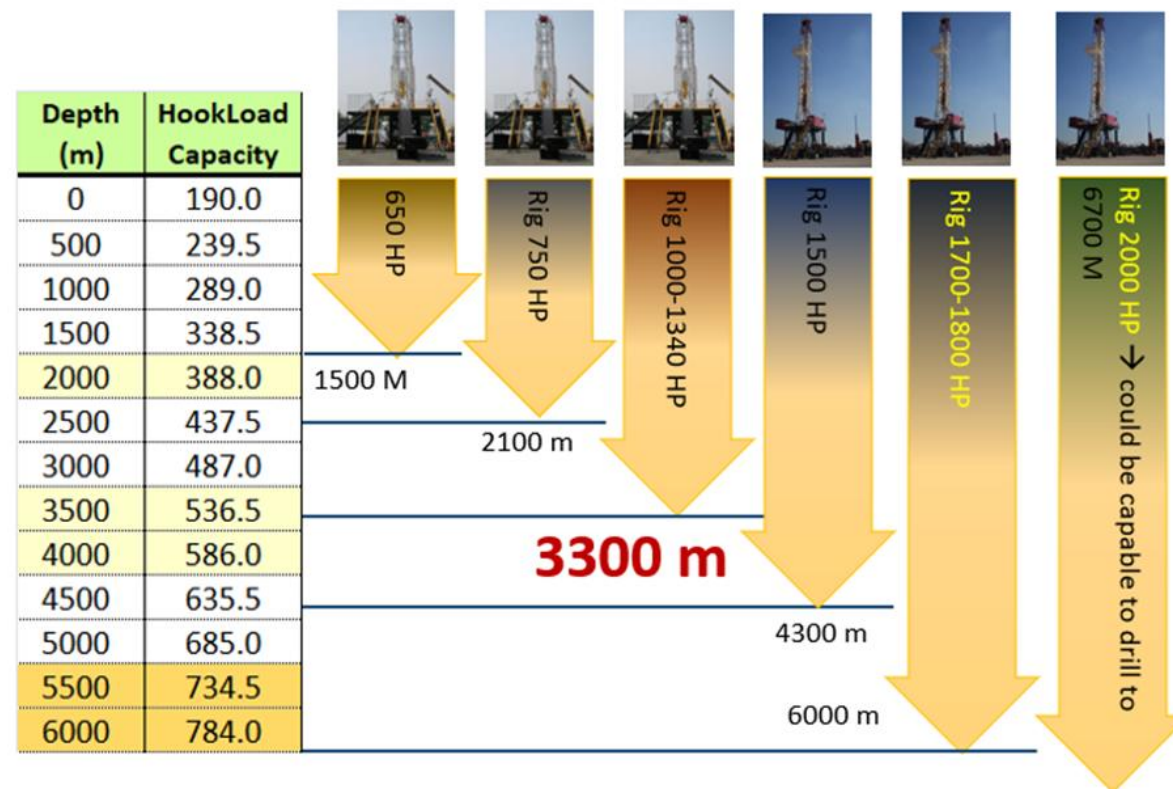
5. Drilling Procurement Strategy



More attention to the major well cost component

5. Drilling Rig Sizing

Capacity	650	700-850	1000-1340	1500	1700-1800	2000
Rated Drilling Depth, M	1800	2500 3993	3000 4100	3500 6095	5100 9144	3500 6700
Mast Capacity, Klbs	350 400	400 500	440 650	675 769	850 1000	975 1250
80% Mast Cap., Klbs	320	400	520	615	800	1000
Corrected Rated Drilling Depth, M	1500	2100	3300	4300	6000	6700



Rig Daily Rental: 25-27 \$/HP

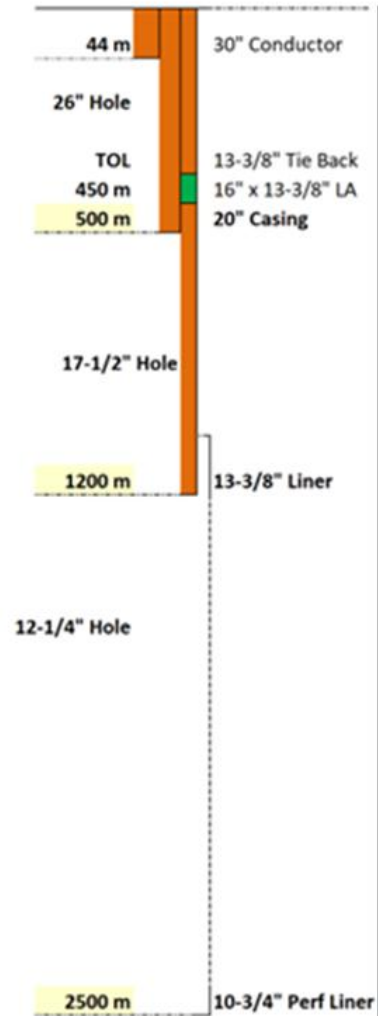
- 1000 HP = \$ 27.000 / day
- 1500 HP = \$ 37.500 / day
- 2000 HP = \$ 50.000 / day

Rig Fuel Consumption:

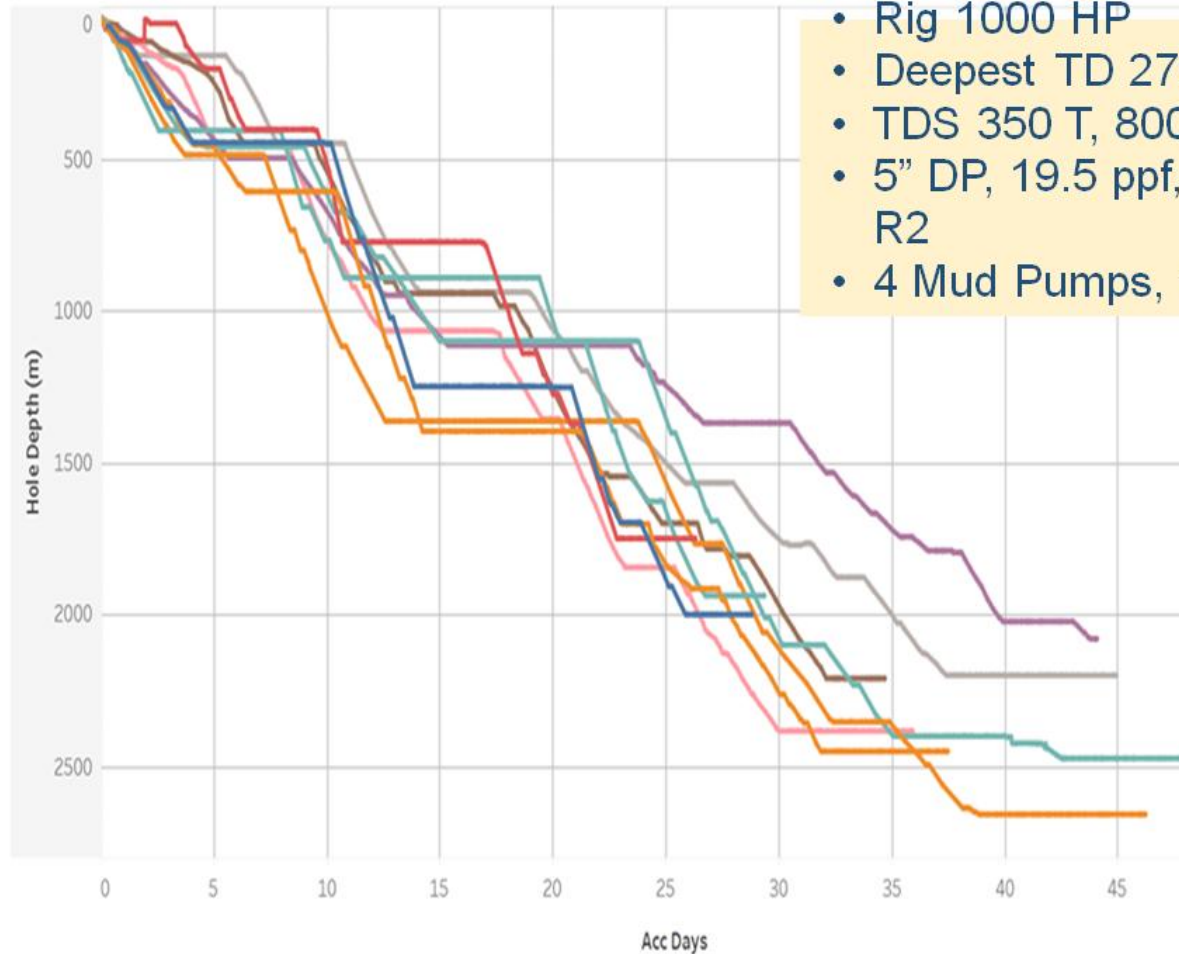
- 1000 HP = 6500 Liter / day
- 1500 HP = 12.300 Liter / day
- 2000 HP = XXXXX Liter / day

5. Drilling Procurement Strategy

**1000 HP, DATI
23**



Overall Depth vs Days

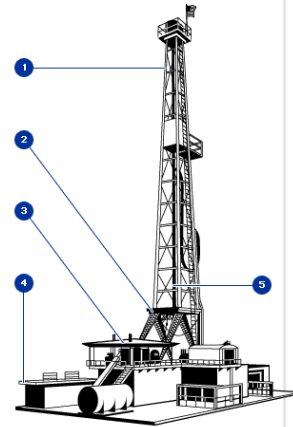
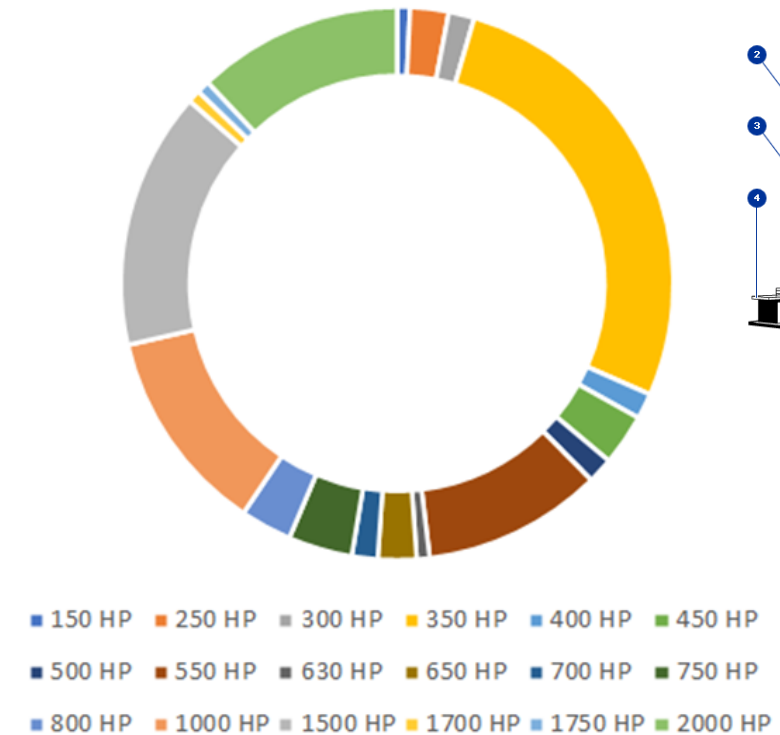


- Rig 1000 HP
- Deepest TD 2700 m-MD
- TDS 350 T, 800 HP
- 5" DP, 19.5 ppf, G-105, 4½ IF, R2
- 4 Mud Pumps, 1000 HP

5. Drilling Rig Sizing

No	Rig Company Name	No Of Rig	Rig Size	No Of Rig	%
1	PT PRA	6	150 HP	1	0.75%
2	PT PDSI	41	250 HP	3	2.26%
3	PT APEXINDO	6	300 HP	2	1.50%
4	PT Bormindo Nusantara	14	350 HP	36	27.07%
5	PT ELNUSA TBK	1	400 HP	2	1.50%
6	PT Harman	1	450 HP	4	3.01%
7	PT EPI (Exspan Petrogas Intranusa)	5	500 HP	2	1.50%
8	PT DATI	1	550 HP	14	10.53%
9	PT SMGP	1	630 HP	1	0.75%
10	PT RDN	2	650 HP	3	2.26%
11	PT APS	42	700 HP	2	1.50%
12	PT GWDC	4	750 HP	5	3.76%
13	PT Binakarindo	4	800 HP	4	3.01%
14	PT Dual OilField	1	1000 HP	16	12.03%
15	PT Tiga Musim Mas Jaya	??	1500 HP	20	15.04%
16	PT Wijaya Bangun Rigutama	4	1700 HP	1	0.75%
			1750 HP	1	0.75%
TOTAL LISTED LAND RIG			2000 HP	16	12.03%
				133	

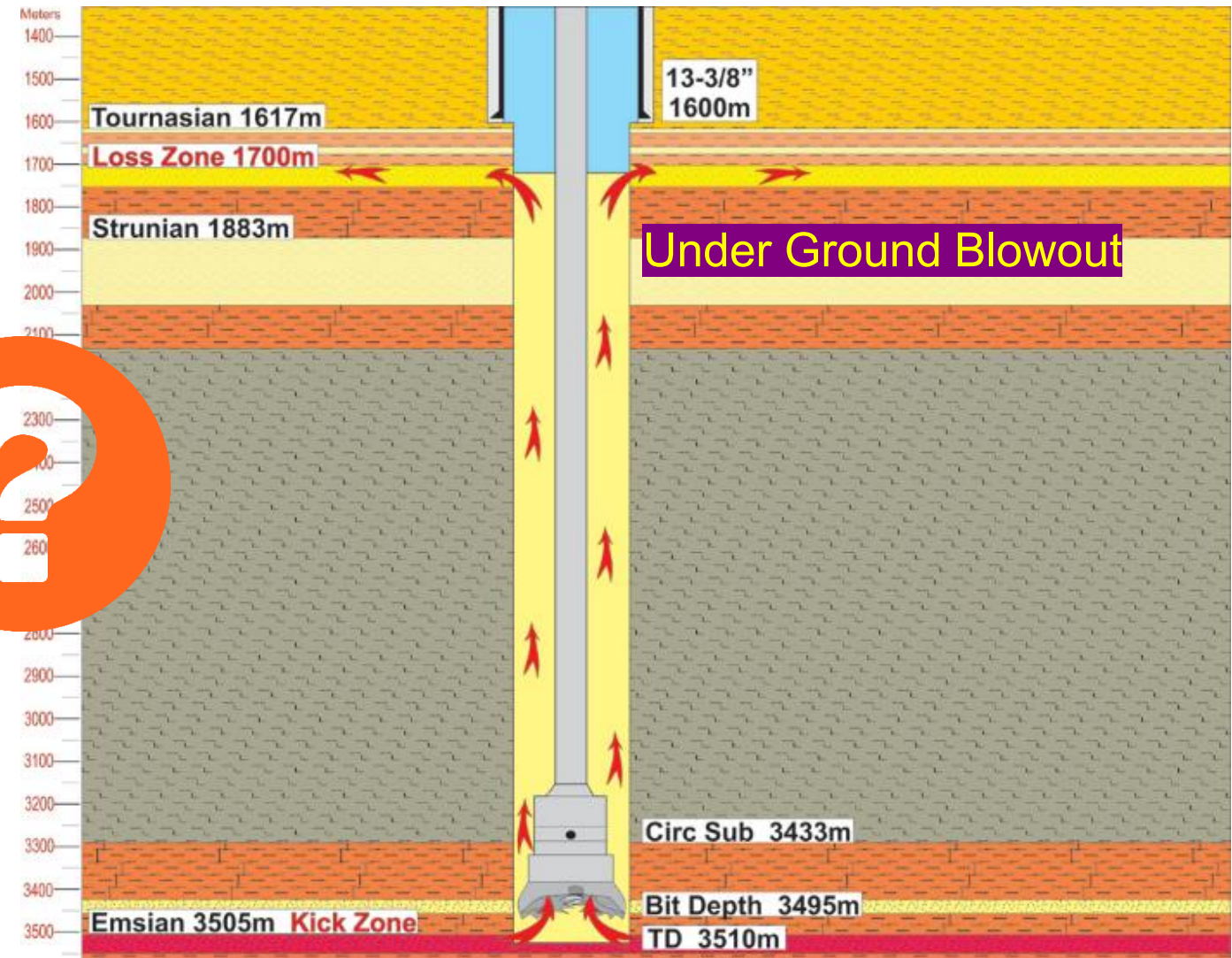
Rig Size Inventory



6

Drilling Risk and Problem

5. Drilling Problem



5. Drilling Risk



5. Drilling Risk



5. Drilling Risk



How to escape



5. Certificate for Offshore Personnel

- Well Control – Sub Sea BOP
- Sea Survival Training
- HUET (Helicopter Underwater Escape Training)
- BOSIET (Basic Offshore Safety Induction and Emergency training)



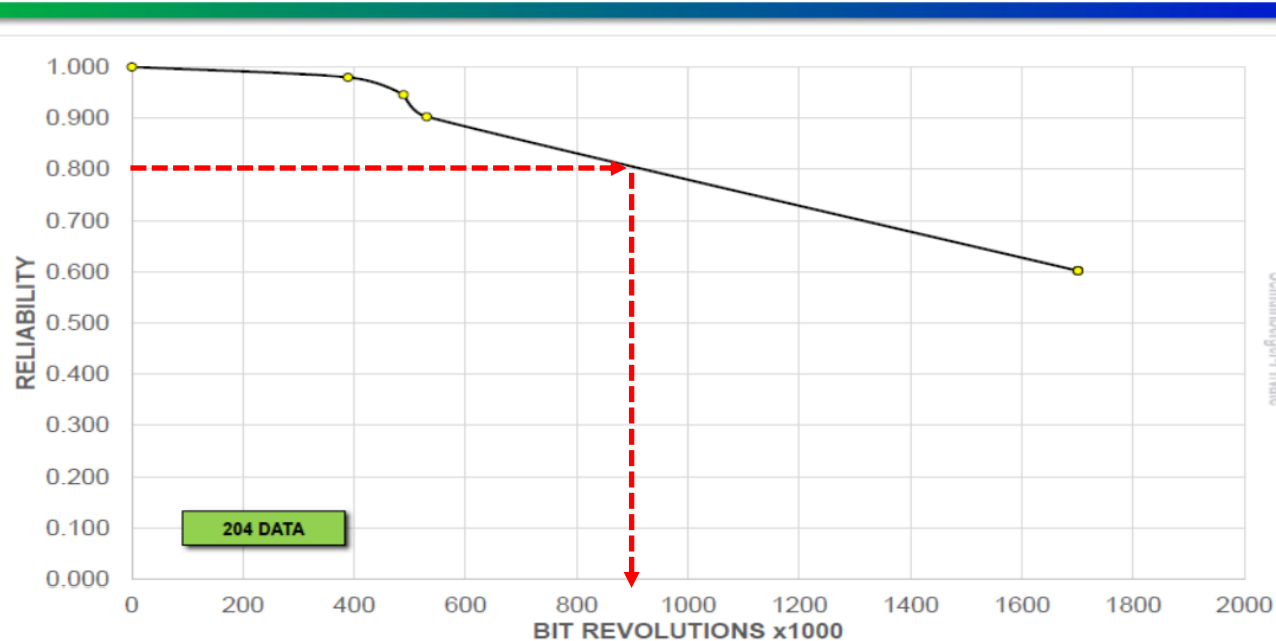


6

Drilling Exercise

6. Drilling Bit Life

26" G04 / GS04 Series
Seal/Bearing Reliability (IADC 415)



- Rotary Drill with 60 rpm, Flow rate 1200 gpm. Use 9-5/8" Mud Motor. See Mud Motor performance
- How long Bit Life ???
- **How far the bit could drill formation, if average drill speed = 15 ft per hour**

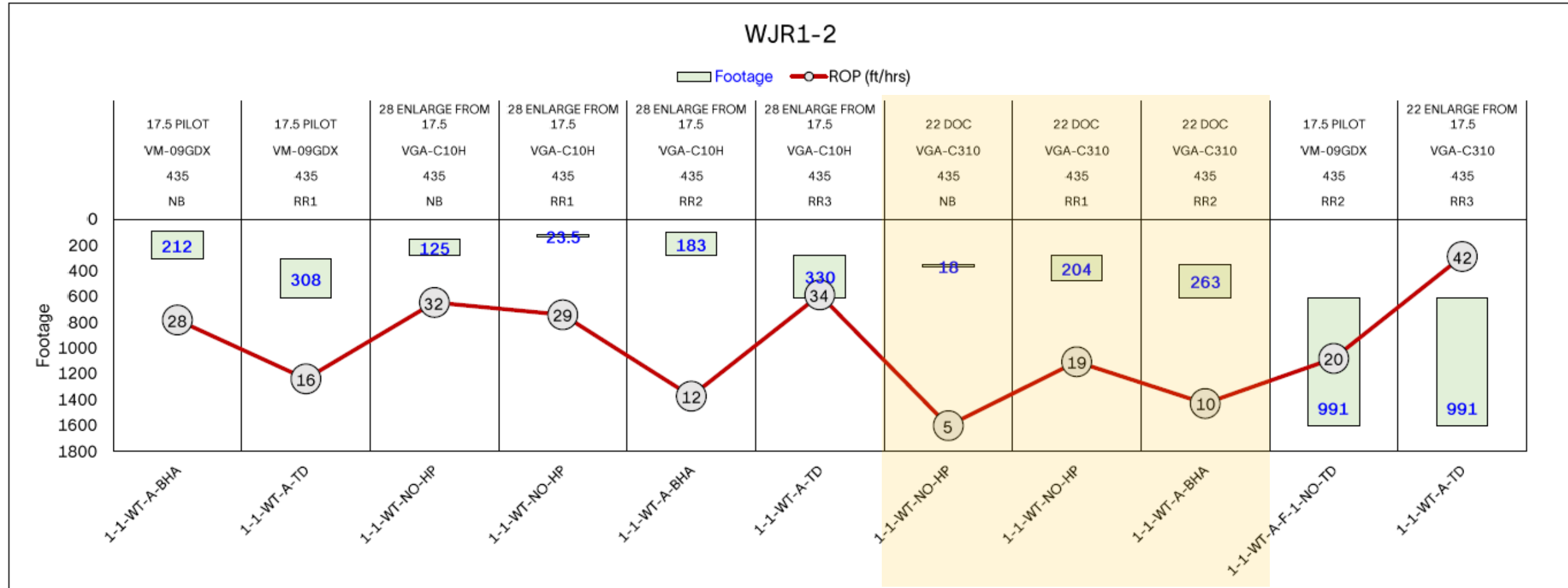
9 5/8" (244 mm) 6-7 5.0 Stage

COUGAR
DRILLING SOLUTIONS

Maximum Pump Rate	1,200 gpm	4,542 Lpm
Revolutions per unit volume	0.130 rev/gal	0.034 rev/L
Pressure at full Load	775 psi	5,343 kPa
Torque at full load	13,302 ft lb	18,035 N m

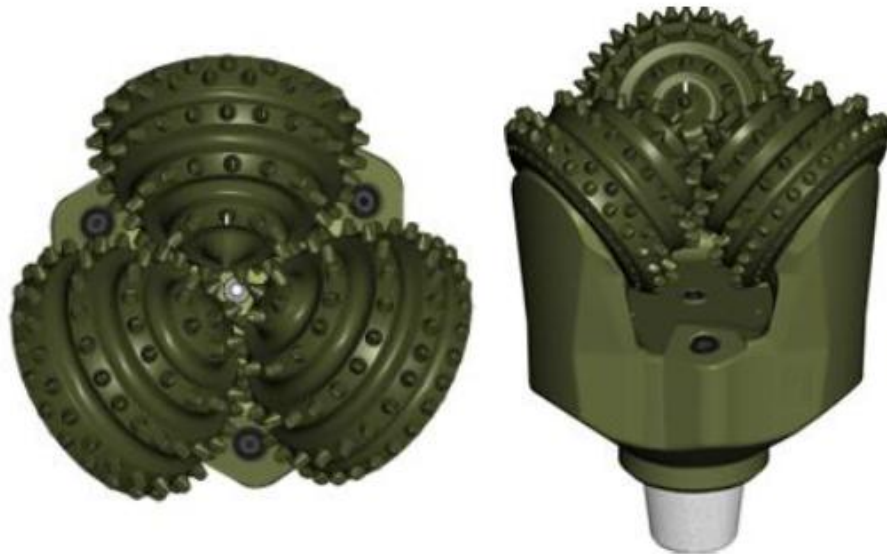
- RPM = Rotation Per Minute
- 80% Reliability, Bit revolution = 900.000 revolution
- Drill string is rotated at 60 RPM
- Down Hole Motor rotation
= 1200 gpm x 0.130 rev/gallon
= 156 RPM
- Total Bit RPM = Drill String RPM + Down Hole Motor
RPM = 60 RPM + 156 RPM = 216 RPM
- Bit Life = 80% Reliability, Bit revolution / Total Bit RPM
= 900.000 Revolution : 216 RPM
= 4166 Minutes
= 69 Hours
- Drilling Footage = ROP X Bit Life
= 15 ft per hour X 69 hours
= 1035 ft

26in Section Bit Records – cont'd



Bit Rate Of Penetration (ROP) from bit record = **Max 19 FPH, Min 10 FPH, Average 14.5 FPH**

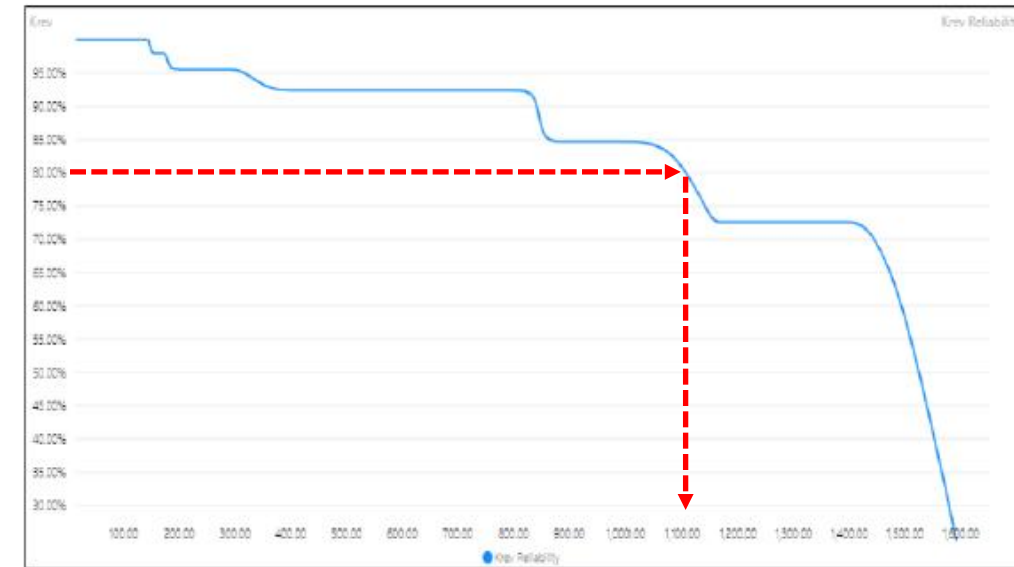
26in Section Bit Recommendation GS09ODCPS IADC: 435



Specifications

FEATURES

- The Gemini twin seal system consists of a primary seal which protects the bearing, and a secondary seal that protects the primary seal. This dual seal system will perform reliably for extended periods of time in high RPM, heavier WOB, high mud weight and severe dogleg applications.
- The sealed precision roller bearing provides a robust, reliable platform capable of excellent performance in a wide range of applications. Tight tolerance surfaces and controlled contour rollers provide significantly improved bearing life and allow the bit to drill at a high ROP for extended periods of time.
- Diamond enhanced heel inserts maintain gauge in abrasive, hard formation drilling and significantly increase cone durability. This feature extends bit life and ensures that a full gauge hole is drilled.



Bit GS09ODCPS IADC: 435, Krev at 80% Reliability = **1.100 KRev**

6. 26" Drilling Bit Life WJR1-3 Drilling Program



Wellbore Geometry

Hole Size (inch)	Casing Point (ftMD)	Casing Diameter (inch)	Schematic	Formation	MD Range(ft)
30"	155	30" Conductor		Upper Rhyolite Tuff	32 - 446
					268 - 608
					615 - 856
					827 - 1818
				Dacite	1251 - 1643
					1844 - 2571
					1048 - 1217
26"	1400	20" Casing			1144 - 1521
				Dacite Tuff	1921 - 2084
					2571 - 3211
17-1/2"	2430 (TOL) 2550	13-3/8" Casing		Middle Rhyolite Tuff	2609 - 3091
					3327 - 3594
				Andesite Lava	4041 - 4146
12-1/4"	5180 (TOL) 5300	10-3/4" Perforated Liner			5063 - 5570
9-7/8" (Contingency 7-7/8")	7200	8-5/8" Perforated Liner (Contingency 7" Liner)		Lower Rhyolite Tuff	

Drilling 26" Hole Section Needs

1. Drilling Footage = 1400 ft – 155 ft

2. Drilling Footage (a) = ft

3. Drilling ROP (b) = FPH

4. Drilling Hours = a/b = Hours

Drilling Parameters:

• Drilling flow rate = 1000 GPM

• Drilling Surface RPM = 60 RPM

• Use 9-5/8" Mud Motor 6-7 Lobes→ rev/gallon

Drilling Bit Life

• 80% Reliability, Bit revolution = revolution

• Drill string is rotated at RPM

• Down Hole Motor rotation
= gpm x 0.130 rev/gallon
= RPM

• Total Bit RPM = Drill String RPM + Down Hole Motor RPM = RPM + RPM = RPM

• Bit Life = 80% Reliability, Bit revolution / Total Bit RPM
= Revolution : RPM
= Minutes
= Hours

• Drilling Footage = ROP X Bit Life
= ft per hour X hours
= ft

6. 26" Drilling Bit Life WJR1-3 Drilling Program

Wellbore Geometry

Hole Size (inch)	Casing Point (ftMD)	Casing Diameter (inch)	Schematic	Formation	MD Range(ft)
30"	155	30" Conductor		Upper Rhyolite Tuff	32 - 446
					268 - 608
					615 - 856
				Dacite	827 - 1818
					1251 - 1643
					1844 - 2571
					1048 - 1217
26"	1400	20" Casing			1144 - 1521
				Dacite Tuff	1921 - 2084
	2430 (TOL)				2571 - 3211
17-1/2"	2550	13-3/8" Casing			
				Middle Rhyolite Tuff	2609 - 3091
					3327 - 3594
				Andesite Lava	4041 - 4146
12-1/4"	5180 (TOL)	10-3/4" Perforated Liner			5063 - 5570
	5300				
				Lower Rhyolite Tuff	
9-7/8" (Contingency 7-7/8")	7200	8-5/8" Perforated Liner (Contingency 7" Liner)			

Drilling 26" Hole Section Needs

1. Drilling Footage = 1400 ft – 155 ft
2. Drilling Footage (a) = 1245 ft
3. Drilling ROP (b) = 14.5 FPH
4. Drilling Hours = a/b = 85 Hours
5. Drilling flow rate = 1000 GPM

Drilling Bit Life

- 80% Reliability, Bit revolution = 1.100.000 revolution
- Drill string is rotated at 60 RPM
- Down Hole Motor rotation
= 1000 gpm x 0.130 rev/gallon
= 130 RPM
- Total Bit RPM = Drill String RPM + Down Hole Motor RPM
= 60 RPM + 130 RPM = 190 RPM
- Bit Life = 80% Reliability, Bit revolution / Total Bit RPM
= 1.100.000 Revolution : 190 RPM
= 5789 Minutes
= 96 Hours
- Drilling Footage = ROP X Bit Life
= 14.5 ft per hour X 96 hours
= 1399 ft

12.25in Section Bit Records

Well Name	Size (inch)	Model	IADC Codes	Run Status	SN	Depth In (ft)	Depth Out (ft)	Footage	ROP (ft/hrs)	Bit Dull
WJR1-2	12.25	KMX525T	HYB	NB	5340314	3028	5300	2272	38.9	TCI: 1-1-WT-A-E-IN-NO-TD PDC: 1-1-WT-A-X-IN-NO-TD



Hole Size (inch)	Casing Point (ftMD)	Casing Diameter (inch)	Schematic	Formation
12-1/4"	5180 (TOL) 5300	10-3/4" Perforated Liner		Middle Rhyolite Tuff Andesite Lava

In WJR1-2, hybrid bit was used to drill the section from 3028 to 5300 ft (2272 footage) in 1 run. With ROP 38.9 ft/hr, the final bit dull grading is 1-1-WT-A-TD for both PDC and TCI parts. Based on the dull grading, bit only experienced minor wear in all area.

For WJR1-3 or WJR1-4, planned interval is from 2550 to 5300 ft, will be drilling through formation middle Rhyolite Tuff, Andesite Lava and Lower Rhyolite Tuff.

12.25in Section Bit Recommendation (for DOC) GFS30 BOD1VCPD IADC 537

There are 1534 bit runs



Specifications

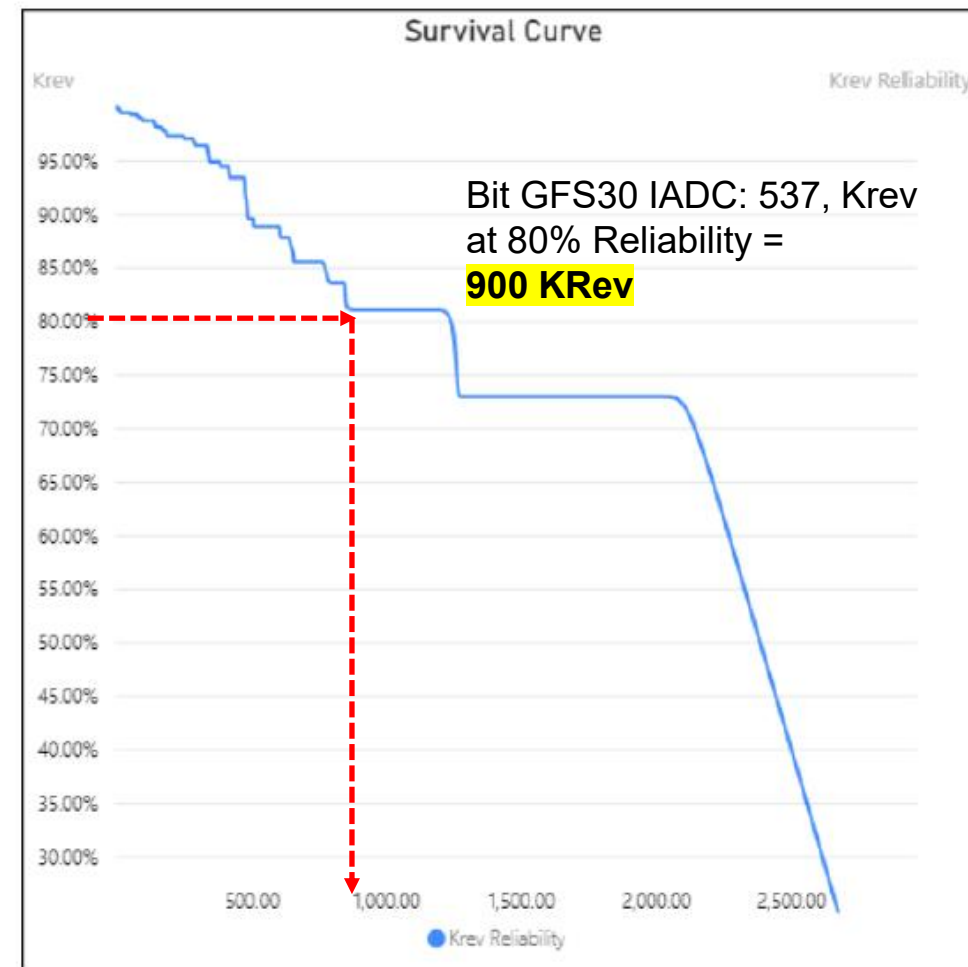
Bearing Type	Spinodal 2 Bearing w/ Silver Plated Components
Seal Type	Gemini Twin Seal System
Bit Connection Type	6 5/8 Reg
Rows	Total: 14 Inner: 11 Gauge: 3
Inserts and Teeth	Total: 195 Inner: 135 Gauge: 60

Operating Parameters

Weight-on-Bit	25000 to 60000 (lbf) 11000 to 27000 (daN) 11 to 27 (Tonnes)
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FEATURES

- The Gemini twin seal system consists of a primary seal which protects the bearing, and a secondary seal that protects the primary seal. This dual seal system will perform reliably for extended periods of time in high RPM, heavier WOB, high mud weight and severe dogleg applications. 
- Spinodal 2 bearing with silver plated components ensures longer runs at higher ROP. This proprietary material offers maximum wear resistance and withstands extreme load forces for longer periods than conventional bearing materials. 
- Binary carbide gauge places wear resistant semi-round top carbide inserts between the primary gauge inserts to enhance gauge retention in transitional drilling. This substantially improves gauge durability and allows longer in-gauge bit runs. 
- 100% diamond enhanced heel row inserts provide maximum durability in extremely abrasive applications. The bit will drill longer intervals of full gauge hole, and eliminate the need for expensive reaming operations. 
- V-Flo™ directs nozzles to the leading side of the following bit cone to maximize cleaning. The resulting upward spiral flow enhances bottom hole cleaning, prevents bit balling and allows the bit to drill at maximum ROP. 
- The Center Jet maximizes fluid flow and increases ROP by 



6. 12-1/4" Drilling Bit Life WJR1-3 Drilling Program



Wellbore Geometry

Hole Size (inch)	Casing Point (ftMD)	Casing Diameter (inch)	Schematic	Formation	MD Range(ft)
30"	155	30" Conductor		Upper Rhyolite Tuff	32 - 446
					268 - 608
					615 - 856
					827 - 1818
				Dacite	1251 - 1643
					1844 - 2571
					1048 - 1217
					1144 - 1521
26"	1400	20" Casing			
				Dacite Tuff	1921 - 2084
					2571 - 3211
17-1/2"	2430 (TOL) 2550	13-3/8" Casing			
				Middle Rhyolite Tuff	2609 - 3091
					3327 - 3594
				Andesite Lava	4041 - 4146
12-1/4"	5180 (TOL) 5300	10-3/4" Perforated Liner			5063 - 5570
9-7/8" (Contingency 7-7/8")	7200	8-5/8" Perforated Liner (Contingency 7" Liner)		Lower Rhyolite Tuff	

Drilling 12-1/4" Hole Section Needs

Drilling Bit Life

1. Drilling Footage = 5300 ft – 2550 ft
2. Drilling Footage (a) = ft
3. Drilling ROP (b) = FPH
4. Drilling Hours = a/b = Hours

Drilling Parameters:

- Drilling flow rate = 1000 GPM
- Drilling Surface RPM = 70 RPM
- Use 8-1/2" Mud Motor 6-7 Lobes→ rev/gallon

- 80% Reliability, Bit revolution = revolution
- Drill string is rotated at RPM
- Down Hole Motor rotation
= gpm x 0.130 rev/gallon
= RPM
- Total Bit RPM = Drill String RPM + Down Hole Motor RPM = RPM + RPM = RPM
- Bit Life = 80% Reliability, Bit revolution / Total Bit RPM
= Revolution : RPM
= Minutes
= Hours
- Drilling Footage = ROP X Bit Life
= ft per hour X hours
= ft

HOW MANY BIT WE NEED TO DRILL 12-1/4" HOLE SECTION???

CASE-1:
DURING DRILLING 12-1/4" HOLE, DRILLING SMOOTHLY, NO HOLE PROBLEM AND DECIDED TO DRILL 12-1/4" TO TD AT 7200 FT-MD. HOW MANY BIT WE NEED TO DRILL 12-1/4" HOLE SECTION???

6. 12-1/4" Drilling Bit Life WJR1-3 Drilling Program

Wellbore Geometry

Hole Size (inch)	Casing Point (ftMD)	Casing Diameter (inch)	Schematic	Formation	MD Range(ft)
30"	155	30" Conductor		Upper Rhyolite Tuff	32 - 446
					268 - 608
					615 - 856
					827 - 1818
				Dacite	1251 - 1643
					1844 - 2571
					1048 - 1217
					1144 - 1521
26"	1400	20" Casing			
				Dacite Tuff	1921 - 2084
					2571 - 3211
17-1/2"	2430 (TOL) 2550	13-3/8" Casing			
				Middle Rhyolite Tuff	2609 - 3091
					3327 - 3594
				Andesite Lava	4041 - 4146
12-1/4"	5180 (TOL) 5300	10-3/4" Perforated Liner			5063 - 5570
9-7/8" (Contingency 7-7/8")	7200	8-5/8" Perforated Liner (Contingency 7" Liner)		Lower Rhyolite Tuff	

Drilling 12-1/4" Hole Section Needs

Drilling Bit Life

1. Drilling Footage = 5300 ft – 2550 ft
2. Drilling Footage (a) = 2750 ft
3. Drilling ROP (b) = 38.9 FPH
4. Drilling Hours = a/b = 70 Hours

Drilling Parameters:

- Drilling flow rate = 1000 GPM
- Drilling Surface RPM = 70 RPM
- Use 8-1/2" Mud Motor 6-7 Lobes→ 0.130 rev/gallon

- 80% Reliability, Bit revolution = 900.000 revolution
- Drill string is rotated at 70 RPM
- Down Hole Motor rotation = 1000 gpm x 0.130 rev/gallon = 130 RPM
- Total Bit RPM = Drill String RPM + Down Hole Motor RPM = 70 RPM + 130 RPM = 200 RPM
- Bit Life = 80% Reliability, Bit revolution / Total Bit RPM = 900.000 Revolution : 200 RPM = 4500 Minutes = 75 Hours
- Drilling Footage = ROP X Bit Life = 38.9 ft per hour X 75 hours = 2917 ft

HOW MANY BIT WE NEED TO DRILL 12-1/4" HOLE SECTION???

CASE-1:
DURING DRILLING 12-1/4" HOLE, DRILLING SMOOTHLY, NO HOLE PROBLEM AND DECIDED TO DRILL 12-1/4" TO TD AT 7200 FT-MD. HOW MANY BIT WE NEED TO DRILL 12-1/4" HOLE SECTION???

Thank You!

